

Number of pages in this package 131 [including additional pages -]
(Fill in when using printed copy as record)

CLIENT INFORMATION	
Company Name	JZPE POWER TRANSFORMER (JIANGSU) CO., LTD.
Address	NO.79, WEST 17 GROUP, JINTAN VILLAGE, YUXI TOWN, JIANGYAN DISTRICT, TAIZHOU CITY, JIANGSU PROVINCE, 225500 CHINA

AUDIT INFORMATION:	
Description of Tests Per Standard Number and Edition/ Revision Date	IEEE C57.12.00, Issue Date 11/09/2021
	IEEE C57.12.90, Issue Date 11/09/2021
	CSA C88, Edition 5, Issue Date 11/2016
☒ Tests Conducted by ¹ Ma guang	
☐ UL Staff supervising UL Staff in training	
☒ UL Staff conducting or witnessing testing (WTDP, CTF Stage 1 or 2 only and Stage 3 or 4 as applicable)	My printed name is my declaration that all results recorded in this datasheet package are factual and accurate, and that I was present during the testing and data acquisition, where such presence is required by UL CB Scheme and/or WTDP rules. ☐ Where Live Stream Video (LSV) is used for remote witness testing of CTF Stage 2, 3 or 4, and/or WTDP, it is clearly declared in this datasheet package. Sam Wu 2024-07-11 Printed Name Include Date of testing completion for Printed Name entry
☐ Authorized Signatory (CTDP, TPTDP, TCP, PPP, CTF Stage 3 or 4)	Printed Name Signature. Include date for CTDP, TPTDP, TCP, PPP, CTF Stage 3 or 4

TESTS TO BE CONDUCTED:			
Test No.	Done ³	Test Name	☒ Comments/Parameters ☐ Tests Conducted by ² ☐ Link to separate data files ⁴ ☐ CTF Stage/LSV declaration ⁵
1	X	Ratio and phase relation tests	PASS, judged by engineer
2	X	Resistance Measurements	PASS, judged by engineer
3	X	Low-frequency test on auxiliary devices and control tests	PASS, judged by engineer
4	X	No-load losses and excitation current	PASS, judged by engineer
5	X	Impedance voltage and load loss	PASS, judged by engineer

TESTS TO BE CONDUCTED:			
Test No.	Done ³	Test Name	☒ Comments/Parameters ☐ Tests Conducted by ² ☐ Link to separate data files ⁴ ☐ CTF Stage/LSV declaration ⁵
6	X	Operation tests of all devices Mechanical tests	PASS, judged by engineer
7	X	Zero-phase-sequence impedance test	PASS, judged by engineer
8	X	Control (auxiliary) and cooling consumption losses	PASS, judged by engineer
9	X	Audible sound levels	PASS, judged by engineer
10	X	Temperature Rise Test	PASS, judged by engineer
11	Dielectric tests		
11-1	X	Lightning impulse	PASS, judged by engineer
11-2	X	Applied-voltage tests	PASS, judged by engineer
11-3	X	Induced-voltage test and Partial discharge Test	PASS, judged by engineer
11-4	X	Insulation resistance	PASS, judged by engineer
11-5	X	Insulation power-factor tests	PASS, judged by engineer
12	X	1 h excitation run at 110% of rated voltage Follow by exciting current and loss at 90%, 100%, and 110% of rated voltage	PASS, judged by engineer
13	X	Dissolved gas in oil analysis	PASS, judged by engineer
14	X	Pressure and Leak	PASS, judged by engineer
15	X	Lifting and moving devices	PASS, judged by engineer
16	X	Test on current transformer	For reference only

Instructions -

- 1 - When all tests are conducted by one person, name can be inserted here instead of including name on each page containing data.
- 2 - When test conducted by more than one person, name of person conducting the test can be inserted next to the test name. Test dates may be recorded here instead of entering test dates on the individual datasheet pages.
- 3 - Use of this field is optional and may be employed differently. If used to include a date instead of entering the testing date on the individual datasheet pages, the date shall be the date the test was conducted.
- 4 - Link to separate data files for a test can be inserted here. The link must be to a server that is accessible to UL staff, that provides for backup, required retention periods and a path, including file name, that does not change and result in a broken link. Not applicable to DAP.

5 - Indication of use of Live Stream Video by entry of "LSV" when limited to specific tests. Otherwise, it is applicable to all tests when "LSV" is selected in the TEST LOCATION table. See table for applicable Note.

Special Instructions -

[] Unless specified otherwise in the individual Methods, the tests shall be conducted under the following ambient conditions. Confirmation of these conditions shall be recorded at the time the test is conducted.

Ambient Temperature, C _____ ? Relative Humidity, % _____ ? Barometric Pressure, mBar _____ ?

[X] No general environmental conditions are specified in the Standard(s) or have been identified that could affect the test results or measurements.

RISK ANALYSIS RELATED TO TESTING PERFORMANCE:

The following types of risks have been identified. Take necessary precautions. This list is not all inclusive.

[X] Electric shock	[] Radiation
[X] Energy related hazards	[] Chemical hazards
[X] Fire	[X] Noise
[X] Heat related hazards	[] Vibration
[X] Mechanical	[] Other (Specify) __?

WITNESS TEST DATA PROGRAM (WTDP) INFORMATION:

Environment:	
Accommodations and Environmental conditions, including proper power source meet the requirements of the test standard or UL default criteria (ISO/IEC 17025:2017 6.3.1, 6.3.2, 6.3.3, 6.3.4)	☒ Yes ☐ No ☐ N/A
Personnel:	
Lab Management shall authorize personnel to operate particular types of equipment used in testing. (ISO/IEC 17025:2017 6.2.3, 6.2.5e)	☒ Yes ☐ No
Technical personnel is knowledgeable in the risks and hazards associated with conducting safety testing, including laboratory safety regulations, safeguards and procedures to reduce laboratory risks. (OSHA Directive Annex A 5.2 A iii)	☒ Yes ☐ No
Equipment:	
Testing is being conducted within the test equipment calibration dates. (See Test Instrument Information Page and ISO/IEC 17025:2017 6.4.1, 6.4.2, 6.4.4, 6.4.5, 6.4.6, 6.4.8, 6.4.13)	☒ Yes ☐ No
Calibrations for testing equipment is traceable to SI Units. Refer to Client/Customer Support Guidance-ULID-000819 (DCS:00-OP-C0032) (Calibration Certificate Analysis). (ISO/IEC 17025:2017 6.5.1, 6.5.2, 6.5.3)	☒ Yes ☐ No
Equipment that does not need to be calibrated against documented specifications and/or procedures must be verified before use. (OSHA Directive Annex A 5.5 E)	☒ Yes ☐ No
Critical Consumables:	
Critical consumables are compliant with test standard requirements. (ISO/IEC 17025:2017 6.6)	☒ Yes ☐ No ☐ N/A
Sample Identification:	
Identification of items to be tested has been made (e.g. model no., Serial No., etc.) (See Test Sample Identification page and ISO/IEC 17025:2017 7.4.2)	☒ Yes ☐ No
Additional Requirements: (Standard Operation Procedures-ULID-000897 (DCS:00-OP-S0854) 14.7)	
If testing is to be performed at an independent lab that is not affiliated with the applicant, confirmation is required that the appropriate agreement is in place. -If the test location is contracted by the customer and does not have an existing UL data acceptance agreement (L56, L53, Third Party by Invitation (TPI), or Subcontractor Services Agreement (SSA)), the Mutual Non-Disclosure and Witnessed Testing Agreement, Form-ULDC-000508 (DCS:00-LE-F0860), is required. The executed agreement shall be stored in Oracle under "Witnessed Testing Agreement". -If the test location is contracted by UL and is not part of the UL DAP (CTDP /TPTDP), the Subcontractor Services	☐ Yes ☐ No ☒ N/A

Agreement (SSA) is required. Contact Global Sourcing for details.		
Test Facility Party Site Number:		
The test facility has an identified party site number	☒ Yes ☐ No If Yes, Enter Party Site Number: [3511272]	
Summary: 2024-07-10		
The test facility ☐ was not deemed to have the environment and capabilities necessary to perform the tests included in this data package.		

~~☐ The CAS Staff as indicated below, (a competent L1, L2 or L3 in a similar CCN/Standard for a similar test method) was utilized to conduct the witnessing of tests on behalf of the project handler. (Please complete the table below to document the rationale and approval.)~~

Name of UL Staff conducting WTDP	CCN/Standard to be witnessed	Test(s) to be witnessed	L1, L2 or L3 Competency	Similar CCN/Standard Competency

~~☐ The Field Services Staff Member, as indicated below, (with a competent program competency as authorized by the FOM) was informed and utilized to conduct the witnessing of tests on behalf of the project handler. (Please complete the table below to document the information and approval.)~~

Name of UL Staff conducting WTDP	CCN/Standard to be witnessed	Test(s) to be witnessed	FOM Approver (name)

TEST LOCATION: (To be completed by Staff Conducting the Testing)					
☒ UL or Affiliate	☒ WTDP	☐ CTDP	☐ TPTDP	☐ TCP	☐ PPP
	☐ RWT (WTDP only)	☐ SCTDP	☐ PWT		
☐ CTF Stage 1	☐ CTF Stage 2	☐ CTF Stage 3	☐ CTF Stage 4	☐ LSV (CTF 2, 3, or 4)	
Note: RWT must be selected with WTDP. LSV must be selected with CTF Stage 2, 3, or 4					
Company Name: JZPE POWER TRANSFORMER (JIANGSU) CO., LTD.					
Address: NO.79, WEST 17 GROUP, JINTAN VILLAGE, YUXI TOWN, JIANGYAN DISTRICT, TAIZHOU CITY, JIANGSU PROVINCE, 225500 CHINA					

TEST EQUIPMENT INFORMATION

☐ UL test equipment information is recorded on Meter Use.

☐ UL test equipment information is recorded on <<insert location and local laboratory equipment system identification.>>

Inst. ID	Instrument Type	Test Number +, Test Title or Conditioning	Function/Range	Last Cal. Date	Next Cal. Date
24KA202406280011	ratio tester	Ratio, Polarity and phase relation tests	1~1000	2024/6/28	2025/6/27
24KA202406280012	DC resistance tester	Resistance Measurements	0mΩ~100Ω	2024/6/28	2025/6/27
		Temperature Rise Test			
24KA202406280010	High voltage insulation resistance tester	Winding insulation resistance	0~2500V, 0.5MΩ~2.5GΩ	2024/6/28	2025/6/27
		Core insulation resistance			
24KA202406280026	Power-Frequency high voltage test equipment	Applied-voltage tests	0~350KV	2024/6/28	2025/6/27
		Low-frequency test on auxiliary devices and control and current transformer circuits			
24KA202406280013	Multi-channel power analyzer	Induced-voltage test	0~600V	2024/6/28	2025/6/27
		Measurement of No-load Current and No-load Loss	0~20A		
		Measurement of short-circuit impedance and load loss	0~5000W		
		Audible sound levels	50/60Hz		
		Zero-phase-sequence impedance			
		1 h excitation run			
		Control (auxiliary) and cooling consumption losses			
24KA202406280018	Sound level meter	Temperature Rise Test			
	Audible sound levels	0~114dB	2024/6/28	2025/6/27	
24KA202406280028	Lightning impulse voltage generator	Lightning impulse	0~1200kV	2024/6/28	2025/6/27
24KA202406280019	Current transformer	Induced-voltage test	200A/1A	2024/6/28	2025/6/27
		Measurement of No-load Current and No-load Loss			
24KA202406280021		Measurement of short-circuit impedance and load loss	400A/2A		
		Audible sound levels	600A/3A		
		Zero-phase-sequence impedance			
24KA202406280022		1 h excitation run	800A/4A		
		Control (auxiliary) and cooling consumption losses			
	Audible sound levels	1000A/5A			
24KA202406280020	Voltage transformer	Induced-voltage test	80kV/0.1kV	2024/6/28	2025/6/27
		Measurement of No-load Current and No-load Loss			

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Only those products bearing the UL Mark should be considered as being covered by UL.

24KA202406280023		Measurement of short-circuit impedance and load loss			
		Audible sound levels			
		Zero-phase-sequence impedance			
24KA202406280024		1 h excitation run			
		Control (auxiliary) and cooling consumption losses			
		Temperature rise test			
J202404303427-0001	temperature and humidity meter	All tests	0°C~60°C, 0~80%RH	2024/5/10	2025/5/9
2JB24000718-0004	Steel measuring tape	Control (auxiliary) and cooling consumption losses	0~5.0m	2024/1/29	2025/1/28
		Audible sound levels			
		Operation tests of all devices			
2JB23008317-0012A	Stop Watch	Temperature Rise Test	1s~1h	2023/11/07	2024/11/06
		Winding insulation resistance			
		Applied-voltage tests			
		1 h excitation run			
		Induced-voltage test			
24KA202406280014	Temperature Recorder	All tests	0°C~200°C	2024/6/28	2025/6/27
24KA202406280030	Oil withstand voltage tester	Dissolved gas in oil analysis	0~80kV	2024/6/28	2025/6/27
24KA202406280031	Oil dielectric loss resistivity tester		0.5kV~10kV, 0.103%~9.845%	2024/6/28	2025/6/27
24KA202406280033	Micro water Analyzer		10µg~5000µg	2024/6/28	2025/6/27
24KA202406280029	Chromatographic analyzer		/	2024/6/28	2025/6/27
2JB24001055-0003	Pressure gauge	Pressure and Leak	0~10kgf/cm ²	2024/2/22	2024/8/21
24KA202406280015	Partial discharge Tester	Partial discharge Test	0~400pC	2024/6/28	2025/6/27
24KA202406280016	Dielectric loss resistivity tester	Insulation power factor and capacitance	0.5kV~10kV, 0.103%~9.845%	2024/6/28	2025/6/27
24KA202406280017	OLTC Tester	Operation tests of all devices	0.5A~1A, 0~20Ω	2024/6/28	2025/6/27
CE24AA022460001	Digital Multimeter		DC50mV~1000V, AC50mV~1000V	2024/04/12	2025/04/11
			DC40µA~10A, AC40µA~10A		
			10Ω~40MΩ		
			50HZ/60HZ		

+ - If Test Number is used, the Test Number must be identified on the data sheet pages or on the Data Sheet Package cover page.

The following additional information is required when using client's or rented equipment, or when a UL ID Number for an instrument number is not used. The Inst. ID No. below corresponds to the Inst. ID No. above.

Inst. ID No.	Make/Model/Serial Number/Asset No.
24KA202406280011	Jiangsu Gaodian Power Equipment Co., Ltd/GD-700
24KA202406280012	Jiangsu Gaodian Power Equipment Co., Ltd/GD200-40WA
24KA202406280010	kyoritsu/3125A
24KA202406280026	Jiangsu Gaodian Power Equipment Co., Ltd/350kV/300pF
24KA202406280013	Yokogawa/WT500
24KA202406280018	Hangzhou Aihua Instrument Co., LTD/AHAI3002
24KA202406280028	Jiangsu Gaodian Power Equipment Co., Ltd/1200kV/90kJ
24KA202406280019	Jiangsu Gaodian Power Equipment Co., Ltd/HL80-1000-0.02

24KA202406280021	Jiangsu Gaodian Power Equipment Co., Ltd/HL80-0.02
24KA202406280022	
24KA202406280020	
24KA202406280023	
24KA202406280024	Firstrate/BRW100-2001D
J202404303427-0001	
2JB24000718-0004	GREENER/5m
2JB23008317-0012A	SMTWTF8/PC396
24KA202406280014	Jiangsu Gaodian Power Equipment Co., Ltd/NJGD700
24KA202406280030	Jiangsu Gaodian Power Equipment Co., Ltd/ZIJJ-III
24KA202406280031	Jiangsu Gaodian Power Equipment Co., Ltd/GD-6100
24KA202406280033	Jiangsu Gaodian Power Equipment Co., Ltd/GD-6400
24KA202406280029	Jiangsu Gaodian Power Equipment Co., Ltd/GD-9860
2JB24001055-0003	Youyi/(0~1)MPa
24KA202406280015	Baoding Tianwei Xinyu Co., Ltd/TWPD-2H
24KA202406280016	Jiangsu Gaodian Power Equipment Co., Ltd/GD-6000
24KA202406280017	Jiangsu Gaodian Power Equipment Co., Ltd/GD-810
CE24AA022460001	FLUKE/17B+

TEST SAMPLE IDENTIFICATION:

The table below is provided to establish correlation of sample numbers to specific product related information. Refer to this table when a test identifies a test sample by "Sample No." only.

Sample Card No.	Date Received	[X] Test No. +	Sample No.	Manufacturer, Product Identification and Ratings
Test at customer's factory	-	ALL test	01	JZPE POWER TRANSFORMER (JIANGSU) CO., LTD. Model: SFSZ-110000/115/34.5

Rating 1 (ONAF with two sets of fans):

Model	SFSZ-110000/115/34.5					
position	1	2	3	4	5	Unit
Primary Voltage	126500	125781	125063	124344	123625	Vac
Primary Current	502.1	504.9	507.8	510.8	513.7	A
position	6	7	8	9	10	Unit
Primary Voltage	122906	122188	121469	120750	120031	Vac
Primary Current	516.7	519.8	522.9	526.0	529.1	A
position	11	12	13	14	15	Unit
Primary Voltage	119313	118594	117875	117156	116438	Vac
Primary Current	532.3	535.5	538.8	542.1	545.4	A
position	16	17	18	19	20	Unit
Primary Voltage	115719	115000	114281	113563	112844	Vac
Primary Current	548.8	552.3	555.7	559.3	562.8	A
position	21	22	23	24	25	Unit
Primary Voltage	112125	111406	110688	109969	109250	Vac
Primary Current	566.4	570.1	573.8	577.5	581.3	A
position	26	27	28	29	30	Unit
Primary Voltage	108531	107813	107094	106375	10565	Vac
Primary Current	585.2	589.1	593.0	597.0	6011.4	A
position	31	32	33	/	/	Unit
Primary Voltage	104938	104219	103500	/	/	Vac
Primary Current	605.2	609.4	613.6	/	/	A
Secondary Voltage	34500	Vac	Secondary Current		1840.9	A
Frequency	60	Hz	Power		110000	KVA
Winding material	Hv	Copper		LV	Copper	
Connection	Y	Y	Ynyn1			
BIL HV/HVN/LV/LVN	550/170/200/200					KV

Rating 2 (ONAF with one set of fans):

Model	88MVA					
position	1	2	3	4	5	Unit
Primary Voltage	126500	125781	125063	124344	123625	Vac
Primary Current	401.6	403.9	406.3	408.6	411.0	A
position	6	7	8	9	10	Unit
Primary Voltage	122906	122188	121469	120750	120031	Vac
Primary Current	413.4	415.8	418.3	420.8	423.3	A
position	11	12	13	14	15	Unit
Primary Voltage	119313	118594	117875	117156	116438	Vac
Primary Current	425.8	428.4	431.0	433.7	436.4	A
position	16	17	18	19	20	Unit
Primary Voltage	115719	115000	114281	113563	112844	Vac
Primary Current	439.1	441.8	444.6	447.4	450.3	A
position	21	22	23	24	25	Unit
Primary Voltage	112125	111406	110688	109969	109250	Vac
Primary Current	453.1	456.1	459.0	462.0	465.1	A
position	26	27	28	29	30	Unit
Primary Voltage	108531	107813	107094	106375	10565	Vac
Primary Current	468.1	471.3	474.4	477.6	4809.1	A
position	31	32	33	/	/	Unit
Primary Voltage	104938	104219	103500	/	/	Vac
Primary Current	484.2	487.5	490.9	/	/	A
Secondary Voltage	34500	Vac	Secondary Current		1472.7	A
Frequency	60	Hz	Power		88000	KVA
Winding material	Hv	Copper		LV	Copper	
Connection	Y	Y	Ynyn1			
BIL HV/HVN/LV/LVN	550/170/200/110					KV

Rating 3 (ONAN) :

Model	66MVA					
position	1	2	3	4	5	Unit
Primary Voltage	126500	125781	125063	124344	123625	Vac
Primary Current	301.2	303.0	304.7	306.5	308.2	A
position	6	7	8	9	10	Unit
Primary Voltage	122906	122188	121469	120750	120031	Vac
Primary Current	310.0	311.9	313.7	315.6	317.5	A
position	11	12	13	14	15	Unit
Primary Voltage	119313	118594	117875	117156	116438	Vac
Primary Current	319.4	321.3	323.3	325.3	327.3	A
position	16	17	18	19	20	Unit
Primary Voltage	115719	115000	114281	113563	112844	Vac
Primary Current	329.3	331.4	333.4	335.6	337.7	A
position	21	22	23	24	25	Unit
Primary Voltage	112125	111406	110688	109969	109250	Vac
Primary Current	339.9	342.0	344.3	346.5	348.8	A
position	26	27	28	29	30	Unit
Primary Voltage	108531	107813	107094	106375	105650	Vac
Primary Current	351.1	353.4	355.8	358.2	360.7	A
position	31	32	33	/	/	Unit
Primary Voltage	104938	104219	103500	/	/	Vac
Primary Current	363.1	365.6	368.2	/	/	A
Secondary Voltage	34500	Vac	Secondary Current		1104.5	A
Frequency	60	Hz	Power		66000	KVA
Winding material	Hv	Copper		LV	Copper	
Connection	Y	Y	Ynyn1			
BIL HV/HVN/LV/LVN	550/170/200/200					KV

+ - If Test Number is used, the Test Number or Numbers the sample was used in must be identified on the data sheet pages or on the Data Sheet Package cover page.

☒ Sampling Procedure -

☒ This document contains data or information using color and if printed, should be printed in color to retain legibility and the information represented by the color.

1. Ratio and phase relation tests**IEEE C57.12.90 Clause 6&7****IEEE C57.12.00 Clause 9.1, Table 17****CSA C88 20.4.2 a), 16.2 b) c)**

1. The turns ratio of a transformer is the ratio of the number of turns in the high-voltage winding to that in the low-voltage winding.

2. When a transformer has taps for changing its voltage ratio, the turns ratio is based on the number of turns corresponding to the normal rated voltage of the respective windings to which operating and performance characteristics are referred.

3. When the transformer has taps, the turns ratio shall be determined for all taps and for the full winding.

4. The ratio test shall be made at rated or lower voltage and rated or higher frequency.

5. In the case of three-phase transformers, when each phase is independent and accessible, single-phase power should be used, although, when convenient, three-phase power may be used.

6. Tolerances for ratio, $\leq \pm 0.5\%$, See 9.1 of IEEE Std C57.12.00- 2015.

7. Ratio test methods: Ratio meter

8. A meter using the basic circuit of Figure 12 may be used to measure ratio. When detector DET is in balance, the transformer ratio is equal to $1/R$.

NOTE 1—Measurement of ratio using circuits of this type has also been described as ratio by resistance potentiometer.

NOTE 2—More accurate results can be obtained using a ratio meter that provides phase-angle correction.

NOTE 3—The ratio meter can also be used to test polarity, phase relation, and phase sequence.

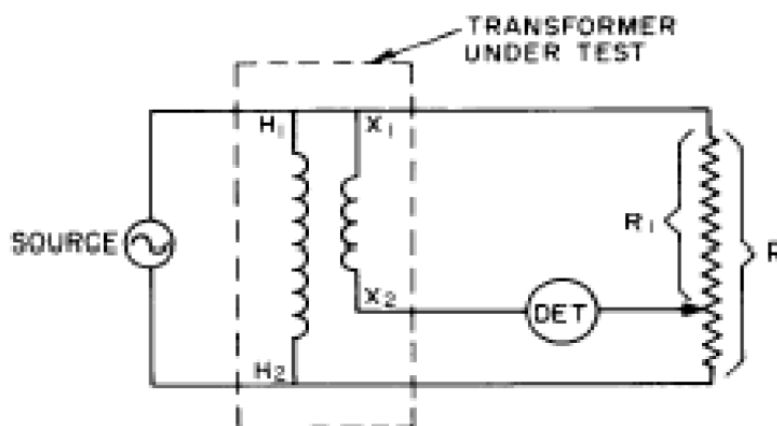


Figure 12—Basic circuit of ratio meter

H.V.		L.V.		Calculated Ratio	Measured ratio			exciting current			Vector Group
Tap	Voltage	Tap	Voltage		H1H2/X1X2	H2H3/X2X3	H1H3/X1X3	H1H2/X1X2	H2H3/X2X3	H1H3/X1X3	Yyn1
Position	(V)	Position	(V)								
1	126500	/	34500	3.667	3.6688	3.6679	3.6679	5.89	3.51	4.82	
2	125781			3.646	3.6474	3.6470	3.6469	5.96	3.52	4.86	
3	125063			3.625	3.6264	3.6261	3.6260	6.05	3.60	4.97	
4	124344			3.604	3.6055	3.6052	3.6052	6.12	3.64	5.04	
5	123625			3.583	3.5847	3.5844	3.5843	6.20	3.71	5.13	
6	122906			3.562	3.5638	3.5635	3.5634	6.26	3.72	5.13	
7	122188			3.542	3.5429	3.5427	3.5425	6.31	3.77	5.21	
8	121469			3.521	3.5221	3.5218	3.5217	6.41	3.85	5.31	
9	120750			3.500	3.5012	3.5010	3.5009	6.48	3.90	5.37	
10	120031			3.479	3.4803	3.4800	3.4800	6.55	3.88	5.37	
11	119313			3.458	3.4595	3.4592	3.4591	6.60	3.90	5.42	
12	118594			3.438	3.4386	3.4384	3.4383	6.66	3.99	5.50	
13	117875			3.417	3.4178	3.4175	3.4174	6.75	4.03	5.55	
14	117156			3.396	3.3969	3.3967	3.3966	6.81	4.07	5.61	
15	116438			3.375	3.3760	3.3757	3.3757	6.87	4.10	5.67	
16	115719			3.354	3.3552	3.3549	3.3548	6.96	4.13	5.72	
17	115000			3.333	3.3344	3.3341	3.3340	7.04	4.20	5.81	
18	114281			3.312	3.3135	3.3133	3.3132	7.14	4.27	5.90	
19	113563			3.292	3.2926	3.2924	3.2923	7.23	4.35	5.97	
20	112844			3.271	3.2718	3.2715	3.2715	7.30	4.36	6.01	
21	112125			3.250	3.2506	3.2503	3.2503	7.40	4.43	6.10	
22	111406			3.229	3.2297	3.2294	3.2294	7.46	4.42	6.13	
23	110688			3.208	3.2089	3.2086	3.2085	7.53	4.50	6.21	
24	109969			3.188	3.1880	3.1877	3.1877	7.62	4.55	6.27	
25	109250			3.167	3.1672	3.1669	3.1669	7.70	4.60	6.34	
26	108531			3.146	3.1463	3.1460	3.1460	7.79	4.67	6.42	
27	107813			3.125	3.1255	3.1252	3.1251	7.88	4.71	6.49	
28	107094			3.104	3.1046	3.1044	3.1043	7.94	4.75	6.54	
29	106375			3.083	3.0838	3.0835	3.0834	8.07	4.81	6.64	
30	105650			3.062	3.0629	3.0626	3.0626	8.16	4.88	6.72	

Date

H.V.		L.V.		Calculated Ratio	Measured ratio			exciting current			Vector Group
Tap	Voltage	Tap	Voltage		H1H2/X1X2	H2H3/X2X3	H1H3/X1X3	H1H2/X1X2	H2H3/X2X3	H1H3/X1X3	Yyn1
Position	(V)	Position	(V)								
30	105650	/	34500	3.062	3.0629	3.0626	3.0626	8.16	4.88	6.72	
31	104938			3.042	3.0421	3.0418	3.0418	8.22	4.91	6.77	
32	104219			3.021	3.0213	3.0210	3.0210	8.31	4.97	6.84	
33	103500			3.000	3.0001	3.0001	3.0001	8.41	5.02	6.91	

Date

Test date:	2024/7/10		Test Sample	01	
Ambient Temp.:	37	°C	Relative Humidity:	69%	
Top Oil Temp. Indicator:	37	°C	Bottom Oil Temp. Indicator:	37	°C
Criterion:					
Test values balance shall be within ±0.5%.					
Measuring instrument:					
temperature and humidity meter		J202404303427-0001			
ratio tester		24KA202406280011			
Temperature Recorder		24KA202406280014			

2. Resistance Measurements**IEEE C57.12.90, Clause 5****IEEE C57.12.00, Table 17****CSA C88 Clause 16.2 a)**

1.The cold temperature of the winding shall be determined as accurately as possible when measuring the cold resistance. The precautions in 5.1.1, 5.1.2, and 5.1.3 shall be observed.

2.Cold-resistance measurements shall be made on a transformer only when the liquid or winding temperature is stable. The temperature is considered stable if the top liquid temperature does not vary more than 2 °C in a 1-h period.

3.The temperature of the windings shall be assumed to be the same as the average temperature of the insulating liquid, provided:

The windings have been under insulating liquid with no excitation and with no current in the windings for a minimum of 3 h for a transformer without pumps and for 1 h for transformer with pumps running before the cold resistance is measured.

The temperature of the insulating liquid has stabilized, and the difference between top and bottom temperature does not exceed 5 °C.

4. Conversion of resistance measurements

Cold winding resistance measurements are normally converted to a standard reference temperature (T) equal to the rated average winding temperature rise plus 20 °C. In addition, it may be necessary to convert the resistance measurements to the temperature at which the impedance loss measurements were made. The conversions are accomplished by Equation (1):

$$R_s = R_m \frac{T_s + T_k}{T_m + T_k} \quad (1)$$

where

R_s is the resistance at desired temperature T_s (Ω)

R_m is the measured resistance (Ω)

T_s is the desired reference temperature (°C)

T_m is the temperature at which resistance was measured (°C)

T_k is 234.5 °C (copper) or 225 °C (aluminum)

NOTE—The value of T_k may be as high as 230 °C for alloyed aluminum.

5. The individual phase- or terminal-to-terminal resistance readings shall be reported along with the sum total winding resistance.

6. For wye windings, the reported resistance measurement may be from terminal to terminal or from terminal to neutral. For the reported total winding resistance, the resistance of the lead from the neutral connection to the neutral bushing may be

Date

excluded. For terminal-to-terminal measurements, the total resistance reported is the sum of the three measurements divided by two.

6. For delta windings, the reported resistance measurement may be from terminal to terminal with the delta closed or from terminal to terminal with the delta open to obtain the individual phase readings. The reported total winding resistance is the sum of the three-phase readings if the delta is open. If the delta is closed, the reported total winding resistance is the sum of the three phase-to-phase readings times 1.5.

7. For delta windings, the reported resistance measurement may be from terminal to terminal with the delta closed or from terminal to terminal with the delta open to obtain the individual phase readings. The reported total winding resistance is the sum of the three-phase readings if the delta is open. If the delta is closed, the reported total winding resistance is the sum of the three phase-to-phase readings times 1.5.

8. For autotransformer winding resistance measurements, the following method or an equivalent method shall be used. For the series winding resistance, the current shall be circulated between the high-voltage and neutral terminals, and the voltage shall be measured between the high-voltage terminal and the low-voltage terminal. For the common winding resistance, the current shall be circulated between the high-voltage and neutral terminals, and the voltage shall be measured between the low-voltage and neutral terminals. If needed, for the resistance of the lead and in-line windings (if any) between the low-voltage terminal and the neutral connection, the current shall be applied between the high-voltage terminal and the low-voltage terminal, and the voltage shall be measured between the low-voltage terminal and the neutral terminal.

High-voltage H1 current transformer				
Terminal	Model specifications	Factory No.	DC. resistance	
			(33℃)	
1~5 (C1)	LRB-115 1200/5	J2405045	X1~X2	121.17mΩ
	MR C800 RF2.0		X1~X3	181.12mΩ
			X1~X4	470.60mΩ
			X1~X5	701.00mΩ
6~10 (C4)	LRB-115 1200/5	J2405047	X1~X2	120.43mΩ
	MR C800 RF2.0		X1~X3	180.41mΩ
			X1~X4	469.70mΩ
			X1~X5	698.80mΩ
11~15 (C7)	LRB-115 1200/5	J2405048	X1~X2	119.83mΩ
	MR C800 RF2.0		X1~X3	179.40mΩ
			X1~X4	468.20mΩ
			X1~X5	699.10mΩ
16~20 (C10)	LRB-115 1200/5	J2405046	X1~X2	121.31mΩ
	MR C800 RF2.0		X1~X3	181.27mΩ
			X1~X4	470.60mΩ
			X1~X5	701.80mΩ
21~22 (C13)	LR-115 600/5	J2405050	X1~X2	275.0mΩ
	SR0.3B1.8 RF2.0			
23~24 (C32)	LR-115 600/1	J2405053	X1~X2	3.764 Ω
	SR0.3B1.8 RF2.0			

High-voltage H2 current transformer				
Terminal	Model specifications	Factory No.	DC. resistance	
			(33℃)	
1~5 (C2)	LRB-115 1200/5	J2405044	X1~X2	121.79mΩ
	MR C800 RF2.0		X1~X3	181.6mΩ
			X1~X4	470.10mΩ
			X1~X5	700.80mΩ
6~10 (C5)	LRB-115 1200/5	J2405037	X1~X2	119.92mΩ
	MR C800 RF2.0		X1~X3	179.16mΩ
			X1~X4	466.70mΩ
			X1~X5	691.90mΩ
11~15 (C8)	LRB-115 1200/5	J2405038	X1~X2	120.39mΩ
	MR C800 RF2.0		X1~X3	179.72mΩ
			X1~X4	466.50mΩ
			X1~X5	692.70mΩ
16~20 (C11)	LRB-115 1200/5	J2405040	X1~X2	120.96mΩ
	MR C800 RF2.0		X1~X3	180.4mΩ
			X1~X4	467.8mΩ
			X1~X5	694mΩ
21~22 (C14)	LR-115 600/5	J2405051	X1~X2	275.2mΩ
	SR0.3B1.8 RF2.0			
23~24 (C16)	LR-115 600/5	J2405052	X1~X2	261.2mΩ
	SR C100 RF2.0			

High-voltage H3 current transformer				
Terminal	Model specifications	Factory No.	DC. resistance	
			(33℃)	
1~5 (C3)	LRB-115 1200/5	J2405039	X1~X2	120.55mΩ
	MR C800 RF2.0		X1~X3	1810.31mΩ
			X1~X4	468.8mΩ
			X1~X5	696.1mΩ
6~10 (C6)	LRB-115 1200/5	J2405042	X1~X2	121.38mΩ
	MR C800 RF2.0		X1~X3	181.53mΩ
			X1~X4	470.9mΩ
			X1~X5	697.8mΩ
11~15 (C9)	LRB-115 1200/5	J2405041	X1~X2	120.81mΩ
	MR C800 RF2.0		X1~X3	181.09mΩ
			X1~X4	469.1mΩ
			X1~X5	696.6mΩ
16~20 (C12)	LRB-115 1200/5	J2405043	X1~X2	119.57mΩ
	MR C800 RF2.0		X1~X3	179.14mΩ
			X1~X4	467.4mΩ
			X1~X5	694.01mΩ
21~22 (C15)	LR-115 600/5	J2405049	X1~X2	275.6mΩ
	SR0.3B1.8 RF2.0			

Low-voltage X1 current transformer				
Terminal	Model specifications	Factory No.	DC. resistance	
			(33℃)	
1~5 (C17)	LRB-34.5 2000/5	J2405057	X1~X2	184.49mΩ
	MR C800 RF2.0		X1~X3	544.4mΩ
			X1~X4	682.1mΩ
			X1~X5	912.4mΩ
6~10 (C20)	LRB-34.5 2000/5	J2405058	X1~X2	183.64mΩ
	MR C800 RF2.0		X1~X3	543.4mΩ
			X1~X4	680.7mΩ
			X1~X5	912.1mΩ
11~12 (C23)	LRB-34.5 2000/5	J2405062	X1~X2	876.6mΩ
	MR C800 RF2.0			
13~14 (C26)	LRB-34.5 1500/5	J2405065	X1~X2	383.1 mΩ
	MR C800 RF2.0			
15~16 (C33)	LR-115 1500/1	J2405067	X1~X2	5.541 Ω
	SR0.3B1.8 RF2.0			

Date

Low-voltage X2 current transformer				
Terminal	Model specifications	Factory No.	DC. resistance	
			(33℃)	
1~5 (C18)	LRB-34.5 2000/5	J2405055	X1~X2	184.44mΩ
	MR C800 RF2.0		X1~X3	544.8mΩ
			X1~X4	683.3mΩ
			X1~X5	914.3mΩ
6~10 (C21)	LRB-34.5 2000/5	J2405059	X1~X2	184.55mΩ
	MR C800 RF2.0		X1~X3	544.01mΩ
			X1~X4	682.3mΩ
			X1~X5	912.5mΩ
11~12 (C24)	LRB-34.5 2000/5	J240563	X1~X2	883.2mΩ
	MR C800 RF2.0			
13~14 (C27)	LRB-34.5 1500/5	J2405064	X1~X2	384.2 mΩ
	MR C800 RF2.0			

Low-voltage X3 current transformer				
Terminal	Model specifications	Factory No.	DC. resistance	
			(33℃)	
1~5 (C19)	LRB-34.5 2000/5	J2405056	X1~X2	185.26mΩ
			X1~X3	546.01mΩ
	MR C800 RF2.0		X1~X4	687.2mΩ
			X1~X5	918.6mΩ
6~10 (C22)	LRB-34.5 2000/5	J2405060	X1~X2	187.63mΩ
			X1~X3	549.9mΩ
	MR C800 RF2.0		X1~X4	690.7mΩ
			X1~X5	922.2mΩ
11~12 (C25)	LRB-34.5 2000/5	J2405061	X1~X2	887.5mΩ
	MR C800 RF2.0			
13~14 (C28)	LRB-34.5 1500/5	J2405064	X1~X2	384.7 mΩ
	MR C800 RF2.0			

High-voltage H0 current transformer				
Terminal	Model specifications	Factory No.	DC. resistance	
			(33℃)	
1~5 (C29)	LRB-115 1200/5	J2405054	X1~X2	120.39mΩ
			X1~X3	180.54mΩ
	MR C800 RF2.0		X1~X4	472.2mΩ
			X1~X5	704.6mΩ

Date

Low-voltage X0 current transformer					
Terminal	Model specifications	Factory No.	DC. resistance		
			(33℃)		
1~5 (C30)	LRB-13.2 1200/5	J2405068	X1~X2	120.22mΩ	
			X1~X3	180.13mΩ	
	MR C800 RF2.0		X1~X4	470.01mΩ	
			X1~X5	699.6mΩ	

Balanced winding current transformer				
Terminal	Model specifications	Factory No.	DC. resistance	
			(33°C)	
1~5 (C31)	LRB-13.2 1200/5	J2405059	X1~X2	695.4mΩ
	MR C800 RF2.0			

Position	H.V. Windings (Ω)		
	H1H0	H2H0	H3H0
1	0.1656	0.1648	0.1648
2	0.1642	0.1637	0.1636
3	0.1632	0.1625	0.1625
4	0.162	0.1614	0.1613
5	0.1608	0.1603	0.16
6	0.1596	0.1591	0.1588
7	0.1585	0.1579	0.1577
8	0.1573	0.1568	0.1565
9	0.1562	0.1557	0.1551
10	0.1549	0.1544	0.154
11	0.1538	0.1532	0.1529
12	0.1526	0.152	0.1517
13	0.1515	0.1509	0.1504
14	0.1502	0.1497	0.1492
15	0.1491	0.1486	0.1481
16	0.1479	0.1475	0.147
17B	0.1458	0.1457	0.145
18	0.1479	0.1474	0.1466
19	0.1491	0.1486	0.1476
20	0.1502	0.1497	0.1488
21	0.1514	0.1509	0.1499
22	0.1526	0.1522	0.1513
23	0.1538	0.1533	0.1524
24	0.1549	0.1545	0.1536
25	0.156	0.1556	0.1547
26	0.1572	0.1569	0.156
27	0.1584	0.158	0.1571
28	0.1597	0.1591	0.1583
29	0.1608	0.1603	0.1594
30	0.162	0.1615	0.1606
31	0.1632	0.1627	0.1618
32	0.1644	0.1639	0.163
33	0.1655	0.165	0.1642
Position	L.V. Windings (Ω)		
	X1X2	X2X3	X1X3
	0.02085	0.02086	0.02091
	X1X0	X2X0	X3X0
	0.01056	0.01061	0.01067

Date

Test date:	2024/7/10		Test Sample	01	
Ambient Temp.:	37	°C	Relative Humidity:	69%	
Top Oil Temp. Indicator:	37	°C	Bottom Oil Temp. Indicator:	37	°C
Criterion:					
Test values balance shall be within ±2% for Hv winding.					
Measuring instrument:					
temperature and humidity meter			J202404303427-0001		
Temperature Recorder			24KA202406280014		
DC resistance tester			24KA202406280012		

Date

3.Low-frequency test on auxiliary devices and control**IEEE C57.12.00, Table 17
CSA C88-16, Clause 16.3n)**

Measured Between	Test Voltage (kV)		Test Frequency(Hz)	Duration (s)	Result
OLTC control circuit	1.5	KV	50	60	Pass
cooling systems control circuit	1.5	KV	50	60	
current transformer circuits	2.5	KV	50	60	

Test date:	2024/7/12		Test Sample	01	
Ambient Temp.:	33	℃	Relative Humidity:	53%	
Top Oil Temp. Indicator:	33	℃	Bottom Oil Temp. Indicator:	33	℃
Measuring instrument:					
temperature and humidity meter		J202404303427-0001			
Temperature Recorder		24KA202406280014			
Power-Frequency high voltage test equipment		24KA202406280026			
Stop Watch		2JB23008317-0012A			

4. Measurement of no-load loss and current**IEEE C57.12.90, Clause 8, Annex B**

UL 1561-XQNX-DataSheet - ULID-012197

Form Issued: 2023-09-05

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Date

IEEE C57.12.00 Table 17, Clause
5.9, 9.3
CSA C88, 16.2 d), k), 16.4.1,
16.4.2

1. The purpose of the no-load loss test is to measure no-load losses at a specified excitation voltage and a specified frequency. The no-load loss determination shall be based on a sine-wave voltage, unless a different waveform is inherent in the operation of the transformer. The average-voltage voltmeter method is the most accurate method for correcting the measured no-load losses to a sine-wave basis and is recommended. This method employs two parallel-connected voltmeters: one is an average-responding [but root mean square (rms) calibrated] voltmeter (a V), and the other is a true rms-responding voltmeter (r V).

The test voltage is adjusted to the specified value as read by the average-responding voltmeter. The readings of both voltmeters are employed to correct the no-load losses to a sine-wave basis, using Equation (2) in accordance with 8.3.

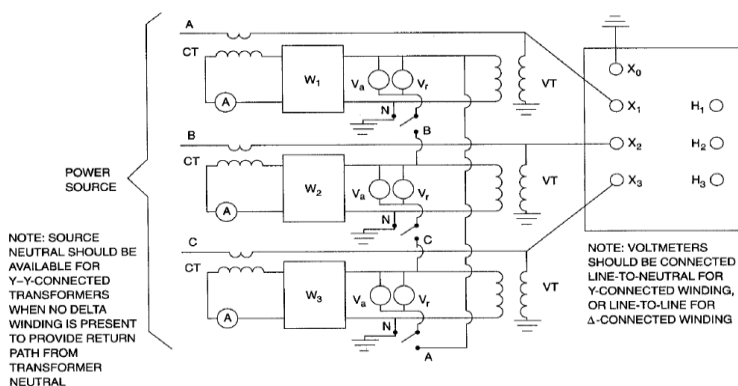


Figure 15—Three-phase transformer connections for no-load loss and excitation current tests using three-wattmeter method

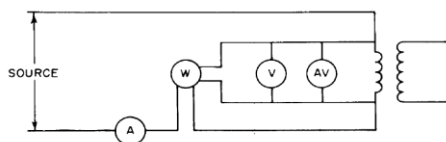


Figure 13—Connections for no-load loss test of a single-phase transformer without instrument transformers

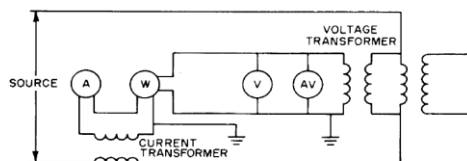


Figure 14—Connections for a no-load loss test of a single-phase transformer with instrument transformers

Date

2.The operating and performance characteristics of a transformer are based on rated voltage and rated frequency, unless otherwise specified. Therefore, the no-load loss test is conducted with rated voltage impressed across the transformer terminals, using a voltage source at a frequency equal to the rated frequency of the transformer under test, unless otherwise specified.

3.To determine no-load losses of a single-phase transformer or a three-phase transformer, the frequency of the test source should be within $\pm 0.5\%$ of the rated frequency of the transformer under test. The voltage shall be adjusted to the specified value as indicated by the average-voltage voltmeter. Simultaneous values of rms voltage, rms current, electrical power, and the average-voltage voltmeter readings shall be recorded. For a three-phase transformer, the average of the three voltmeter readings shall be the desired nominal value.

4. Waveform correction of no-load losses

The eddy-current component of the no-load loss varies with the square of the rms value of excitation voltage and is substantially independent of the voltage waveform. When the test voltage is held at the specified value as read on the average-voltage voltmeter, the actual rms value of the test voltage may not be equal to the specified value. The no-load losses of the transformer corrected to a sine-wave basis shall be determined from the measured value using Equation (2):

$$P_c(T_m) = \frac{P_m}{P_1 + kP_2} \quad (2)$$

where

T_m is the average liquid temperature at the time of test ($^{\circ}\text{C}$)

$P_c(T_m)$ is the no-load losses, corrected for waveform, at temperature T_m

P_m is measured no-load losses at temperature T_m

P_1 is per-unit hysteresis loss

P_2 is per-unit eddy-current loss

k is $\left(\frac{E_r}{E_a} \right)^2$

E_r is the test voltage measured by rms voltmeter

E_a is the test voltage measured by average-voltage voltmeter

5.The actual per-unit values of hysteresis and eddy-current losses should be used if available. If actual values are not available, it is suggested that the two loss components be assumed equal in value and that they each be assigned a value of 0.5 per unit. For more information, see IEEE Std C57.123.

6.Equation (2) is valid only for test voltages with moderate waveform distortion. If waveform distortion in the test voltage causes the magnitude of the correction to be greater than 5%, the test voltage waveform shall be improved for an adequate determination of the no-load losses and currents.

A reference temperature is required when stating no-load losses because the no-load losses vary with core temperature. The standard reference temperature rT for transformer no-load losses is specified in 5.9 of IEEE Std C57.12.00-2015.

- a) The average liquid temperature is within $\pm 10^\circ\text{C}$ of the reference temperature T_r .
- b) The difference between the top and bottom liquid temperatures does not exceed 5°C .

If conducting the test with temperatures outside the specified ranges is necessary, the empirical formula in Equation (3) may be used to correct the measured no-load losses to the reference temperature:

$$P_c(T_r) = P_c(T_m) \{1 + (T_m - T_r) K_T\} \quad (3)$$

where

$P_c(T_r)$ is the no-load losses, corrected to the standard reference temperature T_r

$P_c(T_m)$ is the no-load losses, corrected for waveform, at temperature T_m

T_r is the standard reference temperature ($^\circ\text{C}$)

K_T is an empirically derived per-unit change in no-load loss per degree Celsius

If the actual value of K_T is not available, a value of 0.000 65 per unit change per degrees Celsius should be used. This value is typical for cores constructed of grain-oriented silicon steel and is satisfactory as a correction for no-load losses when the transformer cannot be tested within the specified temperature range.

7. The excitation (no-load) current of a transformer is the current that maintains the rated magnetic flux excitation in the core of the transformer. The excitation current is usually expressed per unit or in percent of the rated line current of the winding in which it is measured. (Where the cooling class of the transformer involves more than one kilovolt ampere rating, the lowest kilovolt ampere rating is used to determine the base current.) Measurement of excitation current is usually carried out in conjunction with the tests for no load losses. Rms current is recorded simultaneously during the test for no-load losses using the average voltage voltmeter method. This value is used in calculating the per-unit or percent excitation current. For a three-phase transformer, the excitation current is calculated by taking the average of the magnitudes of the three line currents.

8. In the event that the no-load loss and excitation current test cannot be done at the rated power frequency, see Annex B for 50/60 Hz frequency conversions.

The following tests shall be conducted on all transformers except as specifically stated below (but not necessarily in the order shown). Where impulse tests are required, the ratio test, **core loss**, and temperature rise test shall be performed before the impulse tests.

Tested Frequency				50	Hz	Tested Ambient Temp			37	°C
Corrected Frequency				60	Hz	Corrected Ambient Temp			20	°C
90%	Vrms (V)				Irm (A)					
	H1		H2	H3	H1		H2	H3	I ₀ (%)	
	25651		25468	25635	0.867		0.501	0.867	/	
	Vavg (V)				P (w)				Corrected Loss	
	H1		H2	H3	H1		H2	H3	38066	W
	25604		25429	25607	9389		6243	12086		
100%	Vrms (V)				Irm (A)				48426	W
	H1		H2	H3	H1		H2	H3		
	28773		28608	28782	1.101		0.633	1.039	0.17%	
	Vavg (V)				P (w)				Corrected Loss	
	H1		H2	H3	H1		H2	H3	48426	W
	28756		28606	28797	12453		7787	15702		
110%	Vrms (V)				Irm (A)					
	H1		H2	H3	H1		H2	H3	I ₀ (%)	
	31631		31434	31610	3.32				/	
	Vavg (V)				P (w)				Corrected Loss	
	H1		H2	H3	H1		H2	H3	61025	W
	31674		31485	31692	15236		9636	20235		

Test date:	2024/7/12		Test Sample	01	
Ambient Temp.:	37	°C	Relative Humidity:	63%	
Top Oil Temp. Indicator:	37	°C	Bottom Oil Temp. Indicator:	37	°C
Measuring instrument:					
temperature and humidity meter	J202404303427-0001				
Temperature Recorder	2JB24000718-0004				
Multi-channel power analyzer	24KA202406280013				
Current transformer	24KA202406280019	24KA202406280021	24KA202406280022		
Voltage transformer	24KA202406280020	24KA202406280023	24KA202406280024		

Date

5.Measurement of short-circuit impedance and load loss**IEEE C57.12.90, Clause 9.4, ~~Annex B~~****IEEE C57.12.00 Table 17, Clause 5.9,
5.8, 9.2, 9.3****CSA C88, 16.2 e), 16.4.3, 16.4.4**

1.The load losses of a transformer are losses incident to a specified load carried by the transformer. Load losses include I²R loss in the windings due to load current and stray losses due to eddy currents induced by leakage flux in the windings, core clamps, magnetic shields, tank walls, and other conducting parts. Stray losses may also be caused by circulating currents in parallel windings or strands. Load losses are measured by applying a short circuit across either the high-voltage winding or the low-voltage winding and applying sufficient voltage across the other winding to cause a specified current to flow in the windings. The power loss within the transformer under these conditions equals the load losses of the transformer at the temperature of test for the specified load current.

2.Regardless of the test method selected, the following preparatory requirements shall be satisfied for accurate test results:

a) To determine the temperature of the windings with sufficient accuracy, the following conditions shall be met, except as stated in the NOTE in 2):

1) The temperature of the insulating liquid has stabilized, and the difference between top and bottom liquid temperatures does not exceed 5 °C.

2) The temperature of the windings shall be taken immediately before and after the load losses and impedance voltage test in a manner similar to that described in 5.1. The average shall be taken as the true temperature.

3) The difference in winding temperature before and after the test shall not exceed 5 °C.

NOTE- For distribution and pad-mounted transformers up to 2500 kVA, where it may not be practical to wait for thermal equilibrium, the method used to determine the winding temperature should take into consideration the lack of thermal equilibrium and the effect of ohmic heating of the winding conductors by load current during the test. The method used can be verified by staging a repeated measurement of the load losses and impedance voltage at a later time when condition 1, condition 2, and condition 3 are met.

b) The conductors used to short-circuit the low-voltage high-current winding of a transformer shall have a cross-sectional area equal to or greater than the corresponding transformer leads.

c) The frequency of the test source used for measuring load losses and impedance voltage shall be within ± 0.5% of the nominal value.

d) The maximum value of correction to the measured load losses due to the test system phase-angle error is limited to ± 5% of measured losses. If more than 5% correction is required, test methods and/or test apparatus should be improved for an adequate determination of load losses.

3. Wattmeter-voltmeter-ammeter method

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Only those products bearing the UL Mark should be considered as being covered by UL.

NOTE-Instrument transformers to be added when necessary.

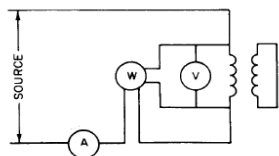


Figure 16—Single-phase transformer connections for load loss and impedance voltage tests without instrument transformers

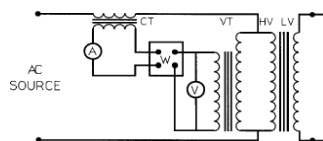


Figure 17—Single-phase transformer connections for load loss and impedance voltage tests with instrument transformers

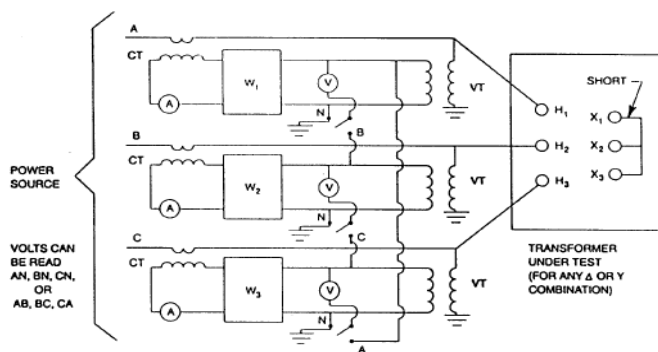


Figure 18—Three-phase transformer connections for load loss and impedance voltage tests using three-wattmeter method

Test of three-phase transformer with single-phase voltage

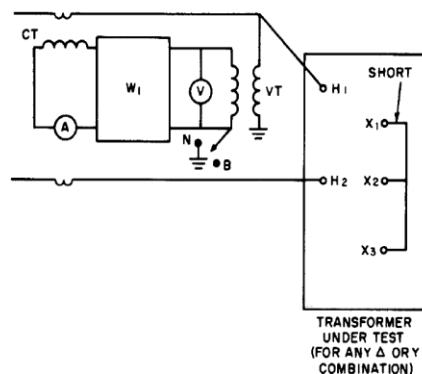


Figure 24—Test of three-phase transformer with single-phase voltage

The three line leads of one winding are short-circuited, and single-phase voltage at rated frequency is applied to two terminals of the other winding. The applied voltage is adjusted to circulate rated line current.

Three successive readings are taken on the three pairs of leads, for example, H1 and H2, H2 and H3, H3 and H1. Then

$$\text{Measured load losses } (W) = 1.5 \left(\frac{P_{12} + P_{23} + P_{31}}{3} \right) \quad (7)$$

$$\text{Measured impedance voltage} = \frac{\sqrt{3}}{2} \left(\frac{E_{12} + E_{23} + E_{31}}{3} \right) \quad (8)$$

where

P is the individual reading of measured load losses as indicated by subscripts
 E is the individual reading of measured impedance voltage as indicated by subscripts

The stray loss component shall be obtained by subtracting the I^2R losses from the measured load losses of the transformer. Let R_1 be the resistance measured between two high-voltage terminals and R_2 be the resistance between two low-voltage terminals; let I_1 and I_2 be the respective rated line currents. Then the total I^2R loss of all three phases will be as in Equation (9):

$$\text{Total } I^2R (\text{watts}) = 1.5 (I_1^2 R_1 + I_2^2 R_2) \quad (9)$$

This formula applies equally well to wye- or delta-connected windings.

Temperature correction shall be made as in 9.4.2.

9.4.2 Temperature correction of load losses

Both I^2R losses and stray losses of a transformer vary with temperature. The I^2R losses $P_r(T_m)$ of a transformer are calculated from the ohmic resistance measurements (corrected to the temperature T_m at which the measurement of load losses and impedance voltage was done) and the current that was used in the impedance measurement. These I^2R losses subtracted from the measured load loss watts $P(T_m)$ give the stray losses $P_s(T_m)$ of the transformer at the temperature at which the load loss test was made, as shown in Equation (13):

$$P_s(T_m) = P(T_m) - P_r(T_m) \quad (13)$$

where

$P_s(T_m)$	is the calculated stray losses at temperature T_m (W)
$P(T_m)$	is the transformer load losses corrected in accordance with 9.4.1, for phase-angle error at temperature T_m (W)
$P_r(T_m)$	is the calculated I^2R loss at temperature T_m (W)

The I^2R component of load losses increases with temperature. The stray loss component diminishes with temperature. Therefore, when it is desirable to convert the load losses from the temperature at which it is measured T_m to another temperature T , the two components of the load losses are corrected separately, as shown in Equation (14) and Equation (15):

$$P_r(T) = P_r(T_m) \left(\frac{T_k + T}{T_k + T_m} \right) \quad (14)$$

$$P_s(T) = P_s(T_m) \left(\frac{T_k + T_m}{T_k + T} \right) \quad (15)$$

and, then, as shown in Equation (16):

$$P(T) = P_r(T) + P_s(T) \quad (16)$$

where

$P_r(T)$	is I^2R loss (W) at temperature T (°C)
$P_s(T)$	is stray losses (W) at temperature T (°C)
$P(T)$	is transformer load losses (W) corrected to temperature T (°C)
T_k	is 234.5 °C (copper) or 225 °C (aluminum; see NOTE below)

NOTE—The temperature 225 °C applies for pure electrical conductor (EC) aluminum. T_k may be as high as 230 °C for alloyed aluminum. Where copper and aluminum windings are employed in the same transformer, a value for T_k of 229 °C should be applied for the correction of stray losses.

9.4.3 Impedance voltage

Impedance voltage and its resistive and reactive components are determined by the use of Equation (17) through Equation (20):

$$E_r(T_m) = \frac{P(T_m)}{I} \quad (17)$$

$$E_x = \sqrt{E_z(T_m)^2 - E_r(T_m)^2} \quad (18)$$

$$E_r(T) = \frac{P(T)}{I} \quad (19)$$

$$E_z(T) = \sqrt{E_r(T)^2 + E_x^2} \quad (20)$$

where

$E_r(T_m)$	is the resistance voltage drop of in-phase component at temperature T_m (V)
$E_r(T)$	is the resistance voltage drop of in-phase component corrected to temperature T (V)
E_x	is the reactance voltage drop of quadrature component (V)
$E_z(T_m)$	is the impedance voltage at temperature T_m (V)
$E_z(T)$	is the impedance voltage at temperature T (V)
$P(T)$	is the transformer load losses corrected to temperature T (W)
$P(T_m)$	is transformer load losses measured at temperature T_m (W)
I	is the current in excited winding (A)

Per-unit values of the resistance, reactance, and impedance voltage are obtained by dividing $E_r(T)$, E_x , and $E_z(T)$ by the rated voltage. Percentage values are obtained by multiplying per-unit values by 100.

9.4.4 Frequency conversion of load losses and impedance

In the event that the load loss and impedance test cannot be done at the rated power frequency, see Annex B for 50/60 Hz frequency conversions.

110MVA								
Tested Frequency	50	Hz	Tested Ambient Temp			33	°C	
Winding material	Copper		234.5		Copper		234.5	
Corrected Frequency	60	Hz	Corrected Ambient Temp			85	°C	
1	Vrms (V)				Arms (A)			
	H1	H2	H3	Average	H1	H2	H3	Average
	17150	17106	17042	17099.333	503.5	494.2	502.5	500.1
	P (w)						Impedance	
	H1	H2	H3	Measured		Measured		
	91353	161258	191204	443815		w	13.57%	
	Corrected impedance			16.28%	60	Hz	85	°C
Corrected Loss		516546	W	60	Hz	85	°C	
17B	Vrms (V)				Arms (A)			
	H1	H2	H3	Average	H1	H2	H3	Average
	14746	14692	14696	14711.333	555.5	544	554.7	551.4
	P (w)						Impedance	
	H1	H2	H3	Measured		Measured		
	91257	160461	168471	420189		w	12.81%	
	Corrected impedance			15.37%	60	Hz	85	°C
Corrected Loss		488293	W	60	Hz	85	°C	
33	Vrms (V)				Arms (A)			
	H1	H2	H3	Average	H1	H2	H3	Average
	13185	13093	13158	13145.333	611.7	599.9	609.4	607.0
	P (w)						Impedance	
	H1	H2	H3	Measured		Measured		
	99709	160520	169606	429835		w	12.84%	
	Corrected impedance			15.41%	60	Hz	85	°C
Corrected Loss		512184	W	60	Hz	85	°C	

88MVA								
Tested Frequency		50	Hz	Tested Ambient Temp			33	℃
Winding material		Copper		234.5		Copper		234.5
Corrected Frequency		60	Hz	Corrected Ambient Temp			85	℃
1	Vrms (V)				Arms (A)			
	H1	H2	H3	Average	H1	H2	H3	Average
	9964	9895	9903	9920.6667	293.7	286.5	292.4	290.9
	P (w)						Impedance	
	H1	H2	H3	Measured		Measured		
	32011	58103	63466	153580		w	10.83%	
	Corrected impedance			12.99%	60	Hz	85	℃
	Corrected Loss		337866	W	60	Hz	85	℃
17B	Vrms (V)				Arms (A)			
	H1	H2	H3	Average	H1	H2	H3	Average
	8876	8813	8853	8847.3333	335.2	329.8	334.1	333.0
	P (w)						Impedance	
	H1	H2	H3	Measured		Measured		
	34193	60362	60314	154869		w	10.21%	
	Corrected impedance			12.25%	60	Hz	85	℃
	Corrected Loss		315621	W	60	Hz	85	℃
33	Vrms (V)				Arms (A)			
	H1	H2	H3	Average	H1	H2	H3	Average
	7913	7832	7886	7877	367.3	357.6	365.2	363.4
	P (w)						Impedance	
	H1	H2	H3	Measured		Measured		
	37087	60834	60143	158064		w	10.28%	
	Corrected impedance			12.34%	60	Hz	85	℃
	Corrected Loss		335990	W	60	Hz	85	℃

66MVA					
Tested Frequency	50	Hz	Tested Ambient Temp		33 °C

Winding material		Copper		234.5		Copper		234.5
Corrected Frequency		60	Hz	Corrected Ambient Temp			85	℃
1	Vrms (V)				Arms (A)			
	H1	H2	H3	Average	H1	H2	H3	Average
	9964	9895	9903	9920.6667	293.7	286.5	292.4	290.9
	P (w)						Impedance	
	H1	H2	H3	Measured			Measured	
	32011	58103	63466	153580		w	8.12%	
	Corrected impedance			9.75%	60	Hz	85	℃
	Corrected Loss		190049	W	60	Hz	85	℃
17B	Vrms (V)				Arms (A)			
	H1	H2	H3	Average	H1	H2	H3	Average
	8876	8813	8853	8847.3333	335.2	329.8	334.1	333.0
	P (w)						Impedance	
	H1	H2	H3	Measured			Measured	
	34193	60362	60314	154869		w	7.65%	
	Corrected impedance			9.19%	60	Hz	85	℃
	Corrected Loss		177537	W	60	Hz	85	℃
33	Vrms (V)				Arms (A)			
	H1	H2	H3	Average	H1	H2	H3	Average
	7913	7832	7886	7877	367.3	357.6	365.2	363.4
	P (w)						Impedance	
	H1	H2	H3	Measured			Measured	
	37087	60834	60143	158064		w	7.71%	
	Corrected impedance			9.25%	60	Hz	85	℃
	Corrected Loss		188994	W	60	Hz	85	℃

Test date:	2024/7/12		Test Sample	01	
Ambient Temp.:	33	°C	Relative Humidity:	63%	
Top Oil Temp. Indicator:	33	°C	Bottom Oil Temp. Indicator:	33	°C
Measuring instrument:					
temperature and humidity meter			J202404303427-0001		
Temperature Recorder			2JB24000718-0004		
Multi-channel power analyzer			24KA202406280013		
Current transformer	24KA202406280019		24KA202406280021	24KA202406280022	
Voltage transformer	24KA202406280020		24KA202406280023	24KA202406280024	

6. Zero-phase-sequence impedance**IEEE C57.12.90, Clause 9.5****CSA C88-16, Clause 16.2r)**

1.The zero-phase-sequence impedance characteristics of three-phase transformers depend on the winding connections and, in some cases, on the core construction. Zero-phase-sequence impedance tests described in this standard apply only to transformers having one or more windings with a physical neutral brought out for external connection. In all tests, one such winding shall be excited at rated frequency between the neutral and the three line terminals connected together. External connection of other windings shall be as described in zero-phase-sequence impedance for various transformer connections. Transformers with connections other than as described in zero-phase-sequence impedance shall be tested as determined by the individuals responsible for design and application.

2.The excitation voltage and current shall be established as follows:

a) If no delta connection is present on the transformer and the transformer is a three leg core design, there is a risk of excessive tank wall heating due to the return flux from the core going into the tank wall. To avoid this, the applied voltage should be such that the current is no more than 20% of the base rating of the winding being excited. This applies to both open-circuit tests and short-circuit tests as described in 9.5.3. If the transformer is a five leg core or a shell form design, the zero-sequence impedance is equal to the positive sequence impedance and the zero-sequence test is generally not needed. However, should the test be done, the applied voltage should not exceed 30% of the rated line-to-neutral voltage of the winding being energized for the open-circuit test, and the phase current should not exceed the base rated value of the winding being excited for the short-circuit test.

b) If a delta connection is present, the applied voltage should be such that the base rated phase current of any delta winding or the maximum allowable neutral current is not exceeded.

3. The percent excitation voltage at which the tests are made shall be shown on the test report. The time duration of the test shall be such that the thermal limits of any of the transformer parts are not exceeded.

4. Single-phase measurements of excitation voltage, total current, and power shall be similar to those described in 9.3. The zero-phase-sequence impedance, in percent, on kilovoltampere base of excited winding for the test connection is as in Equation (21):

$$Z_0(\%) = 300 \left(\frac{E}{E_r} \times \frac{I_r}{I} \right) \quad (21)$$

where

0 Z is the zero sequence impedance, in percent, on kilovoltampere base of excited winding for the test connection.

E is measured excitation voltage (V)

r E is rated phase-to-neutral voltage of excited winding (V)

I is measured total input current flowing in the three parallel-connected phases (A)

r I is rated current per phase of the excited windings (A)

Transformers with one neutral externally available, excluding transformers with interconnected windings.

The zero-phase sequence network, given the external characteristics for transformers of this type, is shown in Figure 25. Winding 1 has the available neutral, whereas windings 2, 3, and so forth do not.

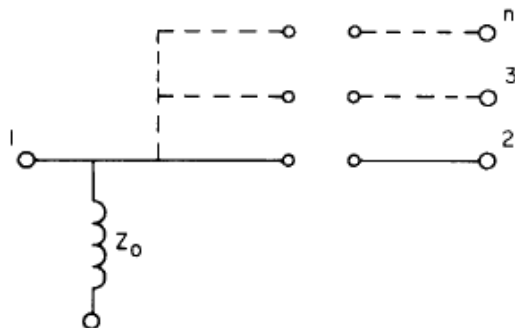


Figure 25—Equivalent zero-phase-sequence network for transformers with one externally available neutral

A zero-sequence test shall be made on the winding with the available neutral. A single-phase voltage shall be applied between the three shorted line terminals and the neutral. The external terminals of all other windings may be open-circuited or shorted and grounded.

The term *interconnected windings* shall be interpreted to mean windings in which one or more electrical phases are linked by more than one magnetic phase.

Test 1				
Short the Low voltage terminals, X1, X2, X3, apply a single-phase voltage to LV winding between the shorted line terminals of Low voltage terminals and its neutral. All other windings are open-circuited. The measured zero-sequence impedance is represented by Z1N0.				
applied current (A)	Measured voltage (V)	Measured loss (W)	test frequency (Hz)	Zero impedance (Ω)
162.28	41	830.47	50	0.758
Rated current (A)	Rated voltage (V)	Rated power (MVA)	Rated frequency	corrected impedance %
1104.528567	34500	66	60	5.043442849
Rated current (A)	Rated voltage (V)	Rated power (MVA)	Rated frequency	corrected impedance %
1472.704756	34500	88	60	6.724590465
Rated current (A)	Rated voltage (V)	Rated power (MVA)	Rated frequency	corrected impedance %
1840.880945	34500	110	60	8.405738081
Test 2				
Short the Low voltage terminals, X1, X2, X3, apply a single-phase voltage to LV winding between the shorted line terminals of Low voltage terminals and its neutral. Short the line terminals and neutral of Hv voltage winding. All other windings may be open circuited or shorted. The measured zero-sequence impedance is represented by Z2N0.				
applied current (A)	Measured voltage (V)	Measured loss (W)	test frequency (Hz)	Zero impedance (Ω)
169.31	29	627.3	50	0.514
Rated current (A)	Rated voltage (V)	Rated power (MVA)	Rated frequency	corrected impedance %
1104.528567	34500	66	60	3.419193147
Rated current (A)	Rated voltage (V)	Rated power (MVA)	Rated frequency	corrected impedance %
1472.704756	34500	88	60	4.558924196
Rated current (A)	Rated voltage (V)	Rated power (MVA)	Rated frequency	corrected impedance %
1840.880945	34500	110	60	5.698655245

Test 3				
Short the HV voltage terminals, Hv Tap on 17, X1, X2, X3, apply a single-phase voltage to HV winding between the shorted line terminals of HV voltage terminals and its neutral. All other windings are open-circuited. The measured zero-sequence impedance is represented by Z3N0.				
applied current (A)	Measured voltage (V)	Measured loss (W)	test frequency (Hz)	Zero impedance (Ω)
134.69	942	12372.27	50	20.982
Rated current (A)	Rated voltage (V)	Rated power (MVA)	Rated frequency	corrected impedance
331.3585701	115000	66	60	12.56511031
Rated current (A)	Rated voltage (V)	Rated power (MVA)	Rated frequency	corrected impedance %
441.8114269	115000	88	60	16.75348041
Rated current (A)	Rated voltage (V)	Rated power (MVA)	Rated frequency	corrected impedance %
552.2642836	115000	110	60	20.94185051
Test 4				
Short the HV voltage terminals, Hv Tap on 17, X1, X2, X3, apply a single-phase voltage to HV winding between the shorted line terminals of HV voltage terminals and its neutral. Short the line terminals and neutral of Lv voltage winding. All other windings may be open circuited or shorted. The measured zero-sequence impedance is represented by Z4N0.				
applied current (A)	Measured voltage (V)	Measured loss (W)	test frequency (Hz)	Zero impedance (Ω)
135.96	591	6255.36	50	13.041
Rated current (A)	Rated voltage (V)	Rated power (MVA)	Rated frequency	corrected impedance %
331.3585701	115000	66	60	7.809569255
Rated current (A)	Rated voltage (V)	Rated power (MVA)	Rated frequency	corrected impedance %
441.8114269	115000	88	60	10.41275901
Rated current (A)	Rated voltage (V)	Rated power (MVA)	Rated frequency	corrected impedance %
552.2642836	115000	110	60	13.01594876

Test date:	2024/7/13		Test Sample	01	
Ambient Temp.:	33	°C	Relative Humidity:	63%	
Top Oil Temp. Indicator:	33	°C	Bottom Oil Temp. Indicator:	33	°C

Measuring instrument:			
temperature and humidity meter	J202404303427-0001		
Temperature Recorder	2JB24000718-0004		
Multi-channel power analyzer	24KA202406280013		
Current transformer	24KA202406280019	24KA202406280021	24KA202406280022
Voltage transformer	24KA202406280020	24KA202406280023	24KA202406280024

7.Operation tests of all devices

Mechanical tests

IEEE C57.12.00, Table 17

CSA C88-16, Clause 16.5

OLTC

1.All electrical and electro-mechanical devices for controlling auxiliary devices such as fans, pumps, motors, LTCs, etc., shall be operated both in auto and manual mode for proper sequencing/staging and function.

2. Tests on on-load tap-changers – Operation test refer to IEC 60076.1 Clause 11.7 With the tap-changer fully assembled on the transformer, the following sequence of operations shall be performed without failure:

- a) with the transformer de-energized, eight complete cycles of operation (a cycle of operation goes from one end of the tapping range to the other, and back again).
- b) with the transformer de-energized, and with the auxiliary voltage reduced to 85 % of its rated value, one complete cycle of operation.
- c) with the transformer energized at rated voltage and frequency at no load, one complete cycle of operation.
- d) with one winding short-circuited and, as far as practicable, rated current in the tapped winding, 10 cycles of tap-change operations across the range of two steps on each side from where a coarse or reversing changeover selector operates, or otherwise from the middle tapping (the tap changer will pass 20 times through the changeover position).

	transformer			OLTC		
a	Voltage	Current	Power	Rated Voltage	Supplied Voltage	cycles
	Vac	A	w	Vac	Vac	
	/			120	120	
b	Voltage	Current	Power	Rated Voltage	Supplied Voltage	cycles
	Vac	A	w	Vac	Vac	
	/			120	102	
c	no load			Rated Voltage	Supplied Voltage	cycles
				Vac	Vac	
				120	120	
d	short-circuited			Rated Voltage	Supplied Voltage	cycles
				Vac	Vac	
				120	120	
Pass						
Operation test on controlling auxiliary devices						
Pass						
Operation test on fans						
Pass						
Operation test on pumps						
N/A						
Operation test on motors						

Date

Pass					
Test date:	2024/7/11		Test Sample	01	
Ambient Temp.:	33	℃	Relative Humidity:	54%	
Top Oil Temp. Indicator:	33	℃	Bottom Oil Temp. Indicator:	33	℃
Measuring instrument:					
temperature and humidity meter		J202404303427-0001			
Temperature Recorder		24KA202406280014			
Digital Multimeter		CE24AA02246001			
Current transformer		24KA202406280019	24KA202406280021	24KA202406280022	
Voltage transformer		24KA202406280020	24KA202406280023	24KA202406280024	
OLTC Tester		24KA202406280017			

Date

8.Control (auxiliary) and cooling consumption losses**IEEE C57.12.00, Table 17
CSA C88-16, Clause 16.2 p)**

	Voltage	frequency	Current	measured losses	Corrected loss
Set 1	199.966	50	5.7271	714	956.76
set 2	200.125	50	5.775	713.1	955.554

Test date:	2023/7/29		Test Sample	01	
Ambient Temp.:	33.9	°C	Relative Humidity:	69%	
Top Oil Temp. Indicator:	33.9	°C	Bottom Oil Temp. Indicator:	33.9	°C
Measuring instrument:					
temperature and humidity meter		J202404303427-0001			
Temperature Recorder		24KA202406280014			
Multi-channel power analyzer		24KA202406280013			
Current transformer		24KA202406280019	24KA202406280021	24KA202406280022	
Voltage transformer		24KA202406280020	24KA202406280023	24KA202406280024	

9. Audible sound levels**IEEE C57.12.90, Clause 13****IEEE C57.12.00 Table 17, Clause 8.2.3****CSA C88 16.2 q), 16.8.1**

The ambient sound pressure level shall be established by averaging the ambient sound pressure levels measured immediately preceding and immediately following the sound measurements with the transformer energized. The ambient sound shall be measured at a minimum of four locations, and the instruments shall be in conformance with 13.2. However, additional measurements may be made if agreed to by the manufacturer and purchaser or if the ambient measurements vary by more than 3 dB around the transformer. At least one of the locations for measuring ambient sound pressure levels shall be on the center of each face of the measurement surface. Ambient sound pressure level corrections may be made at each microphone location before the average transformer sound pressure level is computed. Ambient sound pressure levels shall be measured using the same bandwidth (A-weighted, one-third octave, or discrete frequencies) specified for measuring the transformer sound. The ambient sound corrections shall be made with the identical frequency bandwidths that are used to measure the combined transformer and ambient sound pressure level.

Measurements shall be made in an environment having an ambient sound pressure level at least 5 dB below the combined transformer and ambient sound pressure level. When the ambient sound pressure level is 5 dB or more, below the combined transformer and ambient sound pressure level, the corrections shown in Table 7 shall be applied to the combined transformer and ambient sound pressure level to obtain the transformer sound pressure level. When the difference between the ambient sound pressure level and the combined transformer and ambient sound pressure level is less than 5 dB, and it is desired only to know the sound pressure level that the transformer does not exceed, a correction of -1.6 dB may be used. Alternatively, the manufacturer may attempt to reduce the ambient noise during the sound test. For one third octave or narrowband measurements, the 5 dB difference shall apply to the main frequency components of the transformer noise; refer to NOTES 1, 2, and 3 in 13.5.6.1.

When ambient sound conditions do not comply, suitable corrections may be feasible when the ambient sound conditions are steady and discrete frequency sound levels are measured.

Table 7—Ambient sound corrections

Difference between the ambient sound pressure level and the combined transformer and ambient sound pressure level (dB)	Correction to be added to the combined transformer and ambient sound pressure level to obtain the ambient-corrected sound pressure level of the transformer (dB)
5	-1.6
6	-1.3
7	-1.0
8	-0.8
9	-0.6
10	-0.4
Over 10	0.0

This correction accounts for the influence of undesired sound reflections from test room boundaries (walls, ceiling, and floor) and reflecting objects near the test transformer. The magnitude of K depends principally on the ratio of the transformer sound measuring surface S to the sound absorption area of the test room A. The sound reflection correction K shall be calculated using Equation (34), or obtained from Figure 34 using the calculated value of A/S.

$$K = 10 \log_{10} \left(1 + \frac{4}{(A/S)} \right) \quad (34)$$

The value of “A,” in square meters, is given by Equation (35):

$$A = \alpha S_v \quad (35)$$

where

- α is average acoustic absorption coefficient (from Table 8)
 S_v is total area of surface of test room (walls, ceilings, and floors), in square meters.

The sound measuring surface area S is the vertical area (in square meters) enveloping the transformer (measurement surface) on which the sound measurement points are located plus the horizontal plane bounded by the vertical measurement surface. The measurement surface area is approximately equal to 125 % of the vertical area enveloping the transformer. Therefore, the sound measuring surface area S, is calculated using the following Equation (36):

$$S = 1.25 \times h \times l_m \quad (36)$$

where

- h is the height of the transformer tank, in meters
 l_m is the length of the sound measuring contour, in meters

The 1.25 empirical factor accounts for the sound radiated by the transformer tank cover

The maximum value allowed for this sound reflection correction is 4 dB. This is equivalent to allowing sound reflection equivalent to 1.5 times the sound radiated from the transformer. In cases of high wall sound reflections, the transformer manufacturer should, if applicable/feasible, move the transformer to a larger test area or outdoors.

NOTE—The sound reflection correction described above is based on the IEC 60076-10 (Determination of sound levels) for transformer sound measurements.

Table 8 —Approximate values of the average acoustic absorption coefficient

Description of room	Average acoustic absorption coefficient, α
Nearly, or partly empty, room with smooth hard walls	0.1
Industrial room with machinery and no sound absorbing material on walls or ceiling	0.2
Industrial room with a small amount of sound absorbing material on ceiling or walls	0.3
Room with substantial amount of sound absorbing materials on both ceilings and walls	0.4
Sound room or room with an open side	0.5

13.6.3 Sound power level calculation (L_w)

The A-weighted, one-third octave band, or discrete frequency sound power levels shall be computed using corresponding sound pressure level measurements using Equation (39):

$$L_w = L_p + 10 \cdot \log_{10}(S) \quad (39)$$

where

L_w is the sound power level of the transformer, in dB (A)

L_p is the sound pressure level of the transformer, in dB (A)

Refer to Equation (36) for the calculation of the sound measuring surface area “S.”

The reference sound-producing surface of a transformer is a vertical surface that follows the contour of a taut string stretched around the periphery of the transformer or integral enclosure (see Figure 33). The contour shall include radiators, coolers, tubes, switch compartments, and terminal chambers; but it shall exclude bushings and minor extensions, such as valves, liquid gauges, thermometers, conduit terminal boxes, and projections at or above cover height.

The first microphone locations shall coincide with the main drain valve. Additional microphone locations shall be at 1 m (3 ft) intervals in a horizontal direction, proceeding clockwise as viewed from above along the measurement surface defined in 13.4.1.

No fewer than four microphone locations shall be used. For small transformers, intervals of less than 1 m (3 ft) may result. The microphone shall be located on the measurement surface. When fans are not in operation, the microphone shall be spaced 0.3 m (1 ft) from the reference sound-producing surface (ONAN contour). When fans are in operation, the microphone shall be located 2 m (6 ft) from the reference sound producing surface (ONAF contour), refer to Figure 33.

For transformers having an overall tank or enclosure height of less than 2.4 m (7.9 ft), measurements shall be made at half height. For transformers having an overall tank or enclosure height of 2.4 m (7.9 ft) or more, measurements shall be made at one-third and at two-thirds height.

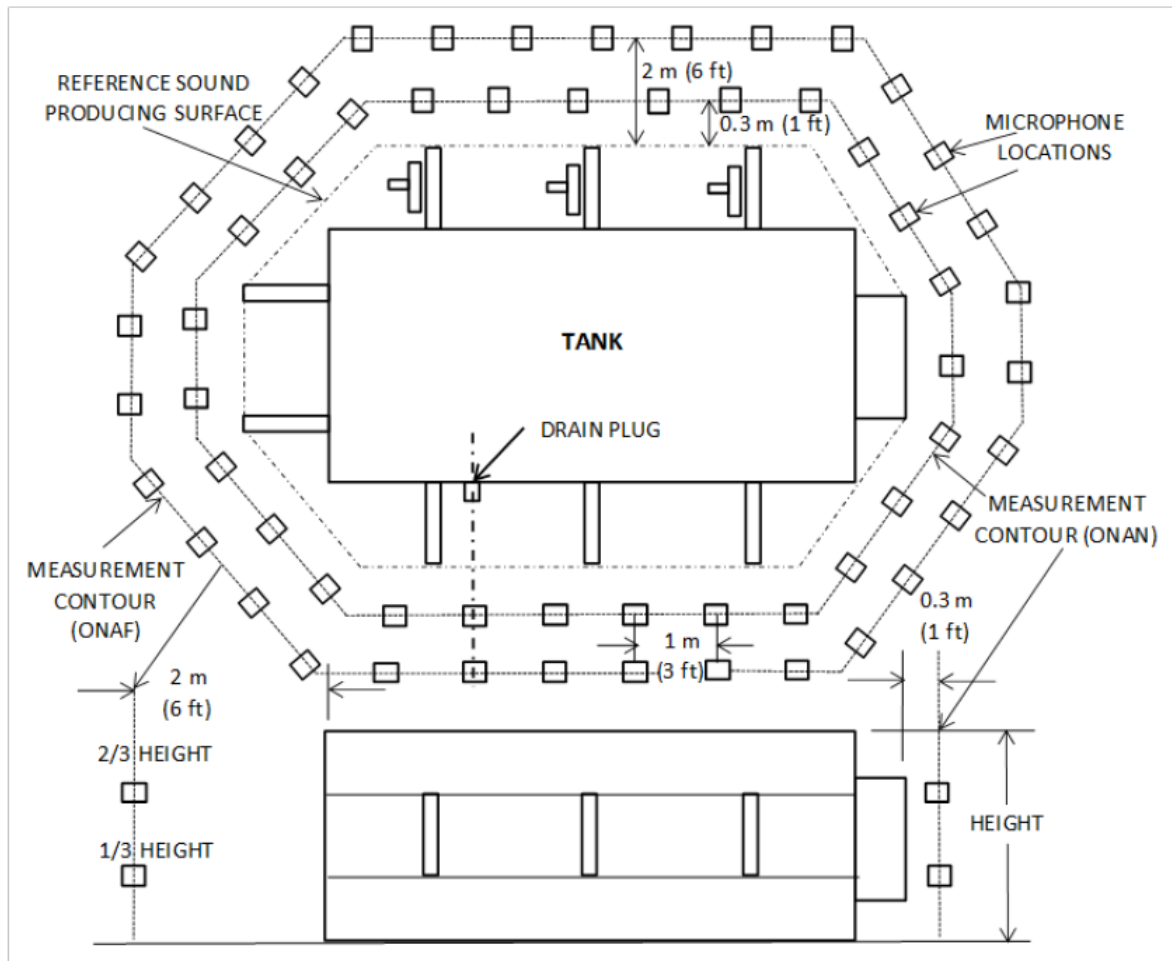


Figure 33—Microphone location for measuring audible sound from transformers

background								
Mic. Position	1	2	3	4	5	6	7	8
Before (1/3h)	44.9	45.3	44.7	44.8	44.1	45.5	46	44.7
Before (2/3h)	44.5	45.6	45.2	45.1	44.8	44.7	44.9	44.5
after (1/3h)	44.4	44.7	44.8	44.9	45.1	45.2	45.1	44.9
after (2/3h)	44.2	44.6	44.3	44.6	45.1	44.3	44.5	44.6
Mic. Position	9	10	11	12	13	14	15	16
Before (1/3h)	43.4	44.2						
Before (2/3h)	45.3	44.5						
after (1/3h)	44.4	44.7						
after (2/3h)	44.2	44.6						

Test at ONAN								
Mic. Position	1	2	3	4	5	6	7	8
Test (1/3h)	55.01	57.41	57.51	54.21	54.11	54.01	54.31	57.71
	5869.56 9376	10200. 143	10437.7 34	4882.0 952	4770.9 651	4662.36 4681	4995.81 3764	10929.649 63
Test (2/3h)	56.41	59.71	54.71	53.01	53.51	54.21	54.61	54.51
	8102.26 1207	17322. 327	5477.80 09	3703.4 479	4155.3 369	4882.09 5159	5353.11 0893	5231.2592 13
Mic. Position	9	10	11	12	13	14	15	16
Test (1/3h)	55.31	55.81	54.71	55.91	58.11	59.61	55.41	55.91
	6289.35 69	7056.7 745	5477.80 09	7221.1 479	11984. 123	16928.0 2299	6435.85 4842	7221.1479 01
Test (2/3h)	54.91	56.31	53.21	55.81	56.81	59.01	56.51	55.21
	5735.96 1665	7917.8 312	3877.98 6	7056.7 745	8883.9 528	14743.6 9167	8290.98 7114	6146.1936 58
Mic. Position	17	18	19	20	21	22	23	24
Test (1/3h)	61.12	58.62	58.42	57.92	57.82	58.02	56.32	59.02
	23966.5 8966	13477. 404	12870.8 21	11471. 131	11210. 016	11738.3 2799	7936.08 3711	14777.679 39
Test (2/3h)	55.11	56.01	55.11	58.31	54.31	57.51	56.51	55.91
	6006.28 921	7389.3 5	6006.28 92	12548. 917	4995.8 138	10437.7 3437	8290.98 7114	7221.1479 01
Mic. Position	25	26	27	Average Noise Level of Trans		Average Backgro und Noise Level dB (A)	Lp Sounding Pressure dB(A)	
Test (1/3h)	52.6	53.8	54.6					
	3369.81 6405	4442.2 832	5340.79 91					
Test (2/3h)	54.5	54.1	55.1					
	5219.22 765	4759.9 922	5992.47 51	56		44.78	56.45	
Total Surface Area of Testing Room	Average Absorption Factor		Absorp tion Amount	Distance to The Reference Sounding surface		Area of Measuring Surface		Environm ental Modifica tion
Sv (m²)	α		A (m²)	(m)		S (m2)		K (dB)
7800	0.2		1794.6	0.3		116.775		1
State of Coolin g Equipm ent	Noise Level Difference Between Trans. and Background dB (A)		Background Noise Level Modification dB (A)		A Sounding Power dB(A)			
OFF	11.22		0		77.1			

Test at ONAF								
Mic. Position	1	2	3	4	5	6	7	8
Test(1/3h)	65.18	65.48	65.28	65.68	64.88	65.38	65.78	65.98
	41201 .2	44147 .9	42160.9	46228 .5	38451 .2	43143	47305 .3	49534.8
Test(2/3h)	64.88	65.18	64.48	65.18	64.68	64.78	65.68	65.38
	38451 .2	41201 .2	35067.9	41201 .2	36720 .6	37576	46228 .5	43143
Mic. Position	9	10	11	12	13	14	15	16
Test(1/3h)	65.48	65.38	65.88	65.88	65.58	65.88	65.78	64.58
	44147 .9	43143	48407.2	48407 .2	45176 .2	48407 .2	47305 .3	35884.8
Test(2/3h)	64.88	65.18	64.48	65.18	64.68	64.78	65.68	65.38
	38451 .2	41201 .2	35067.9	41201 .2	36720 .6	37576	46228 .5	43143
Mic. Position	17	18	19	20	21	22	23	24
Test(1/3h)	63.98	64.78	65.18	65.38	65.88	64.58	64.98	65.08
	31254 .3	37576	41201.2	43143	48407 .2	35884 .8	39346 .9	40263.4
Test(2/3h)	64.78	64.98	65.38	65.58	63.68	64.18	64.78	64.88
	37576	39346 .9	43143	45176 .2	29168 .2	32727 .3	37576	38451.2
Mic. Position	25	26	27	28	29	30	31	32
Test(1/3h)	64.48	65.58	65.98	65.88	68.38	66.68	66.68	64.88
	35067 .9	45176 .2	49534.8	48407 .2	86081 .5	58198 .3	58198 .3	38451.2
Test(2/3h)	64.78	64.88	63.68	64.98	64.88	66.48	66.88	65.58
	37576	38451 .2	29168.2	39346 .9	38451 .2	55578 .9	60941 .1	45176.2
Mic. Position	33	34	35	36	37	38	39	40
Test(1/3h)	66.48	66.08	63.98	64.88	65.48	65.18	65.88	65.78
	55578 .9	50688 .6	31254.3	38451 .2	44147 .9	41201 .2	48407 .2	47305.3
Test(2/3h)	66.48	66.08	63.98	64.88	65.48	65.38	65.58	65.48
	55578 .9	50688 .6	31254.3	38451 .2	44147 .9	43143	45176 .2	44147.9
Total Surface Area of Testing Room	Average Absorption Factor		Absorpti on Amount	Distance to The Reference Sounding surface		Area of Measuring Surface		Environmen tal Modificati on
Sv (m²)	α		A (m²)	(m)		S (m2)		K (dB)
7800	0.2		1794.6	0.3		173		1.42
State of Coolin g Equipm ent	Average Noise Level of Trans		Average Background Noise Level dB(A)			Noise Level Difference Between Trans. and Background dB(A)		
ON	65.31		44.78			20.53		

Date

Background Noise Level Modification dB(A)	Lp, A Sounding pressure dB(A)	A Sounding Power dB(A)
0	65.4	87.8

Test date:	2024/7/12		Test Sample	01	
Ambient Temp.:	33	℃	Relative Humidity:	63%	
Top Oil Temp. Indicator:	33	℃	Bottom Oil Temp. Indicator:	33	℃
Measuring instrument:					
temperature and humidity meter		J202404303427-0001			
Temperature Recorder		2JB24000718-0004			
Sound level meter		24KA202406280018			
Multi-channel power analyzer		24KA202406280013			
Current transformer		24KA202406280019	24KA202406280021	24KA202406280022	
Voltage transformer		24KA202406280020	24KA202406280023	24KA202406280024	
Steel measuring tape		2JB24000718-0004			

10. Temperature Rise Test**IEEE C57.12.90, Clause 11****IEEE C57.12.00 Table 17, Clause 5.11****CSA C88 16.2 f)**

1. A temperature-rise test is defined as a test to determine the temperature rise above ambient of one or more of the transformer's windings, as measured at the terminals. The result for a given terminal pair or winding is the average value of the temperature of the entire circuit; it is not the temperature at any given point in a specific winding. The term average temperature rise refers to the value determined by measurements on a given terminal pair of the winding. It does not refer to the arithmetic average of results determined from different terminal pairs of the transformer.

2. See 4.1.2 and 5.11.2 of IEEE Std C57.12.00-2015 for conditions under which temperature limits apply. All temperature-rise tests shall be made under normal (or equivalent to normal) conditions for the means of cooling, as follows:

a) Temperature-rise tests shall be conducted on transformers that are completely assembled and filled to the proper liquid level.

b) Temperature-rise tests shall be made in a room that is as free from drafts as practicable; defined as a wind speed of 0.5 m/s, or less.

c) When it is not possible or practical to test the transformer as a completed assembly, the transformer shall be tested with the components required to replicate normal means of cooling the transformer during temperature-rise tests. When the transformers are equipped with thermal indicators, bushing current transformers, or the like, such devices shall be assembled with the transformer.

4. Tests shall be made by one of the following methods:

~~a) Actual loading~~

b) Simulated loading

~~1. The loading back (opposition) method, in which rated voltage and current are induced in the transformer under test.~~

~~2. The short-circuit method, in which appropriate total losses are produced by the effect of short circuit current.~~

~~Actual loading is the most accurate method, but energy requirements are excessive for large transformers. Transformers of small output may be tested under actual load conditions by loading them on a rheostat, bank of lamps, water box, and so on.~~

Simulated loading

Loading back method

~~Some smaller transformers (typically distribution class and small power) may be tested by connecting their respective high voltage and low voltage windings in parallel (see Figure 29 and Figure 30). Transformers shall be tested with the combination of connections and taps that give the highest average winding temperature rise. This process will generally involve those connections and taps resulting in the highest losses. In the case of internally grounded transformers, caution should be taken for possible elevated tank voltages.~~

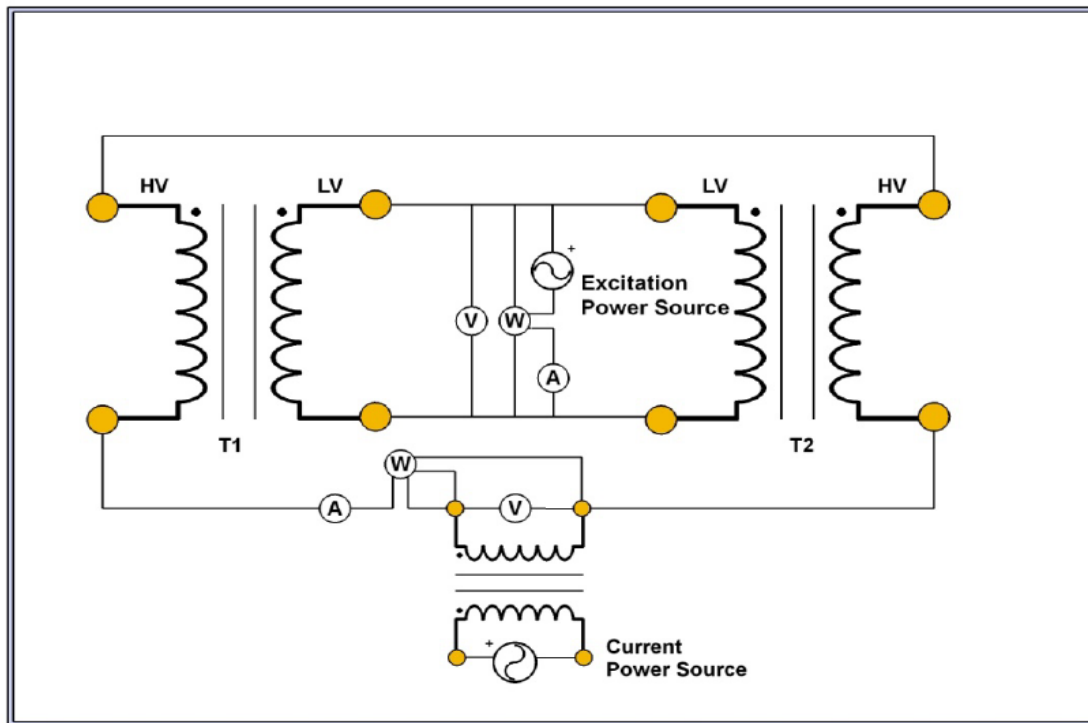


Figure 29—Example of loading back method: single-phase

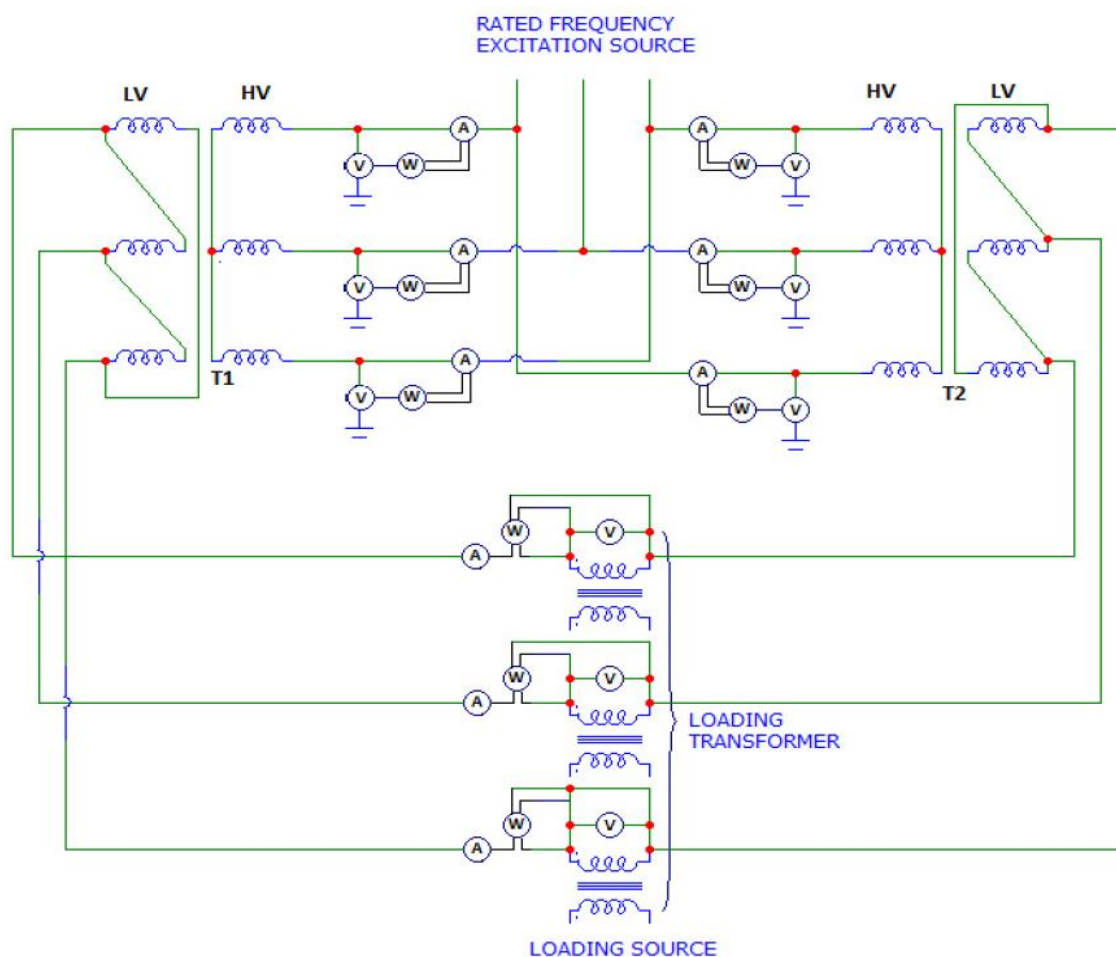


Figure 30—Example of loading back method: three-phase

Short-circuit method

Conduct the short-circuit method as follows:

a) Prior to making the total loss run (see step b), measure load loss at rated current and frequency for the combination of connections and taps that give the highest average winding temperature rise.

This process will generally involve those connections and taps resulting in the highest losses. This thermal connection load loss shall be measured in accordance with Clause 9 and referenced to a temperature equal to rated average winding rise plus 20 °C. The required total losses for the total loss run shall be the sum of thermal connection load loss plus no-load loss measured in accordance with Clause 8.

b) For the total loss run, short-circuit one or more windings and circulate sufficient current at rated frequency to produce the required total losses as determined in step a and in 11.4.2.

c) Determine liquid temperature rises as described in 11.3.2.

d) For the rated current run, reduce the current in the windings to the rated current (or reduced current according to 11.4.1) value for the connection and the loading used. Hold the current constant for 1 h. Measure the liquid temperatures, immediately shut down, and measure the hot resistances in accordance with 11.2.2.

e) Repeat the rated current run (step d) for hot-resistance measurements on additional terminal pairs if needed to meet the time limit criteria of 11.2.2.

f) Determine average winding rises in accordance with 11.3.3.

Cold-resistance measurements

Cold-resistance measurements shall be taken on all terminal pairs in accordance with Clause 5. The same test equipment shall be used for both cold-resistance and hot-resistance measurements. Normally, cold resistance measurements are taken prior to loading the transformer for temperature-rise test. However, it is permissible to allow the transformer to cool to ambient temperature and perform cold-resistance measurements after the loading test. Whenever it is necessary to make cold-resistance measurements following the temperature-rise test, the cool-down time shall be sufficient to allow the criteria in 5.1 to be met.

Hot-resistance measurements

When the transformer is shut down for hot-resistance measurements, fans and cooling water shall be shut off. Insulating liquid pumps may be shut off or left running during shutdown. Hot-resistance measurements shall be taken as soon as possible after shutdown, allowing sufficient time for the inductive effects to disappear as indicated from the cold-resistance measurement. To minimize inductive effects when transferring measuring instrument leads from one terminal pair to another, the same relative polarity should be maintained between measuring leads and transformer terminals.

- a) The time from instant of shutdown, i.e., time zero, shall be recorded for each resistance measurement.
- b) For distribution transformers, a hot resistance measurement shall be made on each terminal pair (corresponding to a phase of a winding), and a series of hot resistance measurements shall be made on at least one terminal pair under test. For power transformers, a series of hot resistance measurements shall be made on each terminal pair (corresponding to a phase of a winding) under test, and all terminal pairs shall be tested at maximum nameplate MVA rating. A series of hot resistance measurements may be made on only one terminal pair under test at lower nameplate ratings. The terminal pair used for those tests should be the one found by prior testing to have the highest winding temperature rise.
- c) The first hot resistance measurement shall be taken as quickly as possible after shutdown and after the inductive effects disappear and in any case, no later than 4 min after shutdown. In order to achieve core saturation in a short time period, the dc power supply voltage should be large enough to saturate the cores in a short time. The dc source should be selected such that it can also provide sufficient current to keep the core saturated without causing excessive heating of windings. A series resistor added to the circuit will decrease time to stabilization. When a transformer has a winding connected in delta, all involved core legs shall be saturated, unless the delta is open during shutdown, for stable and accurate readings. When performing winding resistance measurements on a transformer that utilizes series transformers for load tap changing, care is required to increase the likelihood that the whole circuit, consisting of the main and series transformer windings, is saturated before stable readings of the combined winding resistances can be obtained.

- d) The hot resistance versus time data shall be used as the basis for correction to time zero. The cooling curve format shall consist of a series of hot resistance measurements in accordance with c). For distribution transformers, at least ten resistance measurements shall be made on one terminal pair corresponding to a phase of a winding. All resistance measurements shall be recorded at no longer than 30-s intervals and no less than 10-s intervals. For power transformers, the last hot resistance measurement shall be taken no sooner than 10 min from shutdown. All hot resistance measurements taken within the 10-min period after shutdown shall be recorded at no longer than 15-s intervals.
- e) The hot resistance-time data collected in d) shall be extrapolated to the instant of shutdown using a computerized curve fitting program to determine the hot resistance-time cooling curve. The hot resistance-time data collected shall be fitted to an exponential decay curve using the method of least squares.

NOTE 1— Comparative to large power transformers, the time constant of small distribution transformers may be short due to transformer thermal characteristics and measuring equipment, which may affect the appropriateness of an exponential decay curve.

NOTE 2— The mean liquid temperature surrounding the winding and the actual location of the winding may result in top and bottom liquid temperatures not giving the intended average winding liquid temperature, which could affect the appropriateness of the exponential decay curve.

NOTE 3— The Working Group is initiating an investigation to define appropriate curves concerning NOTE 1 and NOTE 2 for a future modification to this clause

- f) When the series of hot resistance measurements on a distribution transformer is made on only one terminal pair under test, the resistance-time data obtained on one phase of a winding shall be used to determine the correction to shutdown for the other phases of the same winding, provided the first measurement of each of the other phases has been taken within 4 mins after shutdown.

5. Temperature measurements

5.1 Ambient temperature measurements

5.1.1 Air-cooled transformers

For air-cooled transformers, the ambient temperature shall be taken as that of the surrounding air, which should not be less than 10 °C nor more than 40 °C. For temperatures within this range, no correction factor shall be applied. Tests may be made at temperatures outside this range when suitable correction factors are available. The temperature of the surrounding air shall be determined by at least three thermocouples or thermometers in containers spaced uniformly around the transformer under test. They shall be located about mid-height of the transformer and 1 m to 2 m (3 ft to 6 ft) from the transformer. They shall be protected from drafts and radiant heat from the transformer under test or other sources.

When the liquid time constant of the transformer (calculated at rated kVA according to IEEE Std C57.91) is 2 h or less, the time constant of the containers shall be between 50% and 150% of that of the transformer under test. When the liquid time constant of the transformer under test is more than 2 h, the time constant of the containers shall be within 1 h of the liquid time constant of the transformer under test.

The time constant of a container shall be taken as the time necessary for its temperature to change 6.3 °C when the ambient temperature is abruptly changed 10 °C.

5.1.2 Water-cooled transformers

For water-cooled transformers, the flow rate in liters per minute and the temperature of the incoming and outgoing water shall be measured. The ambient temperature shall be taken as that of the incoming water and should not be less than 20 °C nor more than 30 °C. For temperatures within this range, no correction factor shall be applied. Tests may be made at temperatures outside this range when suitable correction factors are available.

Liquid temperature rise determination

Liquid temperature rise is the difference between liquid temperature and ambient temperature. The ultimate liquid temperature rise above ambient shall be considered to be reached when the top liquid temperature rise does not vary more than 2.5% or 1 °C, whichever is greater, during a consecutive 3-h period.

It is permissible to shorten the time required for the test by the use of initial overloads, restricted cooling, and so on. The top liquid temperature shall be measured by a thermocouple or suitable thermometer immersed approximately 50 mm (2 in) below the top liquid surface. The bottom liquid temperature shall be measured by one of the following methods:

- a) Thermocouples may be attached to an insulated rod and located inside the tank so that the thermocouples are in the liquid flow path from the external cooling means to the bottom of the windings.
- b) If heat exchangers or radiators are mounted on a common manifold with a single entrance to the tank, the thermocouples may be located in the piping of the single entrance.
- c) If heat exchangers or radiators have multiple entrances into the tank, thermocouples may be installed in the bottom of one radiator or heat exchanger. For accuracy, a radiator or heat exchanger located in the middle of the bank is preferred.
- d) If it is not possible to measure the temperature of the liquid inside the tank, radiators, or heat exchangers, surface temperature measurements may be used with the results corrected to account for the temperature difference between the surface and the liquid inside the tank. If surface temperature measurements are made on radiator headers, choose headers one third or one half the way in from either end of a bank of radiators. For transformers without radiators, locate the thermocouples on the tank wall at the elevation of the bottom of the winding. Thermocouples located on external cooling surfaces, for the purpose of determining internal liquid temperatures, shall be shielded and insulated so that their readings are not significantly affected by the air movement from fans or thermally induced air currents.

The average liquid temperature shall be determined as equal to top liquid temperature minus half the difference in temperature of the moving liquid at the top and bottom of the cooling means. When bottom liquid temperature cannot be measured directly, the temperature difference may be taken as the difference between the surface temperature of the liquid inlet and the outlet. A thermocouple is the preferred method of measuring surface temperature (see 11.3.4 for method of measurement). Infrared measurement devices may also be used to measure surface temperatures.

Average winding temperature-rise determination

The average winding temperature of a terminal pair corresponding to a winding phase shall be determined from the terminal pair's hot resistance at shutdown. When it is not possible to determine the hot resistance (e.g., with extremely low resistance windings), other methods may be used. The average winding temperature of a terminal pair shall be determined by Equation (28):

$$\theta_w = \frac{R_h}{R_c} (\theta_k + \theta_{rc}) - \theta_k \quad (28)$$

The average temperature rise of a terminal pair corresponding to a winding phase shall be determined by Equation (29):

$$\Delta\theta_w = \Delta\theta_l + \theta_w - \theta_l \quad (29)$$

where

$\Delta\theta_w$	is the average winding temperature rise of a terminal pair (°C)
$\Delta\theta_l = \theta_{l,TL} - \theta_a$	is the average liquid temperature rise as determined from the total loss run (°C)
θ_w	is the average winding temperature of a terminal pair corresponding to hot resistance R_h (°C)
$\theta_{l,TL}$	is the average liquid temperature at end of total loss run (°C)
θ_l	is the average liquid temperature at shutdown (°C)

θ_a	is the ambient temperature (°C)
θ_{rc}	is the temperature at which cold resistance R_c was measured (°C)
R_c	is the cold resistance, measured according to Clause 5 (Ω)
R_h	is the hot resistance of a terminal pair (Ω)
θ_k	is 234.5 °C for copper and 225.0 °C for aluminum

NOTE—The value for θ_k may be as high as 230 °C for alloyed aluminum.

Average winding rise shall be calculated by using either top liquid rise or average liquid rise. When other-than-rated winding current is used, the average liquid rise method shall be used to determine winding rises.

- In the top liquid rise method, the average winding temperature rise is equal to the top liquid rise, measured during the total loss run, plus the quantity (average winding temperature at shutdown minus top liquid temperature at shutdown).
- In the average liquid rise method, the average winding temperature rise is the average liquid rise, measured during the total loss run, plus the quantity (average winding temperature at shutdown minus average liquid temperature at shutdown).

The average winding temperature rise for each terminal pair corresponding to a winding phase shall be corrected for actual test currents, test losses, and altitude as prescribed in 11.4. The corrected average winding temperature rise shall be reported for each terminal pair of the transformer.

Other temperature measurements

When measured, the temperature rise of metal parts other than windings shall be determined by use of a thermocouple, suitable thermometer, fiber optic temperature sensor, or other appropriate temperature measurement techniques. A thermocouple is the preferred method of measuring surface temperature. When used for this purpose, the thermocouple should be soldered to the surface. When this option is not practical, the thermocouple should be soldered to a thin metal plate or foil approximately 645 mm² (1 in²) in size. The plate should be held firmly and snugly against the surface. The thermocouple should be thoroughly insulated thermally from the surrounding medium. The surface temperature of metal parts surrounding, or adjacent to, outlet leads or terminals carrying heavy current may be measured at intervals or immediately after shutdown.

Correction of temperature-rise test results

For any of the loading methods adopted, temperature-rise test results shall be corrected for the predictable effects caused by:

- Difference in winding rated current and the winding test current
- Difference in required loss and test loss
- Difference in altitude of operation

11.4.1 Correction for differences between winding rated current and test current

When test equipment limitations dictate, it is permissible to hold winding current at a value lower than rated current for the winding, but not less than 85% of rated winding current. When the current held in any of the windings under test differs from the rated current, the observed differences between the average winding temperature at shutdown and the average liquid temperature at shutdown shall be corrected to give the average temperature rise of the windings at the rated current by using Equation (30):

$$\Delta\theta_{w,c} = \Delta\theta_{w,o} \left(\frac{\text{rated current}}{\text{test current}} \right)^{2m} \quad (30)$$

where

$\Delta\theta_{w,c}$ is the corrected difference between average winding temperature corrected to shutdown, and the average liquid temperature at shutdown (°C)

$\Delta\theta_{w,o}$ is the observed difference between average winding temperature corrected to shut own, and the average liquid temperature at shutdown (°C)

m is 0.8 for Classes ONAN, ONAF, OFAF, and OFWF
is 1.0 for Classes ODAF and ODWF

The corrected average winding rise is the average liquid rise plus $\Delta\theta_{w,c}$.

11.4.2 Correction of liquid temperature rise for differences in required total loss and actual loss

This method may be used when actual loss is within 20% of the required total loss, as in Equation (31):

$$\Delta\theta_{l,c} = \Delta\theta_{l,o} \left[\left(\frac{P_r}{P_T} \right)^n - 1 \right] \quad (31)$$

where

$\Delta\theta_{l,c}$ is the liquid temperature rise correction (°C)

$\Delta\theta_{l,o}$ is the observed liquid temperature rise (°C)

P_r is the required total loss (W)

P_T is the actual test total loss (W)

n is 0.8 for Class ONAN
is 0.9 for Classes ONAF, OFAF, and OFWF
is 1.0 for Classes ODAF and ODWF

Corrected liquid temperature rise = observed liquid temperature rise + $\Delta\theta_{l,c}$.

Corrected average winding temperature rise = observed winding temperature rise + $\Delta\theta_{w,c}$.

Correction of liquid temperature rises for differences in altitude

When tests are made at an altitude of 1000 m (3280 ft) or less, no altitude correction shall be applied to the temperature rises.

When a transformer tested at an altitude less than 1000 m (3280 ft) is to be operated at an altitude above

1000 m (3280 ft), it shall be assumed that the liquid temperature rise will increase in accordance with

Equation (32):

$$\Delta\theta_A = \Delta\theta_o \left(\frac{A}{A_o} - 1 \right) F \quad (32)$$

where

A $\Delta\theta$ is the increase in liquid temperature rise (°C) at altitude A meters (ft)

o $\Delta\theta$ is the observed liquid temperature rise (°C)

A is altitude meters (ft)

o A is 1000 m (3300 ft)

F is 0.04 for self-cooled mode or 0.06 for forced-air-cooled mode

NOTE-Winding temperature rise above liquid temperature is not affected by altitude

Frequency conversion of temperature-rise test results

In the event that the temperature-rise test cannot be done at the rated power frequency, see Annex B for 50/60 Hz frequency conversions.

110MVA Tap 33							
Test group and conditions							
cooling type	ONAF			Hv position		33	
Quantity of Radiators	Group			Lv position		/	
Quantity of Fans	2		Unit	Running Radiators		/	
Method	Short circuit			Running Fans		/	
Rated Frequency	60		Hz	Actual Frequency	50		Hz
test position	Hv	103500	V	test position	Hv	613.6	A
Rated Voltage	Lv	34500	V	Rated Current	Lv	1840.9	A
Applied loss	No load	48426	W	Actual Current	661.0		A
	load	512184	W	Actual Load loss	498803		W
	Total	560610	w	Actual Total loss	561940		W
test date	start	10:20	2024/7/29		End	21:50	2024/7/29
Top oil Temperature rise(total loss)		Corrected	47.5	℃	tested	47.6	℃
Top oil Temperature (total loss)						85.8	℃
Top oil Temperature rise (shut down)		Corrected	46.0	℃	tested	45.0	℃
Top oil Temperature (shut down)						82.8	℃
diff liquid Temperature rise(total loss)			Corrected	16.0	℃	tested	16.0
diff oil Temperature rise (shut down)			Corrected	15.6	℃	tested	15.3
average oil Temperature Rise(total loss)			Corrected	31.6	℃	tested	31.6
average oil Temperature (total loss) 01 TL						70.5	℃

average oil Temperature rise(shut down)θl	Corrected	30.4	℃	tested	29.7	℃
average oil Temperature (shut down)θl				tested	67.5	℃
position	HV		LV		cold ambinet	
conduct(θk)	Copper	234.5	Copper	234.5	37	℃
Δθl	31.6	℃	hot ambinet		37.8	℃
Cold resistance	H1H0		H2H0		H3H0	
	0.1655	(Ω)	0.165	(Ω)	0.1642	(Ω)
	X1X0		X2X0		X3X0	
	0.01056	(Ω)	0.01061	(Ω)	0.01067	(Ω)
Hot resistance	H1H0		H2H0		H3H0	
	0.19951	(Ω)	0.2002	(Ω)	0.20028	(Ω)
	X1X0		X2X0		X3X0	
	0.012892	(Ω)	0.012942	(Ω)	0.012995	(Ω)
average Winding Temperature θw	H1H0		H2H0		H3H0	
	92.8	℃	94.9	℃	96.7	℃
	X1X0		X2X0		X3X0	
	97.0	℃	96.7	℃	96.2	℃
Winding Temperature rise (AVG) Δθw	H1H0		H2H0		H3H0	
	56.9	℃	59.0	℃	60.8	℃
	X1X0		X2X0		X3X0	
	61.1	℃	60.8	℃	60.3	℃
Corrected Winding Temperature rise (AVG)	H1H0		H2H0		H3H0	
	58.1	℃	60.3	℃	62.1	℃
	X1X0		X2X0		X3X0	
	62.4	℃	62.1	℃	61.6	℃
Winding Temperature rise (TOP)	H1H0		H2H0		H3H0	
	57.6	℃	59.8	℃	61.5	℃
	X1X0		X2X0		X3X0	
	61.8	℃	61.5	℃	61.0	℃
Corrected Winding Temperature rise (TOP)	H1H0		H2H0		H3H0	
	58.9	℃	61.0	℃	62.8	℃
	X1X0		X2X0		X3X0	
	63.1	℃	62.8	℃	62.3	℃

Time	Voltage (V)	Current (A)	Total losses (W)	Temperature on top of oil-T1		Bottom oil		output oil		Environment temperature (℃)				AVERAGE (℃)	Top oil rise (K)
				1	2	1	2	1	2	1	2	3	4		
10:20	15404	712.5	561637	30.8	30.5	29.4	29.5	29.2	30.6	29.4	29.7	29.9	29.7	29.7	1.1
13:00	14706	678.8	561817	67.6	67.5	30.9	32.5	54.9	49.7	30.9	31.2	31.4	31.1	31.2	36.4

13:30	14660	676.4	561683	77	75.9	31.9	42	65.6	67.8	31.9	32.3	32.2	31.8	32.1	44.9
14:00	14726	679.2	561562	79.7	77.8	33.1	43.4	67.8	69.9	33.1	33.6	33.3	32.8	33.3	46.4
14:30	14705	677.9	561552	81.6	79.5	34.4	44.4	69.7	71.2	34.4	35	34.7	34.1	34.7	46.9
15:00	14703	678.1	561747	82.6	81	35.2	45.2	70.2	72.2	35.2	35.9	35.7	35	35.6	47.0
15:30	14589	673.4	561442	83.3	82	36.1	45.9	71.4	72.9	36.1	36.8	36.7	35.9	36.5	46.8
16:00	14617	673.2	561524	84	82.9	36.7	46.5	72	73.9	36.7	37.4	37.3	36.5	37.1	46.9
16:30	14632	674.4	561568	85	83.8	37.3	47	72.8	74.5	37.3	38	38	37.1	37.8	47.2
17:00	14602	672.9	561737	85.5	84.3	37.6	47.5	73.6	75	37.6	38.3	38.5	37.5	38.1	47.4
17:30	14399	664.4	561812	85.9	84.9	38	47.9	74.3	75.6	38	38.7	38.9	37.8	38.5	47.4
18:00	14379	663.3	561853	86.2	85.1	38.3	48.2	74.2	75.7	38.3	39	39.2	38.1	38.8	47.4
18:30	14357	662.2	561650	86.3	85.1	38.5	48	74.1	75.7	38.5	39.2	39.6	38.4	39.1	47.2
19:00	14326	661	561896	86.6	85.1	38.5	48	74.4	75.4	38.5	39.3	39.8	38.5	39.2	47.4
19:30	14279	659.6	561839	86.5	85.2	38.5	48.1	74.4	75.7	38.5	39.2	40	38.5	39.2	47.3
20:00	14368	662.4	561847	86.5	85.1	38.3	47.8	74.1	75.5	38.3	39	40.1	38.4	39.1	47.4
20:30	14330	661	561872	86.5	85.3	38.1	47.6	73.9	75.3	38.1	38.8	40	38.3	39.0	47.5
20:50	14331	662	561940	86.5	85.3	37.9	47.6	74.2	75.3	37.9	38.7	40	38.2	38.9	47.6
20:50	13286	613.5	504020	86.5	85.3	37.9	47.6	74.2	75.3	37.9	38.7	40	38.2	38.9	47.6
20:55	13287	613.6	504178	86.5	85.3	37.9	47.4	74.2	75.2	37.9	38.7	40	38.2	38.9	47.6

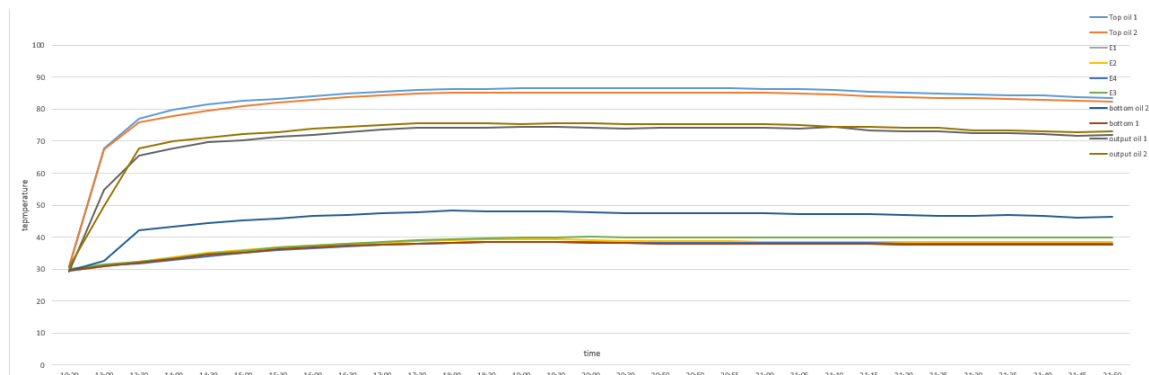
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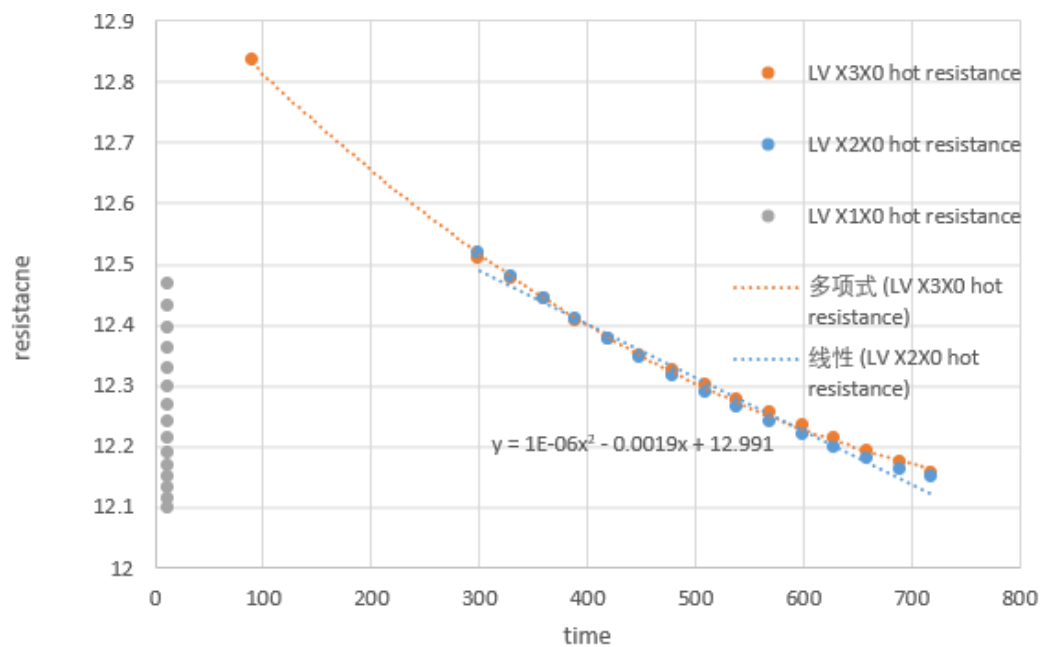
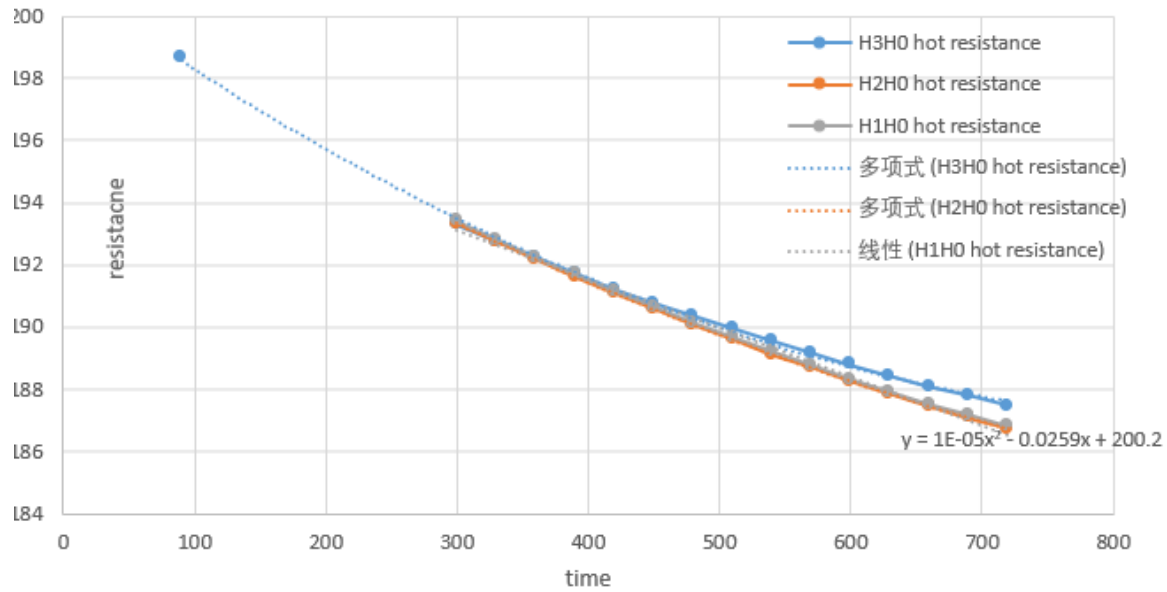
21:00	13305	613.7	504336	86.4	85.2	37.9	47.4	74.2	75.3	37.9	38.6	40	38.1	38.8	47.6
21:05	13298	613.3	503705	86.2	85	37.9	47.3	73.8	75	37.9	38.6	40	38.1	38.8	47.4
21:10	13303	613.4	503863	86	84.6	37.8	47.3	74.5	74.5	37.8	38.6	40	38.1	38.8	47.2
21:15	13297	613.7	504336	85.6	84.1	37.8	47.1	73.4	74.5	37.8	38.6	40	38.1	38.8	46.8
21:20	13296	613.4	500523	85.2	83.8	37.7	46.8	73.1	74.2	37.7	38.5	40	38	38.7	46.5
21:25	13298	613.6	499933	84.9	83.6	37.7	46.6	73.1	74.1	37.7	38.5	40	38	38.7	46.2
21:30	13300	613.5	499332	84.7	83.4	37.7	46.7	72.6	73.5	37.7	38.5	40	38	38.7	46.0
21:35	13286	613.4	499324	84.4	83.2	37.7	46.8	72.5	73.5	37.7	38.5	40	37.9	38.7	45.7
21:40	13291	613.6	499208	84.2	82.9	37.6	46.7	72.1	73.1	37.6	38.4	39.9	37.9	38.6	45.6
21:45	13280	613.2	499885	83.9	82.7	37.6	46.1	71.8	72.9	37.6	38.4	39.9	37.9	38.6	45.3
21:50	13276	613.5	498803	83.6	82.5	37.5	46.4	71.9	73.1	37.5	38.4	39.9	37.8	38.6	45.0

cold temperature		37	°C	Copper	234.5	Copper	234.5	hot temperature		38	°C
Time	H1H0	Time	H2H0	Time	H3H0	Time	X1X0	Time	X2X0	Time	X3X0
Rise	96.1	K	53.9	K	55.4	K	59	K	58.7	K	58.2
cold	145.8	mΩ	145.7	mΩ	145	mΩ	10.56	mΩ	10.61	mΩ	10.67
0	197.93	mΩ	175.14	mΩ	175.14	mΩ	12.892	mΩ	12.942	mΩ	12.995
30	/	30	/	30	/	30	/	30	/	30	/
60	/	60	/	60	/	60	/	60	/	60	/
90	198.68	100		90	173.07	60		90		90	12.835
120		120		120		120		120			
150		150		150		150		150			
180		180	195.87	180		180		180	12.688		
210		210		210		210		210			
240		240		240		240	12.55	240		240	
270		270		270		270		270			

Date

300	191.83	300	193.41	300	169.2 9	300	12.462	300	12.518	300	12.462
330	191.24	330	192.83	330	168.9 2	330	12.433	330	12.48	330	12.433
360	190.67	360	192.26	360	168.4 8	360	12.392	360	12.443	360	12.392
390	190.12	390	191.7	390	168.1 7	390	12.365	390	12.409	390	12.365
420	189.59	420	191.17	420	167.8 2	420	12.34	420	12.376	420	12.34
450	189.07	450	190.65	450	167.4 7	450	12.315	450	12.345	450	12.315
480	188.57	480	190.16	480	167.1 5	480	12.29	480	12.316	480	12.29
510	188.09	510	189.68	510	166.8 5	510	12.268	510	12.289	510	12.268
540	187.63	540	189.21	540	166.5 3	540	12.245	540	12.263	540	12.245
570	187.19	570	188.77	570	166.2 5	570	12.224	570	12.24	570	12.224
600	186.76	600	188.34	600	165.9 7	600	12.203	600	12.218	600	12.203





110MVA Tap 17B								
Test group and conditions								
cooling type	ONAF			Hv position		17B		
Quantity of Radiators	Group			Lv position		/		
Quantity of Fans	2		Unit	Running Radiators		/		
Method	Short circuit			Running Fans		/		
Rated Frequency	60		Hz	Actual Frequency		50		Hz
test position	Hv	103500	V	test position	Hv	613.6	A	
Rated Voltage	Lv	34500	V	Rated Current	Lv	1840.9	A	
	No load	48426	W	Actual Current	596.1		A	
	load	488293	W	Actual Load loss	492434		W	
	Total	536719	w	Actual Total loss	536162		W	
	Applied loss							
test date	start	23:00	2024/7/29		End	6:00	2024/7/30	
Top oil Temperature rise(total loss)			Corrected	47.2	℃	tested	47.1	℃
Top oil Temperature (total loss)							85.1	℃
Top oil Temperature rise (shut down)			Corrected	43.8	℃	tested	44.1	℃
Top oil Temperature (shut down)							82.1	℃
diff liquid Temperature rise(total loss)			Corrected	13.5	℃	tested	13.5	℃
diff oil Temperature rise (shut down)			Corrected	14.5	℃	tested	14.6	℃
average oil Temperature Rise(total loss)			Corrected	33.7	℃	tested	33.7	℃
average oil Temperature (total loss) 01 TL						tested	72.4	℃
average oil Temperature rise(shut down)01			Corrected	29.2	℃	tested	29.5	℃
average oil Temperature (shut down)01						tested	67.5	℃
position			HV		LV		cold ambinet	
conduct(θk)			Copper	234.5	Copper	234.5	37	℃
Δ01			33.7	℃	hot ambinet		38	℃
Cold resistance			H1H0		H2H0		H3H0	
			0.1655	(Ω)	0.165	(Ω)	0.1642	(Ω)
			X1X0		X2X0		X3X0	
			0.01056	(Ω)	0.01061	(Ω)	0.01067	(Ω)
Hot resistance			H1H0		H2H0		H3H0	
			0.17523	(Ω)	0.17523	(Ω)	0.17514	(Ω)
			X1X0		X2X0		X3X0	
			0.012837	(Ω)	0.012905	(Ω)	0.012971	(Ω)
average Winding Temperature θw			H1H0		H2H0		H3H0	
			53.0	℃	53.8	℃	55.1	℃
			X1X0		X2X0		X3X0	
			95.5	℃	95.7	℃	95.5	℃
Winding Temperature rise (AVG) Δθw			H1H0		H2H0		H3H0	
			19.2	℃	20.0	℃	21.3	℃
			X1X0		X2X0		X3X0	
			61.7	℃	61.9	℃	61.7	℃
Corrected Winding Temperature rise(AVG)			H1H0		H2H0		H3H0	
			19.0	℃	19.9	℃	21.1	℃

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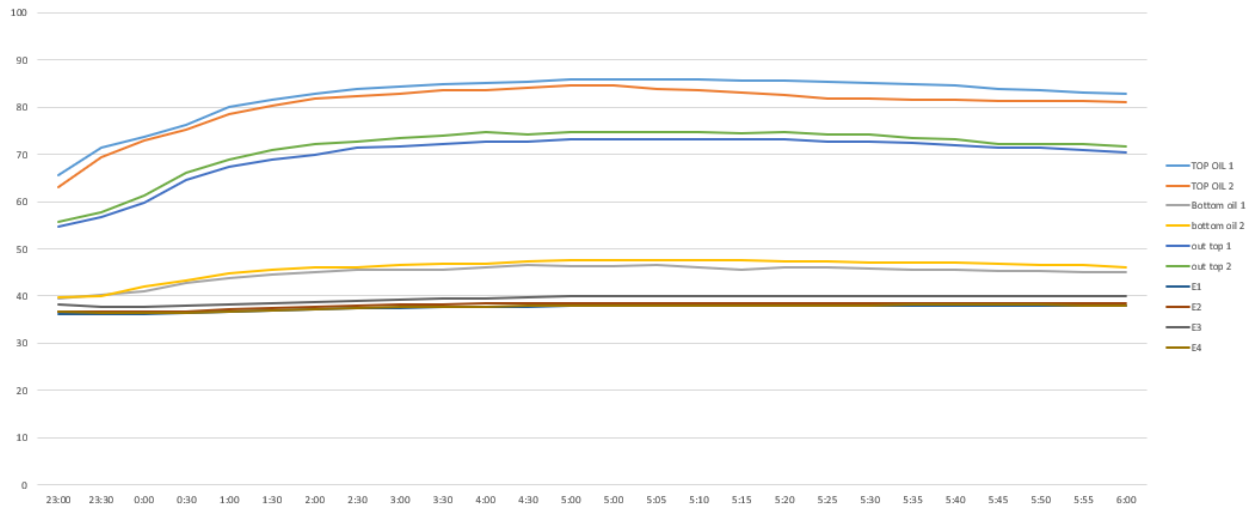
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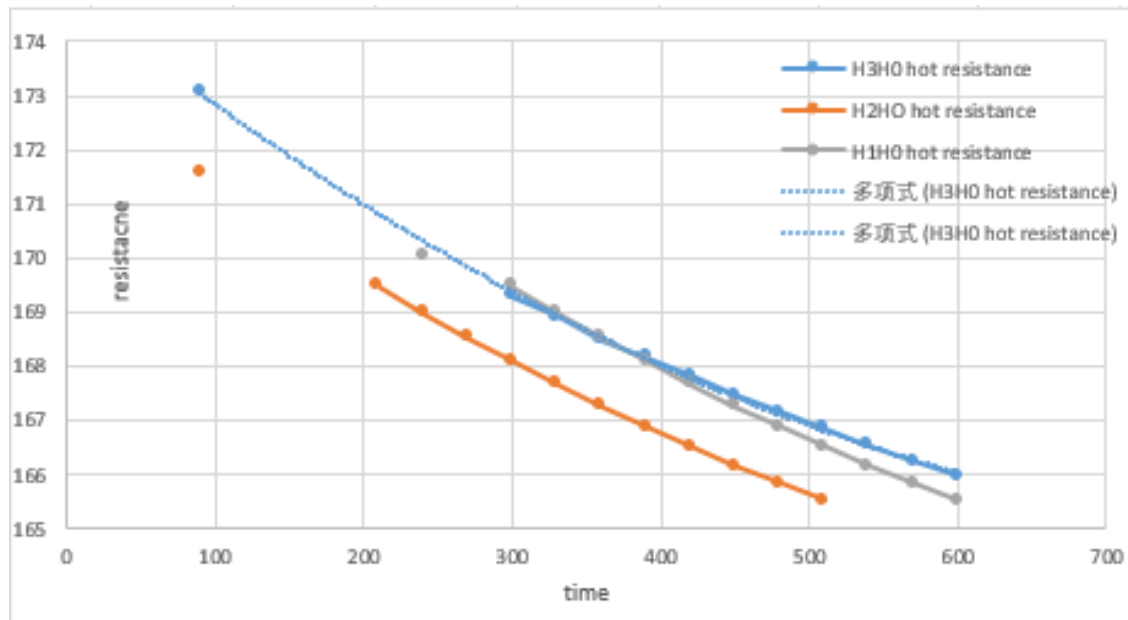
	X1X0		X2X0		X3X0	
	61.3	°C	61.5	°C	61.3	°C
	H1H0		H2H0		H3H0	
	18.0	°C	18.9	°C	20.2	°C
Winding Temperature rise (TOP)	X1X0		X2X0		X3X0	
	60.6	°C	60.8	°C	60.6	°C
Corrected Winding Temperature rise (TOP)	H1H0		H2H0		H3H0	
	17.9	°C	18.8	°C	20.0	°C
	X1X0		X2X0		X3X0	
	60.2	°C	60.4	°C	60.2	°C

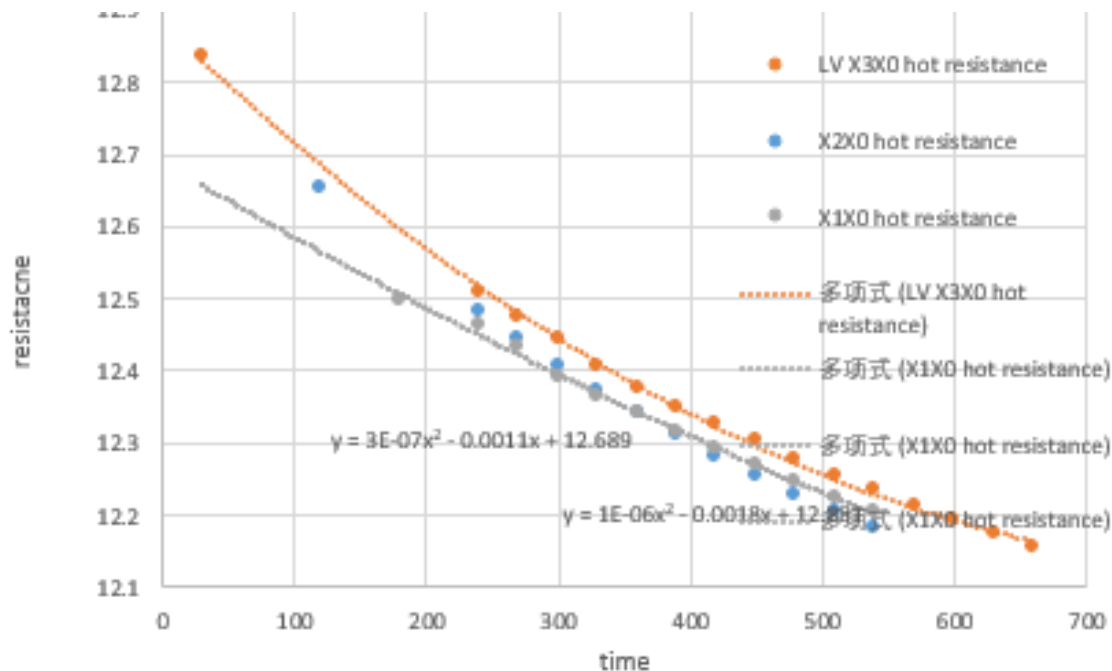
Time	Voltage (V)	Current (A)	Total loss (W)	Temperature on top of oil-T1		Bottom oil		output oil		Environment temperature (°C)				AVERAGE (°C)	Top oil rise (K)
				1	2	HV	LV	HV	LV	1	2	3	4		
23:00	15572	589	535836	75.5	73.1	39.6	39.7	58.8	59.6	36.3	36.8	36.1	36.8	37.1	38.4
23:30	16161	608	535951	71.4	69.4	40.3	39.9	56.7	57.9	36.3	36.7	37.8	36.5	36.9	34.5
0:00	16736	628.8	536315	70.6	69.9	41.1	42	59.8	61.2	36.2	36.7	37.8	36.4	36.9	33.7
0:30	16689	626.7	536323	76.3	75.2	42.8	43.3	64.7	66	36.5	36.8	37.9	36.5	37.1	39.2
1:00	16677	625.3	535199	90	79.4	43.9	44.7	67.3	69	36.7	37.1	38.1	36.7	37.3	42.7
1:30	16650	625.5	536129	81.6	80.2	44.5	45.5	68.9	70.8	36.9	37.4	38.4	37	37.6	44.0
2:00	16637	624.7	536041	82.9	81.8	45.1	46	69.9	72.1	37.1	37.7	38.7	37.2	37.8	45.1
2:30	16615	624.5	535989	83.8	82.4	45.5	46.1	71.4	72.8	37.4	37.9	39	37.4	38.1	45.7
3:00	16622	624.2	536010	84.4	82.9	45.6	46.5	71.7	73.5	37.5	38.1	39.2	37.6	38.3	46.1
3:30	16613	623.7	536037	84.9	83.5	45.6	46.8	72.3	73.9	37.7	38.2	39.4	37.7	38.4	46.5
4:00	16601	623.7	536049	85.1	83.7	46	46.9	72.6	74.8	37.8	38.4	39.6	37.8	38.6	46.5
4:30	16605	623.3	536101	85.4	84.2	46.8	47.3	72.8	74.2	37.8	38.4	39.9	37.9	38.7	46.1
5:00	16593	622.6	536102	85.9	84.5	46.4	47.5	73.2	74.6	37.9	38.5	39.9	38	38.8	47.1
5:00	15590	596.4	491124	85.9	84.5	46.4	47.5	73.2	74.6	37.9	38.5	39.9	38	38.8	47.1
5:05	15590	596.4	491124	85.9	84.4	47.5	47.5	73.2	74.6	37.9	38.5	39.9	38	38.8	47.1
5:10	15874	596.1	499875	85.8	46.7	37.9	47.5	73.3	74.6	37.9	38.5	39.9	38	38.8	47.0
5:15	15883	596.4	491519	85.7	46.1	37.9	47.7	73.1	74.4	37.9	38.5	39.9	38	38.8	46.9
5:20	15870	596.3	492081	85.5	45.6	37.8	47.4	73.3	74.7	37.9	38.5	39.9	38	38.8	46.7
5:25	15966	596.3	492150	85.3	46	37.8	47.3	72.6	74.1	37.9	38.5	39.9	38	38.8	46.5
5:30	15967	596.2	492031	85.2	46	37.7	47.2	72.7	74.2	37.9	38.6	40	38	38.8	46.4
5:35	15770	596.2	492460	84.9	45.9	37.7	47.1	72.4	73.5	37.9	38.6	40	38.1	38.8	46.1
5:40	15872	596.3	492582	84.5	45.6	37.7	47	71.9	73.3	37.9	38.6	40	38.1	38.8	45.7
5:45	15877	596.2	492083	83.9	45.6	37.7	46.8	71.3	72.1	38	38.6	40	38.1	38.9	45.0
5:50	15969	596.1	491648	83.6	45.2	37.6	46.7	71.3	72.1	38	38.6	40	38.1	38.9	44.7
5:55	15966	596.3	492525	83.2	45.2	37.6	46.5	71	72.2	38	38.6	40	38	38.9	44.3
6:00	15874	596.1	492434	82.9	45.2	37.5	46.2	70.4	71.7	37.9	38.6	40	38	38.8	44.1



cold temperature		37	°C	Copper	234.5	Copper	234.5	hot temperature		38	°C
Time	H1H0	Time	H2H0	Time	H3H0	Time	X1X0	Time	X2X0	Time	X3X0
Rise	53.8	K	54	K	55.4	K	57.5	K	57.7	K	57.5
cold	145.8	mΩ	145.7	mΩ	145	mΩ	10.56	mΩ	10.61	mΩ	10.67
0	175.23	mΩ	175.23	mΩ	175.14	mΩ	12.837	mΩ	12.905	mΩ	12.971
30		30	/	30	/	30	/	30	/	30	/

60		60	/	60	/	60	/	60	/	60	/
90		100		90	173.07	60		90		90	12.835
120		120		120		120		120			
150		150		150		150		150			
180		180	171.56	180		180		180	12.651		
210		210		210		210		210			
240	170.04	240		240		240	12.495	240		240	
270		270		270		270		270			
300	169.47	300	169.47	300	169.29	300	12.462	300	12.481	300	12.462
330	168.99	330	168.99	330	168.92	330	12.433	330	12.443	330	12.433
360	168.53	360	168.53	360	168.48	360	12.392	360	12.407	360	12.392
390	168.09	390	168.09	390	168.17	390	12.365	390	12.372	390	12.365
420	167.67	420	167.67	420	167.82	420	12.34	420	12.339	420	12.34
450	167.27	450	167.27	450	167.47	450	12.315	450	12.309	450	12.315
480	166.88	480	166.88	480	167.15	480	12.29	480	12.279	480	12.29
510	166.51	510	166.51	510	166.85	510	12.268	510	12.252	510	12.268
540	166.16	540	166.16	540	166.53	540	12.245	540	12.227	540	12.245
570	165.83	570	165.83	570	166.25	570	12.224	570	12.203	570	12.224
600	165.51	600	165.51	600	165.97	600	12.203	600	12.181	600	12.203





Test group and conditions							
cooling type	ONAF			Hv position	33		
Quantity of Radiators	Group			Lv position	/		
Quantity of Fans	2	Unit		Running Radiators	/		
Method	Short circuit			Running Fans	/		
Rated Frequency	60	Hz		Actual Frequency	50	Hz	
test position	Hv	103500	V	test position	Hv	368.2	A
Rated Voltage	Lv	34500	V	Rated Current	Lv	1104.5	A
Applied loss	No load	48426	W	Actual Current	431.2		A
	load	188994	W	Actual Load loss	178570		W
	Total	237421	w	Actual Total loss	237007		W
test date	start	8:30	2024/7/30	End	18:00	2024/7/30	
Top oil Temperature rise(total loss)		Corrected	40.5	℃	tested	40.5	℃
Top oil Temperature (total loss)						77.6	℃
Top oil Temperature rise (shut down)		Corrected	40.6	℃	tested	38.8	℃
Top oil Temperature (shut down)						75.9	℃
diff liquid Temperature rise(total loss)		Corrected	8.3	℃	tested	8.3	℃
diff oil Temperature rise (shut down)		Corrected	8.3	℃	tested	8.0	℃
average oil Temperature Rise(total loss)		Corrected	32.2	℃	tested	32.2	℃
average oil Temperature (total loss) @1 TL					tested	69.3	℃
average oil Temperature rise(shut down)@1		Corrected	32.4	℃	tested	30.8	℃
average oil Temperature (shut down)@1					tested	67.9	℃

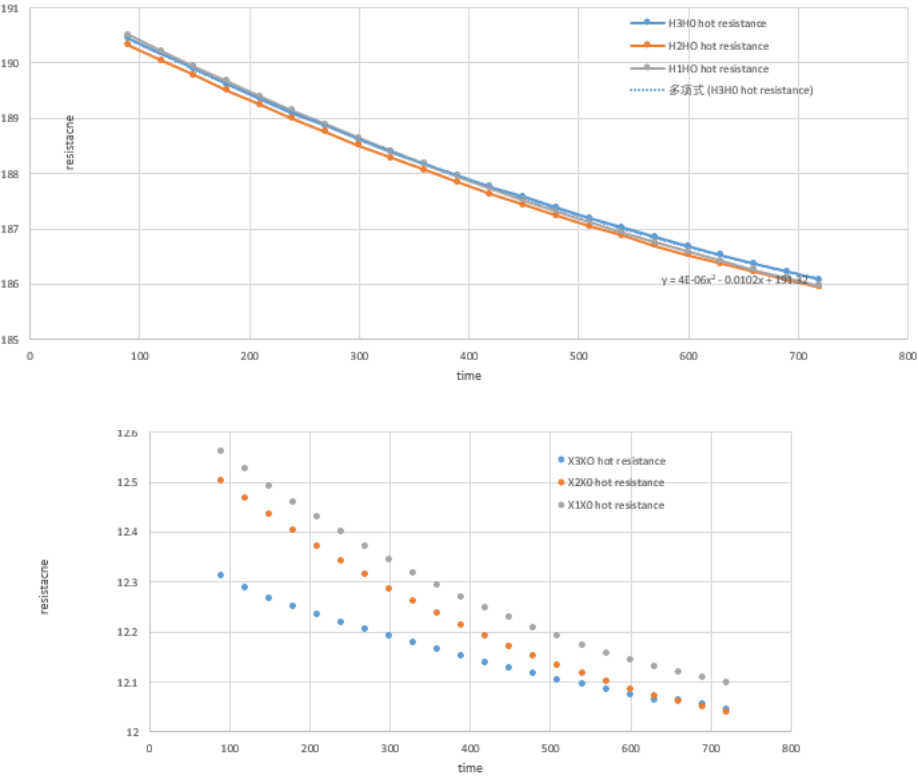
Date

position	HV		LV		cold ambinet	
conduct(θ_k)	Copper	234.5	Copper	234.5	37	°C
$\Delta\theta_l$	32.2	°C	hot ambinet		37.1	°C
Cold resistance	H1H0		H2H0		H3H0	
	0.1655	(Ω)	0.165	(Ω)	0.1642	(Ω)
	X1X0		X2X0		X3X0	
	0.01056	(Ω)	0.01061	(Ω)	0.01067	(Ω)
Hot resistance	H1H0		H2H0		H3H0	
	0.1908	(Ω)	0.19121	(Ω)	0.19135	(Ω)
	X1X0		X2X0		X3X0	
	0.012633	(Ω)	0.012613	(Ω)	0.012681	(Ω)
average Winding Temperature θ_w	H1H0		H2H0		H3H0	
	78.5	°C	80.1	°C	81.9	°C
	X1X0		X2X0		X3X0	
	90.3	°C	88.3	°C	88.2	°C
Winding Temperature rise (AVG) $\Delta\theta_w$	H1H0		H2H0		H3H0	
	42.8	°C	44.4	°C	46.2	°C
	X1X0		X2X0		X3X0	
	54.6	°C	52.5	°C	52.4	°C
Corrected Winding Temperature rise(AVG)	H1H0		H2H0		H3H0	
	44.8	°C	46.5	°C	48.3	°C
	X1X0		X2X0		X3X0	
	57.1	°C	55.0	°C	54.9	°C
Winding Temperature rise (TOP)	H1H0		H2H0		H3H0	
	43.1	°C	44.7	°C	46.5	°C
	X1X0		X2X0		X3X0	
	54.9	°C	52.8	°C	52.7	°C
Corrected Winding Temperature rise (TOP)	H1H0		H2H0		H3H0	
	45.1	°C	46.8	°C	48.6	°C
	X1X0		X2X0		X3X0	
	57.4	°C	55.3	°C	55.2	°C

Date

Time	Voltage (V)	Current (A)	Total loss (W)	Temperature on top of oil-T1		Bottom cooling means		Top cooling means		Environment temperature (°C)				AVERAGE (°C)	Top oil rise (K)
				1	2	1	2	1	2	1	2	3	4		
8:30	9424	441.3	237021	62.3	61.6	38.2	38.3	49.7	51.2	24.6	24.7	25.3	35	24.9	27.9
9:00	9286	430.8	236977	61.2	60	38.5	39	49.7	51.3	24.5	24.6	25.2	24.9	24.8	26.4
9:30	9286	435.6	237096	59.3	59.1	41.3	41	51.7	53.6	24.5	24.6	25	24.9	24.7	24.6
10:00	9392	435.4	237038	61.7	61.3	43.1	43	54.3	56.3	24.6	24.7	25.1	24.9	24.8	26.9
10:30	9436	437.4	237072	64.4	64.5	44.5	45.5	56.5	58.7	24.8	24.8	25.3	35	25.0	29.4
11:00	9407	435.6	2370432	66.6	66.9	45.7	46.6	58.5	60.9	25	25	25.2	25.2	25.1	31.5
11:30	9394	424.9	236925	68.7	69	46.4	47.6	60.2	62.4	25.1	25.1	25.7	25.4	25.3	33.4
12:00	9392	434.2	237013	70.1	70.3	47.7	48.4	61.5	63.7	25.4	25.3	25.9	25.6	25.5	34.6
12:30	9403	435.7	237029	71.6	71.6	49.7	49.4	62.6	64.9	25.5	25.5	26.2	25.9	25.8	35.8
13:00	9235	233.5	236979	72.7	72.7	49.5	50.3	63.6	65.6	25.9	25.7	26.4	26	26.0	36.7
13:30	9249	432.9	237050	73.7	73.7	49.1	51.6	64.5	66.7	26.1	25.9	26.7	26	26.2	37.5
14:00	9267	439.5	236932	74.7	74.7	49.8	52.3	65	68.3	26	26.2	26.8	26.2	26.2	38.4
14:30	9344	431	236965	75.3	75.3	49.1	52.6	65.5	68.5	26.2	26.4	26.9	26.4	26.5	38.8
15:00	9208	41.9	236956	75.7	75.7	49.4	52.5	65.9	69.1	26.3	26.6	26.9	26.5	26.6	39.1
15:30	9222	421.5	236927	76.2	76.3	49.7	52.1	66.2	69.6	26.5	26.7	27.1	26.7	26.8	39.5
16:00	9316	421.9	236965	76.6	76.6	50	52.3	67	69.9	26.7	26.8	27.2	26.9	26.9	39.7
16:30	93122	431.2	236955	77.2	76.9	51.9	52.1	67.5	70.3	26.8	26.9	27.3	26.9	27.0	40.2
17:00	9216	431.2	237007	77.5	77.3	52.5	52.3	67.8	70.4	26.9	27.1	27.4	27.1	27.1	40.4
17:00	7965	365.4	173295	77.6	77.3	52.5	52.5	67.9	70.4	26.5	27.2	27.4	27.1	27.1	40.5
17:05	7934	367.5	175285	77.4	77.4	52.7	51.6	67.5	70.3	26.5	27.1	27.4	27.1	27.1	40.3
17:10	7942	368	176533	77.2	77.5	52.7	51.7	67.9	70.3	26.5	27.1	27.4	27.1	27.1	40.1
17:15	7923	367.5	176626	77	77.6	52.8	51.8	68	69.9	26.8	27.1	27.4	27.1	27.1	39.5
17:20	7940	369	177045	76.9	77.5	52.9	52.3	67.7	70	26.8	27.1	27.4	27.1	27.1	39.7
17:25	7949	367.5	177450	76.8	77.4	52.3	52.1	67.7	69.7	26.8	27.1	27.4	27.1	27.1	39.4
17:30	7940	367.5	175931	76.6	77.2	52.5	51.9	67.3	69.6	26.8	27.1	27.5	27.1	27.1	39.5
17:35	7949	3677.9	178301	76.5	77	52.6	51.7	67.3	69.4	26.8	27.2	27.5	27.1	27.2	39.3
17:40	7944	369.2	178525	76.2	76.4	52.2	51.6	67.2	69	26.9	27.2	27.5	27.1	27.2	39.1
17:45	7940	369.1	177421	76.2	76.2	52.2	51.6	67.1	69.1	26.9	27.2	27.5	27.1	27.2	39.0
17:50	7934	369.1	177934	76.2	76	51.6	51.5	66.9	66.7	26.9	27.2	27.5	27.1	27.2	39.0
17:55	7934	368.2	177952	76.1	75.8	51.7	51.6	66.7	66.9	26.9	27.2	27.5	27.1	27.2	38.9
18:00	7950	369.2	178570	76	75.6	51.6	51.6	66.5	66.6	26.9	27.2	27.5	27.1	27.2	38.8

cold temperature		37	°C	Copper	234.5	Copper	234.5	hot temperature		37.1	°C
Time	H1H0	Time	H2H0	Time	H3H0	Time	X1X0	Time	X2X0	Time	X3X0
Rise	41.4	K	43	K	44.8	K	53.2	K	51.2	K	51.1
cold	165.5	mΩ	165	mΩ	164.2	mΩ	10.56	mΩ	10.61	mΩ	10.67
0	190.8	mΩ	191.21	mΩ	191.35	mΩ	12.633	mΩ	12.613	mΩ	12.681
30	/	30	/	30	/	30	/	30	/	30	/
60	/	60	/	60	/	60	/	60	/	60	/
90	190.5	100	190.32	90	190.44	90	12.56	90	12.502	90	12.311
120	190.21	120	190.04	120	190.17	120	12.525	120	12.467	120	12.287
150	189.93	150	189.77	150	189.89	150	12.492	150	12.434	150	12.267
180	189.66	180	189.5	180	189.61	180	12.46	180	12.402	180	12.25
210	189.39	210	189.24	210	189.35	210	12.429	210	12.371	210	12.235
240	189.13	240	188.99	240	189.09	240	12.399	240	12.341	240	12.219
270	188.88	270	188.75	270	188.85	270	12.371	270	12.313	270	12.204
300	188.64	300	188.51	300	188.62	300	12.344	300	12.286	300	12.19
330	188.4	330	188.28	330	188.39	330	12.318	330	12.26	330	12.177
360	188.17	360	188.06	360	188.17	360	12.294	360	12.236	360	12.164
390	187.95	390	187.84	390	187.96	390	12.27	390	12.212	390	12.151
420	187.73	420	187.63	420	187.76	420	12.248	420	12.19	420	12.139
450	187.52	450	187.43	450	187.57	450	12.228	450	12.17	450	12.127
480	187.32	480	187.24	480	187.38	480	12.208	480	12.15	480	12.116
510	187.13	510	187.05	510	187.19	510	12.19	510	12.132	510	12.104
540	186.94	540	186.87	540	187.02	540	12.173	540	12.115	540	12.094
570	186.76	570	186.7	570	186.85	570	12.157	570	12.099	570	12.083
600	186.59	600	186.53	600	186.68	600	12.143	600	12.085	600	12.073
630	186.42	630	186.37	630	186.53	630	12.13	630	12.072	630	12.064
660	186.26	660	186.22	660	186.37	660	12.118	660	12.06	660	12.064
690	186.11	690	186.08	690	186.22	690	12.107	690	12.049	690	12.054
720	185.97	720	185.94	720	186.08	720	12.098	720	12.04	720	12.045



Measuring instrument:			
temperature and humidity meter	J202404303427-0001		
Temperature Recorder	2JB24000718-0004		
Multi-channel power analyzer	24KA202406280013		
Current transformer	24KA202406280019	24KA202406280021	24KA202406280022
Voltage transformer	24KA202406280020	24KA202406280023	24KA202406280024
Stop Watch	2JB23008317-0012A		
DC resistance tester	24KA202406280012		

Test date:	2024/7/29		Test Sample	01	
Ambient Temp.:	35	℃	Relative Humidity:	63%	
Top Oil Temp. Indicator:	35	℃	Bottom Oil Temp. Indicator:	36	℃

Test date:	2024/7/30		Test Sample	01	
Ambient Temp.:	36	℃	Relative Humidity:	63%	
Top Oil Temp. Indicator:	36	℃	Bottom Oil Temp. Indicator:	36	℃

11.1 Lightning impulse**IEEE C57.12.90 Clause 10.3****IEEE C57.12.00 5.10.7.1, Table 5, Table 17****CSA C88 16.2 i), 16.3.4**

1.Lightning impulse tests, when required as a routine test or when otherwise specified, shall consist of and be applied in the following order: one reduced full wave, one full wave, two chopped waves, and two full waves. The time interval between applications of the last chopped wave and the following full wave should be minimized, without intentional delays, to avoid recovery of dielectric strength.

2.When front-of-wave tests are also specified, the front-of-wave impulses should be inserted within the impulse test sequence as follows: one reduced full wave, one full wave, two front-of-waves, two chopped waves, and two full waves.

3.The order of the chopped wave and front-of-wave tests is not mandatory. However, a reduced full wave shall be applied first, and a full wave shall be the last wave to be applied to the terminal under test. Other reduced front-of-waves, reduced chopped waves, or reduced full waves may be applied at any time during the intervening sequence.

4.For information on impulse testing techniques, interpretation of oscillograms, and failure detection criteria, see IEEE Std C57.98. See IEEE Std 4 for more information on high voltage testing techniques.

5.Impulse tests shall be made without excitation.

6.Full-wave test

The test wave rises to crest in 1.2 μ s and decays to half of crest value in 50 μ s from the virtual time zero. The crest value shall be in accordance with the assigned basic impulse insulation level (BIL), subject to a tolerance of $\pm 3\%$, and no flashover of the bushing or test gap shall occur. The tolerance on virtual front time should be $\pm 30\%$, and the tolerance on time to half of crest should be $\pm 20\%$. However, as a practical matter, once the manufacturer has proven that they have test equipment limitations, the following shall be considered:

a) The virtual front time shall not exceed 2.5 μ s except for windings of large impulse capacitance (low-voltage, high-kilovolt ampere and some high-voltage, high-kilovolt ampere windings). To demonstrate that the large capacitance of the winding causes the long front, the impulse generator series resistance may be reduced. The reduction should cause superimposed oscillations. Only the inherent generator and lead inductances should be in the circuit.

b) The impedance of some windings may be so low that the desired time to the 50% voltage point on the tail of the wave cannot be obtained with available test equipment. For such cases, shorter waves are considered acceptable provided that the optimum impulse generator connection is used (i.e., using parallel stages, if applicable, and the largest available capacitance). However, if by using the optimum impulse generator connection, the minimum tail time specified (40 μ s) cannot still be achieved, the following alternative methods shall be used in the following preferred order:

1) Use of resistor(s) on the nonimpulsed winding(s). The resistor value shall be the minimum necessary to achieve the required minimum tail time of 40 μ s and shall not exceed the values given in Table 3. If the transformer is to be connected to a cable

or to a gas-insulated switchgear or line (GIS or GIL), the resistor value shall not exceed 30 Ω for cable-connected terminals or 75 Ω for GIS- or GIL-connected terminals. 2) Use of resistor on the grounded terminal of the impulsed winding. Here also, the resistor value shall be the minimum necessary to achieve the required minimum tail time of 40 μ s and shall not exceed the values given in Table 3. If the transformer is foreseen to be connected to a cable or to a GIS or GIL, the resistor value shall not exceed 30 Ω for cable-connected terminals or 75 Ω for GIS- or GIL-connected terminals.

NOTE 1—The impulse voltage applied to the resistor should not exceed 80% of the rated lightning impulse level of the terminal(s) on which the resistor(s) is(are) connected, unless the manufacturer has consented. For delta-connected windings, the impulse voltage applied to the resistor, if used on nonimpulsed terminals of the delta, should not exceed 50% of the rated lightning impulse level of the terminal(s) on which the resistor(s) is(are) connected.

NOTE 2—IEEE Std C57.98 and IEC 60076-4 [B15] give background information regarding the effect of added resistors to the dielectric stresses applied to the transformer.

NOTE 3—In general, the voltage peak appearing across the resistor is considerably delayed compared to the instant of the voltage peak of the applied lightning impulse. Thus, the resulting difference between the applied impulse and the voltage across the resistor (e.g., voltage across the winding) is similar to the one that would appear across the winding if the resistor were not used and the terminal were directly grounded. If the resistor applied voltage peak coincides within 10 μ s of the voltage impulse peak, then the voltage drop across the winding is significantly reduced, and a special procedure should be agreed upon between the manufacturer and the purchaser.

NOTE 4—The use of the Glaninger circuit (see IEEE Std C57.98) is also an effective method to increase the tail time. If such circuit is used, care should be exercised on the overswing in the opposite polarity. Overswing in opposite polarity up to 75% is common.

NOTE 5—If the specific application (line, cable, GIS, or GIL) is not indicated at the time of order, the transformer is deemed to be overhead line-connected.

If the calculated tail time for a particular connection and transformer design is such that the minimum time to 50% (e.g., 40 μ s) cannot be achieved, the manufacturer shall notify the purchaser of this possibility. The manufacturer shall also state the strategy to be taken to obtain the best achievable wave shape. Notification should be given during the bidding stage for cases where the minimum tail time cannot be obtained for a particular transformer design and/or because of test laboratory limitations. In such cases, shorter wave shapes may be agreed upon between manufacturer and purchaser.

NOTE 6—The minimum impulse generator energy required to meet the minimum tail time (40 μ s) during an impulse test on a particular transformer design and connection can be estimated by

$$E_{\min.} = \frac{2\pi f (t_2)^2}{zU^2} \left(\frac{U_{BIL}}{\eta} \right)^2 VA \quad (26)$$

where

$E_{\min.}$ is the minimum energy required from the impulse generator (J)

f is the power frequency, 60 Hz

t_2 is the tail time (s) (t_2 equals 40 μ s)

z is the impedance (in per unit) seen from the impulsed terminal

U is the winding rated voltage (V , phase-to-phase)

U_{BIL} is the rated BIL of the tested winding (V)

η is the impulse generator efficiency (in per unit) ($\eta = 1.0$)

UL 1561-X VA is the three-phase power rating (VA) for which the impedance z is defined

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FORM REVISED: 2023-09-05

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Date

2. When front-of-wave tests are also specified, the front-of-wave impulses should be inserted within the impulse test sequence as follows: one reduced full wave, one full wave, two front-of-waves, two chopped waves, and two full waves.

NOTE 7—This equation has been derived from the equations given in IEC 60076-4 [B15]. More information about wave shape control can be found in IEEE Std C57.98 and IEC 60076-4. For single-phase transformers, the three-phase bank power rating and associated phase-to-phase voltage should be used in the above equation.

7. Reduced full-wave test

A reduced full wave is the same as a full wave, except that the crest value shall be between 50% and 70% of the full-wave value.

8. Chopped-wave test

A chopped wave is inherently a full lightning impulse wave, except that the crest value shall be at the required level and the voltage wave shall be chopped at or after the required time to flashover (time to chopping) but not later than 6 μ s after virtual origin. The virtual front time of the chopped wave may be different than the virtual front during a full-wave test because of the presence of the chopping gap. Nevertheless, the tolerance on the virtual front time for the chopped-wave test should remain as defined for full-wave test.

The gap or other equivalent chopping device shall be located as close as possible to the terminals of the transformer without disrupting its electrical field distribution. The distance between the chopping device and the test object shall not exceed a lead length greater than the total height of the transformer (tank + bushing). The impedance between the tested terminal and the grounded end of the chopping device shall be limited to that of the necessary leads. The voltage zero following the instant of chopping shall occur within 1 μ s. However, for some winding designs (particularly low-voltage windings of high stray capacitance and some layer windings), the circuit response after chopping may not be oscillatory; it may be overdamped. For such cases, the time interval to the first voltage zero after the instant of chopping may be significantly greater than 1 μ s.

Only for cases where the overswing to the opposite polarity is greater than 30% is it permissible to add a series-connected resistor in the chopping circuit to limit the amount of overswing. When a resistor is added in the chopping circuit, the resistor shall not decrease the overswing below 30% of the amplitude of the chopped wave.

The use of a resistor in the chopping circuit may increase the time interval to the first voltage zero after the instant of chopping. If the use of a resistor in the chopping circuit does conflict with the above requirement of the maximum time interval to the first voltage zero after the instant of chopping, the priority shall be given to the maximum limit of the time interval. For such cases, it may not be possible to reduce the overswing to the opposite polarity to 30%.

NOTE 1—This method will increase the likelihood that the steepness of the voltage collapse (dv/dt) is as high as possible.

NOTE 2—The use of a chopping gap made of sphere gap(s) (single or multiple sphere gaps) is the preferred chopping method since it usually gives faster voltage collapse. The use of a rod-rod chopping gap is also permissible since this method more

accurately replicates in-service flashover of an air insulator. Notably, the rod-rod gap requires a greater distance between its electrode for a given operating voltage than does a sphere gap. The extended arc length of the rodrod gap provides more natural circuit damping than the shorter arc length of a sphere gap.

NOTE 3—If the above prescribed maximum lead length to the chopping gap cannot be achieved because of the presence of transformer accessories such as coolers or any other transformer accessories, then the shortest possible lead length should be used during tests.

9. Wave polarity

For liquid-immersed transformers, the test waves are normally of negative polarity to reduce the risk of erratic external flashover in the test circuit.

10. Impulse oscillograms

All impulses applied to a transformer shall be recorded by an oscilloscope or by a suitable digital transient recorder, unless their crest voltage is less than 40% of the full-wave level. These oscillograms shall include voltage oscillograms for all impulses and ground-current oscillograms for all full-wave and reduced fullwave impulses. Sweep times should be on the order of 2 μ s to 5 μ s for front-of-wave tests, 5 μ s to 10 μ s for chopped-wave tests, 50 μ s to 100 μ s for full-wave tests, and 100 μ s to 600 μ s for ground-current measurements.

When reports require oscillograms, those of the first reduced full-wave voltage and current, the last two chopped waves, and the last full wave of voltage and current shall represent a record of the successful application of the impulse test to the transformer.

11. Connections for impulse tests of line terminals

In general, the tests shall be applied to each terminal, one at a time.

12. Terminals not being tested

Neutral terminals shall be solidly grounded. Line terminals, including those of autotransformers and regulating transformers, shall be either solidly grounded or grounded through a resistor with an ohmic value not in excess of the values given in Table 3.

Table 3—Grounding resistor values

Nominal system voltage (kV)	Resistance (Ω)
345 and below	450
500	350
765	300
NOTE—These values are representative of typical transmission-line surge impedances.	

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The two extreme
ion for testing
ree single-phase

transformers of a three-phase bank. When fewer than three single-phase transformers are tested (e.g., for a single-phase application or for a replacement unit or as a spare, etc.), the tap position for the first unit shall be set to achieve the minimum effective turns in the winding under test. In case of class II transformers, for additional single-phase units (greater than one or three units), the tap position

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shall be changed from one unit to the other, taking into account the requirements given above.

NOTE 1—As an example, if two single-phase units are to be tested, the first unit is to be tested with a tap position giving the minimum effective turns. The second unit may be tested either at the middle tap position or the maximum tap position.

NOTE 2—The above required tap connections may not necessarily produce the maximum stress conditions. If different tap positions are proposed to be used during test, it should be agreed upon between manufacturer and purchaser prior to design of the transformer.

As an example, a transformer having a load tap changer on the (primary) HV winding and a de-energized tap changer on the (secondary) LV winding, possible tap changer position combinations during an impulse test are as follows. Other combinations may be used.

- Impulse on HV winding phase A: HV: max. tap; LV min. tap (maximum ratio);
- Impulse on HV winding phase B: HV: nom. tap; LV nom. tap (nominal ratio);
- Impulse on HV winding phase C: HV: min. tap; LV max. tap (minimum ratio);
- Impulse on LV winding phase A: LV: max. tap; HV min. tap (minimum ratio);
- Impulse on LV winding phase B: LV: nom. tap; HV nom. tap (nominal ratio);
- Impulse on LV winding phase C: LV: min. tap; HV max. tap (maximum ratio).

14. Impulse tests on transformer neutrals

Impulse tests on the neutral of a transformer or a separate regulator connected in the neutral of a transformer require one reduced full wave and three full waves to be applied directly to the neutral or regulator winding with an amplitude equal to the insulation level of the neutral. The winding being tested shall be either on the minimum voltage connection or on the maximum voltage connection. A wave having a front of not more than 10 μ s and a tail of 50 μ s ($\pm 20\%$) to half-crest shall be used, except that, when the inductance of the winding is so low that the desired voltage magnitude and duration to the 50% point on the tail of the wave cannot be obtained, a shorter wave tail may be used, taking into account the considerations given in 10.3.1.1.

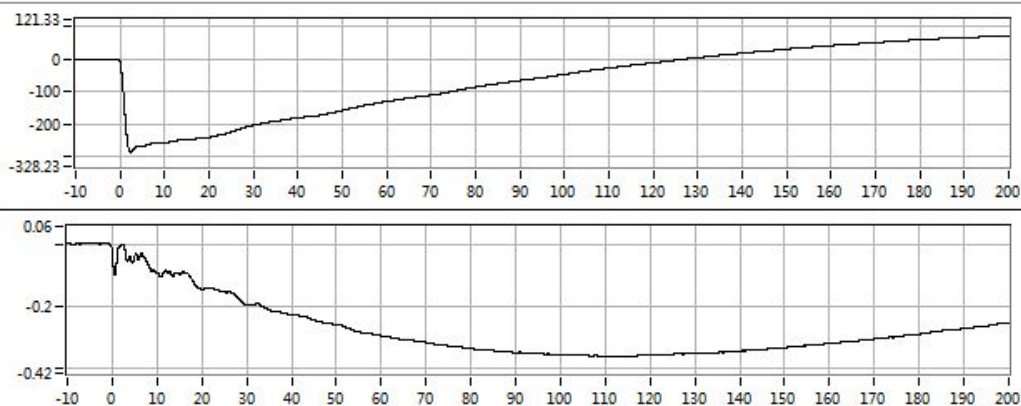
Date

HV	Wave No.	Wave Name	Test voltage (kV)		Wave time (μs)	Flashbver time (μs)
			Expected value	Actual	T1 × T2	
H1 Tap1	1	50%~70%FW	275~385	285.2	1.775/55.137	-
H1 Tap1	2	100% FW	550	544.96	1.796/55.499	-
H1 Tap1	3	100% CW	605	589.58	-	4.426
H1 Tap1	4	100% CW	605	590.59	-	4.439
H1 Tap1	5	100% FW	550	547.71	1.734/54.759	-
H1 Tap1	6	100% FW	550	546.34	1.771/55.55	-
H2 Tap17	1	50%~70%FW	275~385	285.48	1.812/56.245	-
H2 Tap17	2	100% FW	550		1.796/55.499	-
H2 Tap17	3	100% CW	605	590.54	-	4.376
H2 Tap17	4	100% CW	605	587.03	-	4.385
H2 Tap17	5	100% FW	550	545.97	1.790/55.465	-
H2 Tap17	6	100% FW	550	542.9	1.782/54.792	-
H3 Tap33	1	50%~70%FW	275~385	286.02	1.796/56.296	-
H3 Tap33	2	100% FW	550	548.27	1.791/54.986	-
H3 Tap33	3	100% CW	605	595.35	-	4.347
H3 Tap33	4	100% CW	605	598.23	-	4.361
H3 Tap33	5	100% FW	550	544.82	1.782/55.549	-
H3 Tap33	6	100% FW	550	546.19	1.798/55.502	-
HV (N)	Wave No.	Wave Name	Test voltage (kV)		Wave time (μs)	Flashbver time (μs)
			Expected value	Actual	T1 × T2	
H0	1	50%~70%FW	85~119	92.48	2.359/52.18	-
H0	2	100% FW	170	168.86	2.376/51.865	-
H0	3	100% FW	170	167.98	2.287/52.108	-
H0	4	100% FW	170	169.36	2.394/51.436	-
LV (N)	Wave No.	Wave Name	Test voltage (kV)		Wave time (μs)	Flashbver time (μs)
			Expected value	Actual	T1 × T2	
X0	1	50%~70%FW	100~140	104.76	2.095/44.556	-
X0	2	100% FW	200	194.43	2.107/46.142	-
X0	3	100% FW	200	194.56	2.119/46.041	-
X0	4	100% FW	200	196.14	2.112/46.103	-

Date

LV	Wave No.	Wave Name	Test voltage (kV)		Wave time (μs) T1 × T2	Flashbver time (μs)
			Expected value	Actual		
X1	1	50%~70%FW	100~140	116.37	1.664/45.317	-
X1	2	100% FW	200	196.79	1.659/45.393	-
X1	3	100% CW	220	218.01	-	4.616
X1	4	100% CW	220	214.71	-	4.632
X1	5	100% FW	200	196.82	1.643/45.776	-
X1	6	100% FW	200	195.13	1.647/45.86	-
X2	1	50%~70%FW	100~140	116.77	1.656/46.097	-
X2	2	100% FW	200	196.29	1.664/46.823	-
X2	3	100% CW	220	217.94	-	4.631
X2	4	100% CW	220	218	-	4.655
X2	5	100% FW	200	197.08	1.652/46.69	-
X2	6	100% FW	200	196.32	1.656/46.723	-
X3	1	50%~70%FW	100~140	113.08	1.648/47.738	-
X3	2	100% FW	200	195.22	1.652/48.428	-
X3	3	100% CW	220	216.89	-	4.679
X3	4	100% CW	220	218.04	-	4.677
X3	5	100% FW	200	196.18	1.634/48.327	-
X3	6	100% FW	200	196.41	1.647/48.379	-

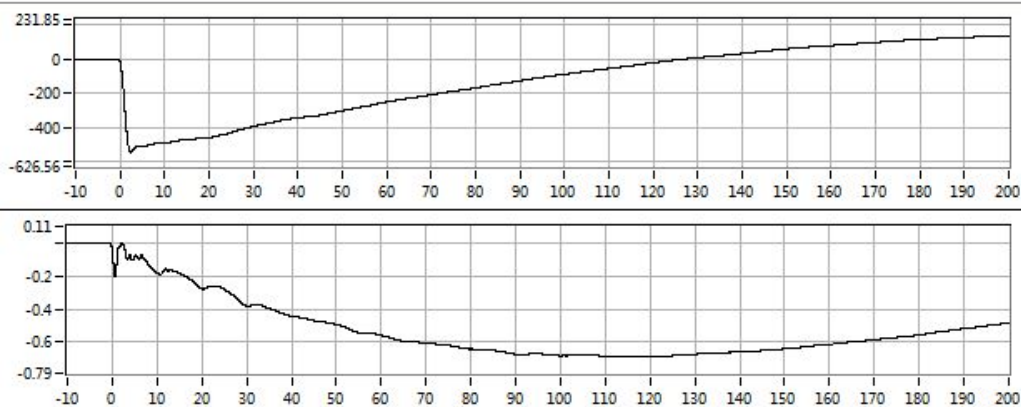
Test date:	2023/7/30		Test Sample	01	
Ambient Temp.:	30	℃	Relative Humidity:	55%	
Top Oil Temp. Indicator:	30.6	℃	Bottom Oil Temp. Indicator:	32.5	℃
Measuring instrument:					
temperature and humidity meter			J202404303427-0001		
Temperature Recorder			2JB24000718-0004		
Lightning impulse voltage generator			24KA202406280028		



高压A 50%雷电全波_1

Upk=-285.20 kV; T1=1.775 us; T2=55.137 us;

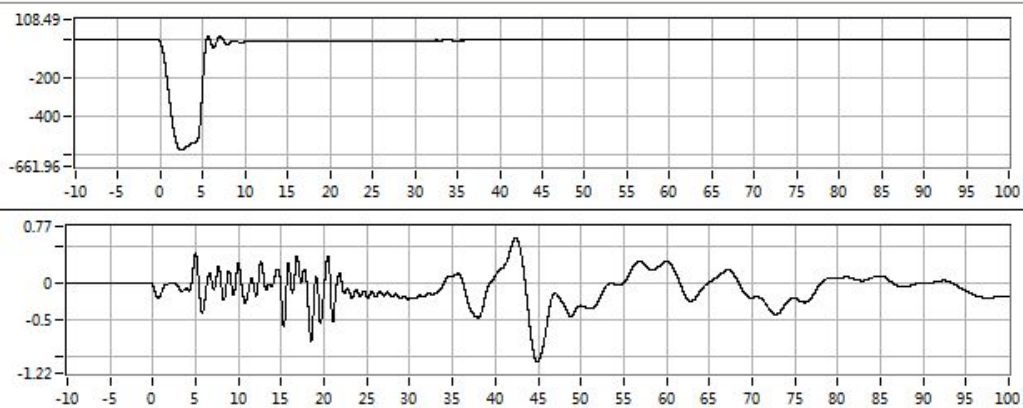
Max=0.001 kA; Min=-0.364 kA;



高压B 100%雷电全波_2

Upk=-544.96 kV; T1=1.796 us; T2=55.499 us;

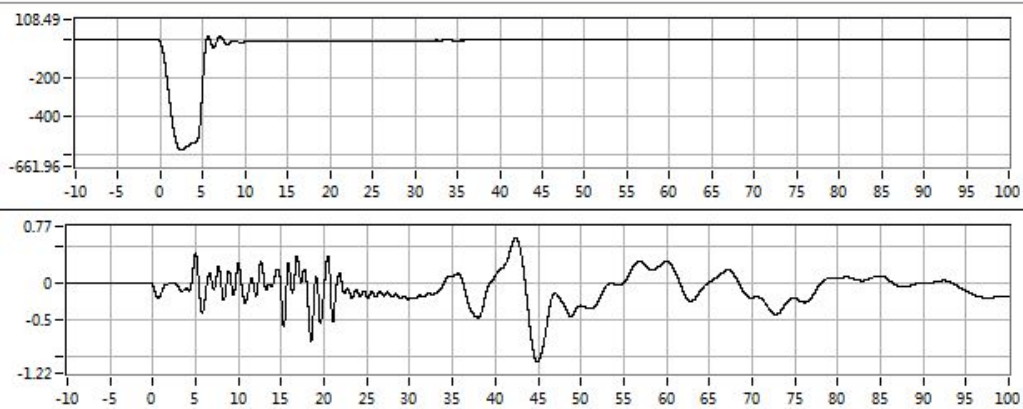
Max=0.002 kA; Min=-0.689 kA;



高压A 100%雷电截波_3

Upk=-589.58 kV; T1=1.800 us; Tc=4.426 us; Oz=3.85%;

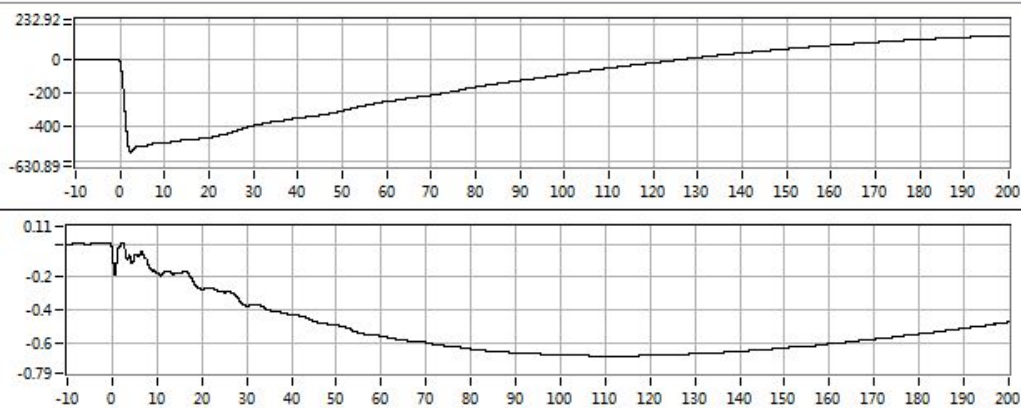
Max=0.565 kA; Min=-1.042 kA;



高压A 100%雷电截波_4

Upk=-590.59 kV; T1=1.813 us; Tc=4.439 us; Oz=3.91%;

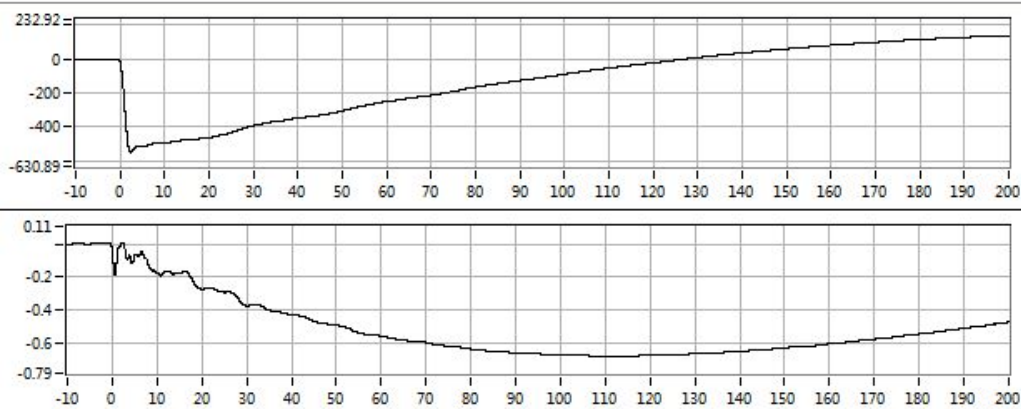
Max=0.569 kA; Min=-1.046 kA;



高压A 100%雷电全波_5

Upk=-547.71 kV; T1=1.734 us; T2=54.759 us;

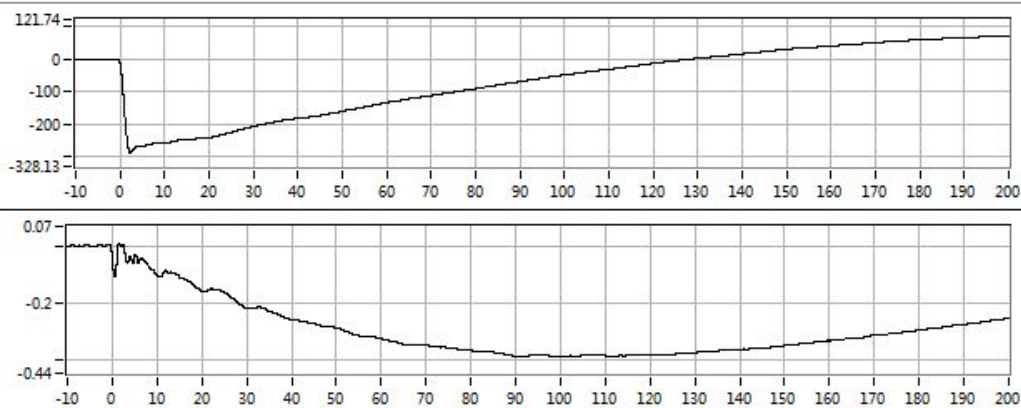
Max=0.003 kA; Min=-0.695 kA;



高压A 100%雷电全波_6

Upk=-546.34 kV; T1=1.771 us; T2=55.550 us;

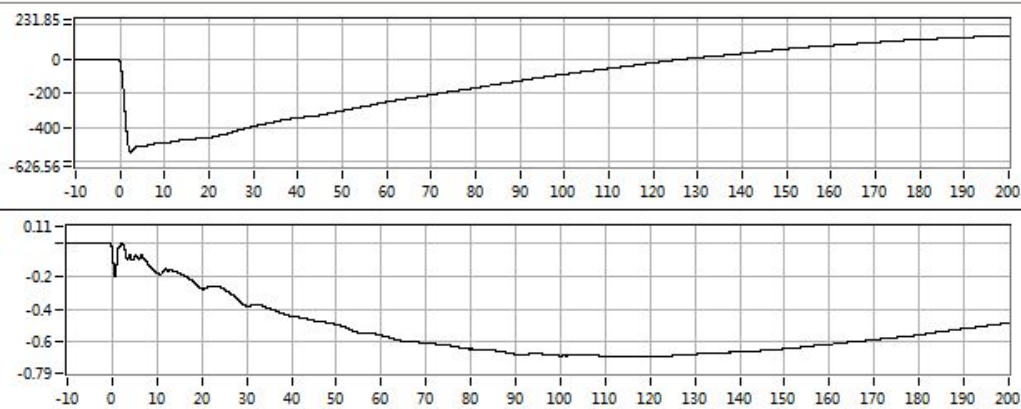
Max=0.002 kA; Min=-0.690 kA;



高压B 50%雷电全波_1

Upk=-285.48 kV; T1=1.812 us; T2=56.245 us;

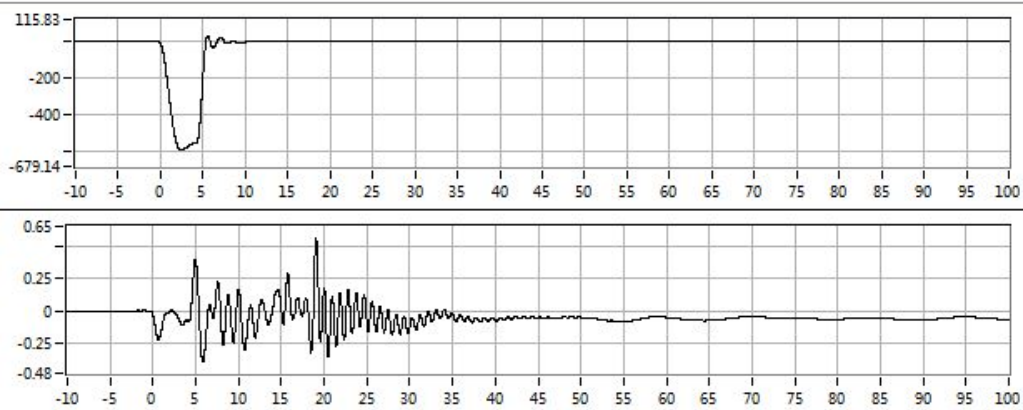
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高压B 100%雷电全波_2

Upk=-544.96 kV; T1=1.796 us; T2=55.499 us;

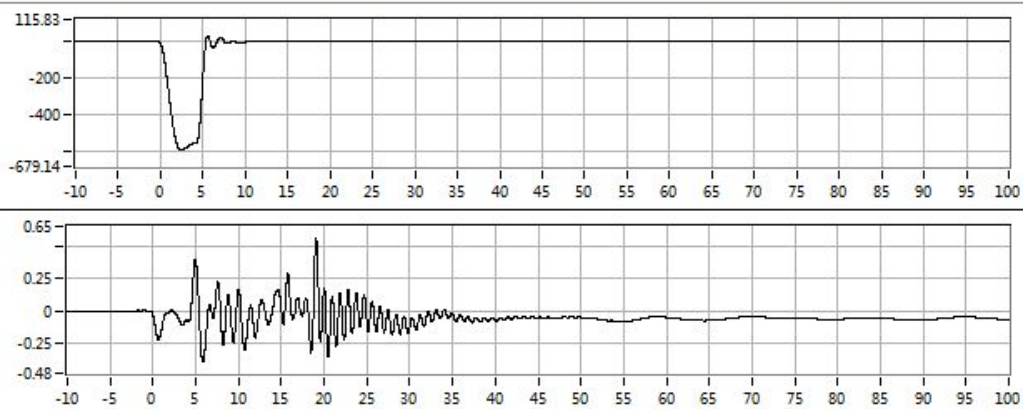
Max=0.002 kA; Min=-0.689 kA;



高压B 100%雷电截波_3

Upk=-590.54 kV; T1=1.802 us; Tc=4.376 us; Oz=4.62%;

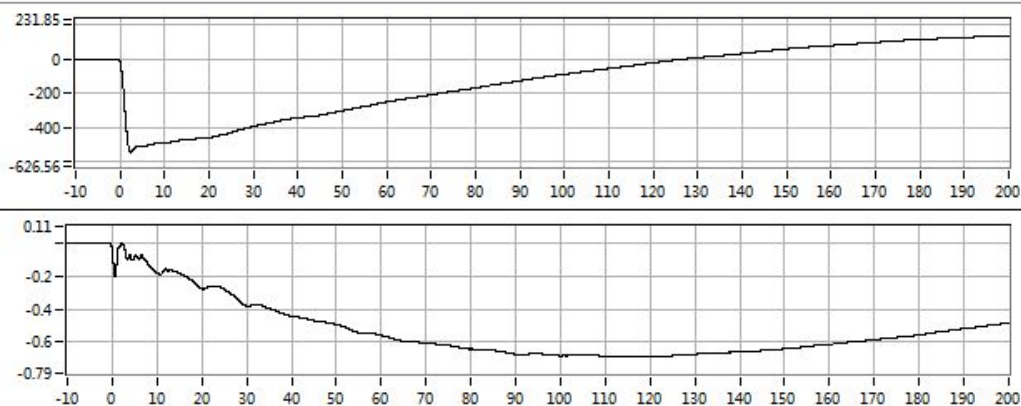
Max=0.214 kA; Min=-0.211 kA;



高压B 100%雷电截波_4

Upk=-587.03 kV; T1=1.796 us; Tc=4.385 us; Oz=4.59%;

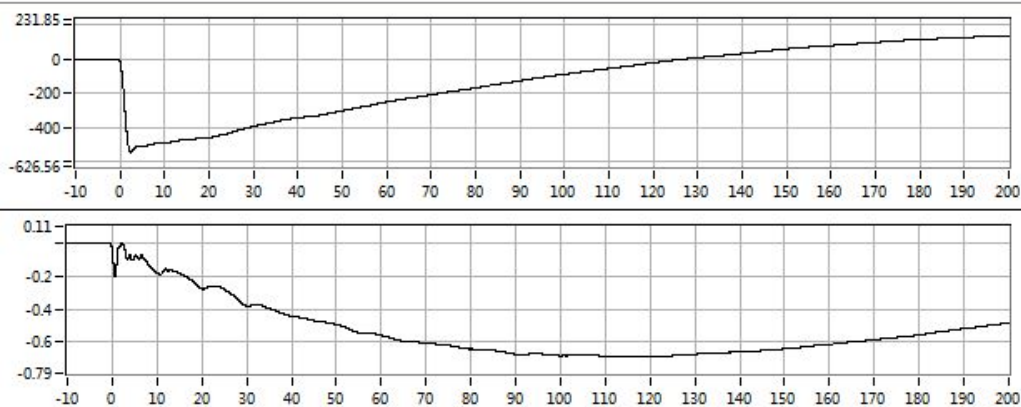
Max=0.217 kA; Min=-0.209 kA;



高压B 100%雷电全波_5

Upk=-545.97 kV; T1=1.790 us; T2=55.465 us;

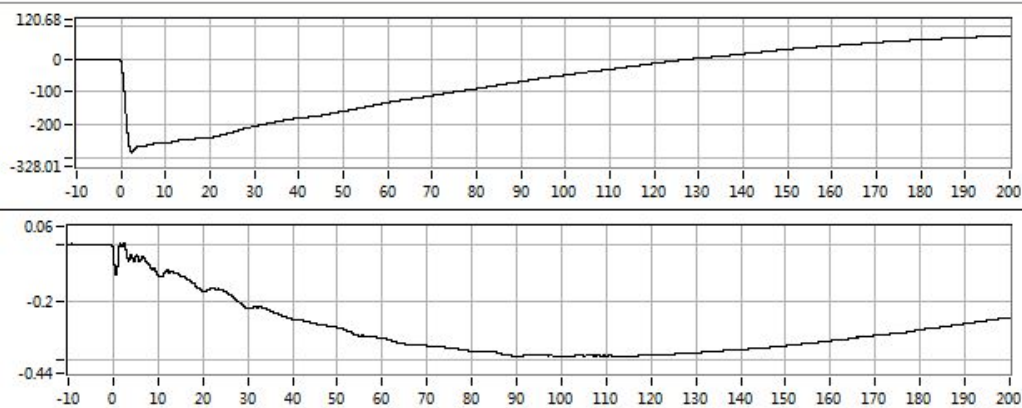
Max=0.003 kA; Min=-0.692 kA;



高压B 100%雷电全波_6

Upk=-542.90 kV; T1=1.782 us; T2=54.759 us;

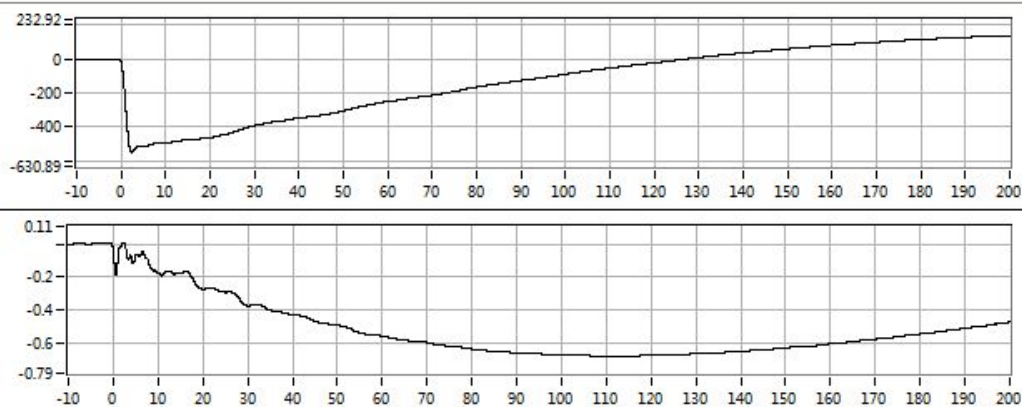
Max=0.003 kA; Min=-0.709 kA;



高压C 50%雷电全波_1

Upk=-286.02 kV; T1=1.796 us; T2=56.296 us;

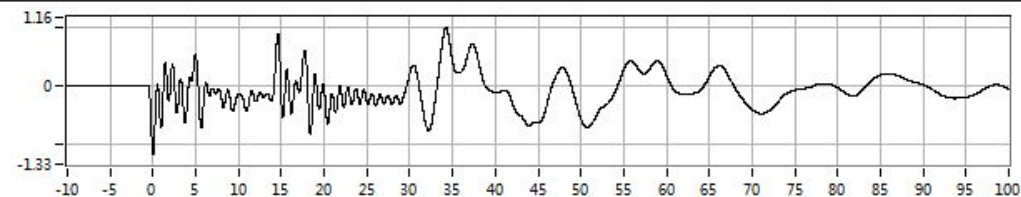
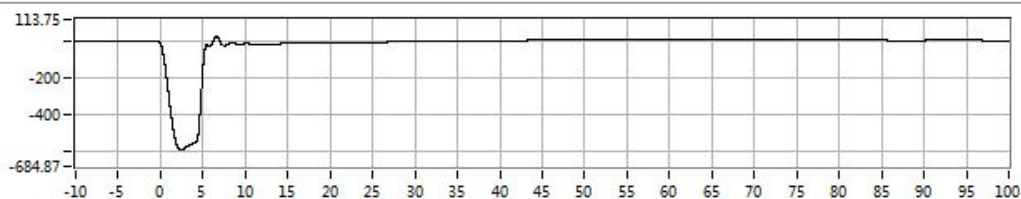
Max=0.001 kA; Min=-0.391 kA;



高压C 100%雷电全波_2

Upk=-548.27 kV; T1=1.791 us; T2=54.986 us;

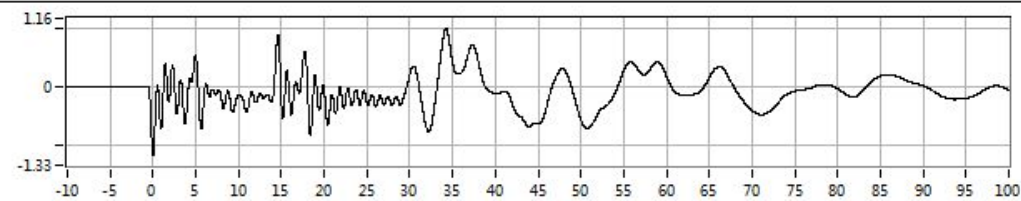
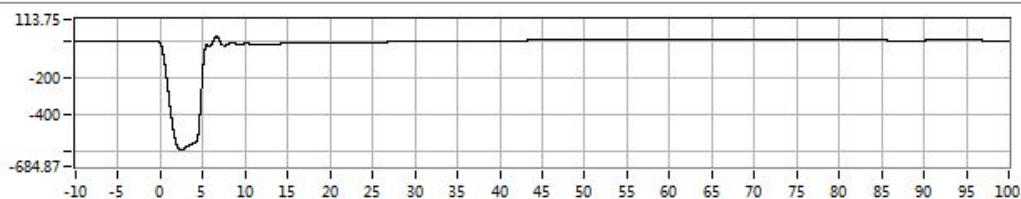
Max=0.002 kA; Min=-0.690 kA;



高压C 100%雷电截波_3

Upk=-595.38 kV; T1=1.785 us; Tc=4.347 us; Oz=4.13%;

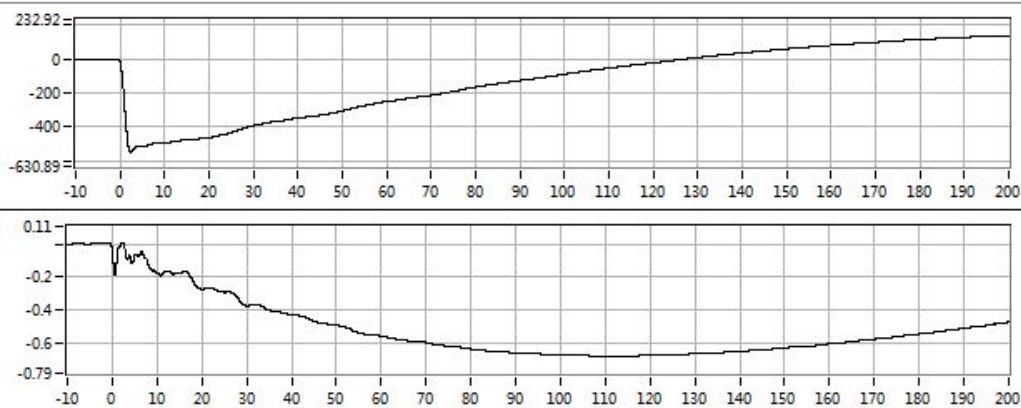
Max=0.878 kA; Min=-0.698 kA;



高压C 100%雷电截波_4

Upk=-598.23 kV; T1=1.766 us; Tc=4.361 us; Oz=4.27%;

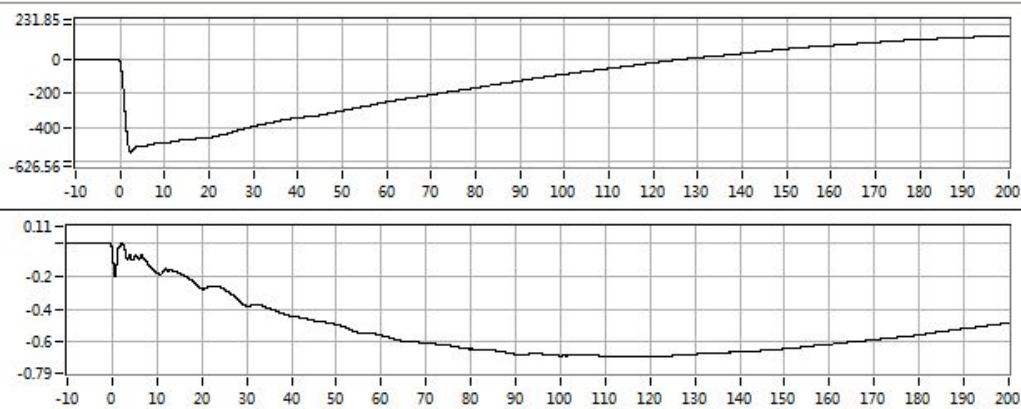
Max=0.893 kA; Min=-0.694 kA;



高压C 100%雷电全波_5

Upk=-544.82 kV; T1=1.782 us; T2=55.549 us;

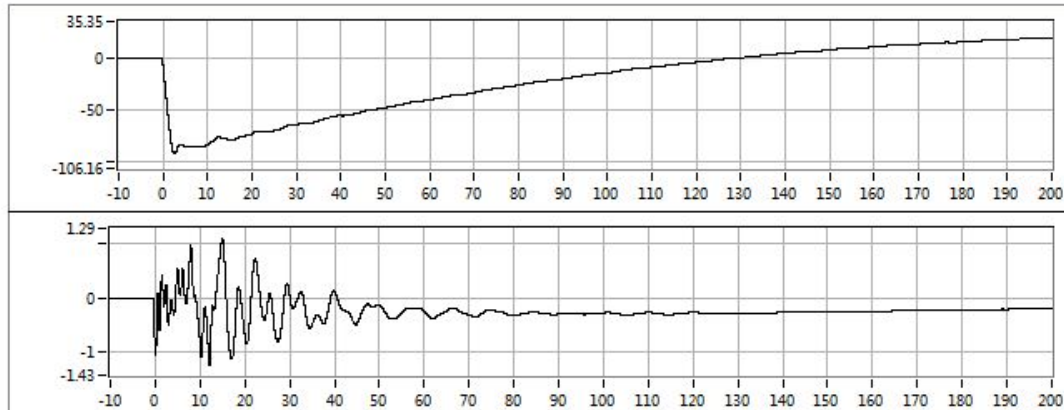
Max=0.003 kA; Min=-0.701 kA;



高压C 100%雷电全波_6

Upk=-546.19 kV; T1=1.798 us; T2=55.502 us;

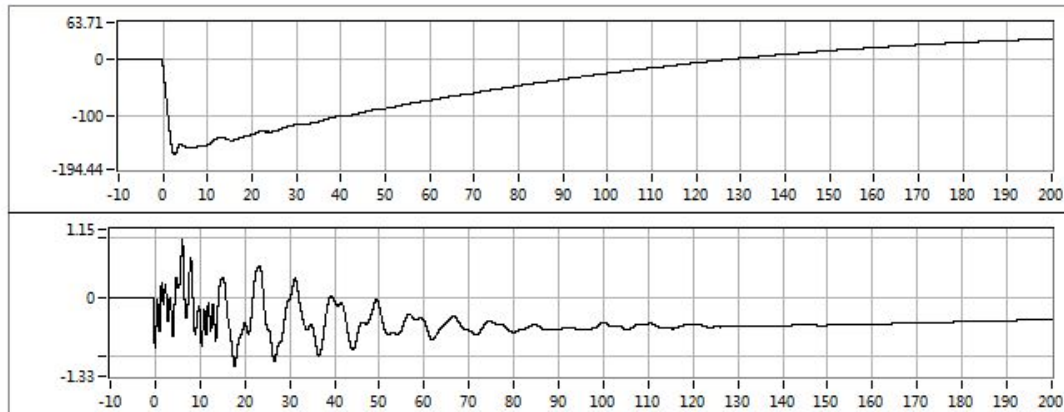
Max=0.003 kA; Min=-0.702 kA;



高压O 50%雷电全波_1

Upk=-92.48 kV; T1=2.359 us; T2=52.180 us;

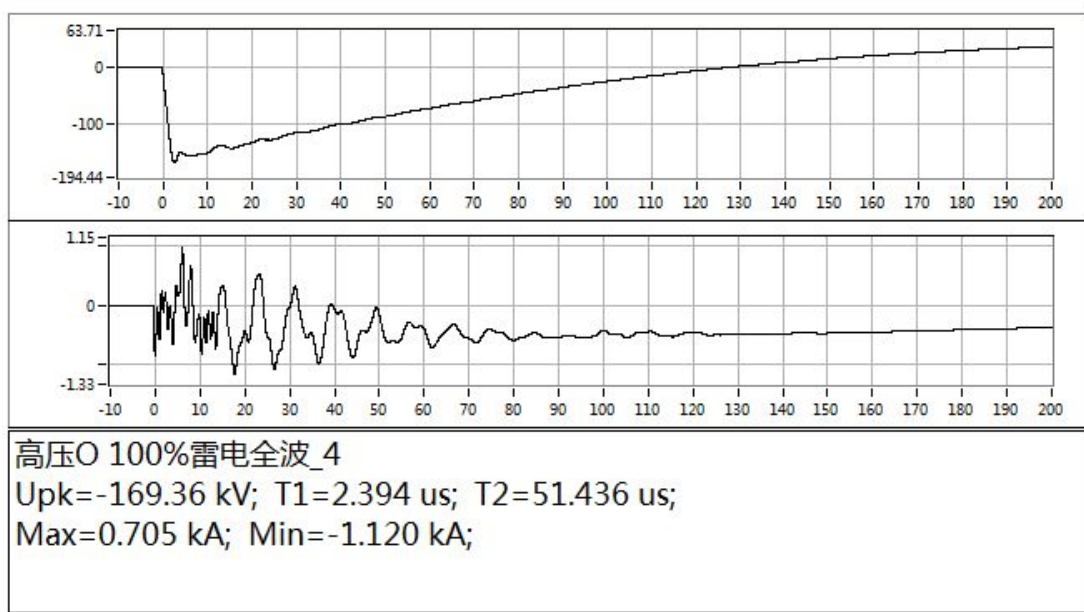
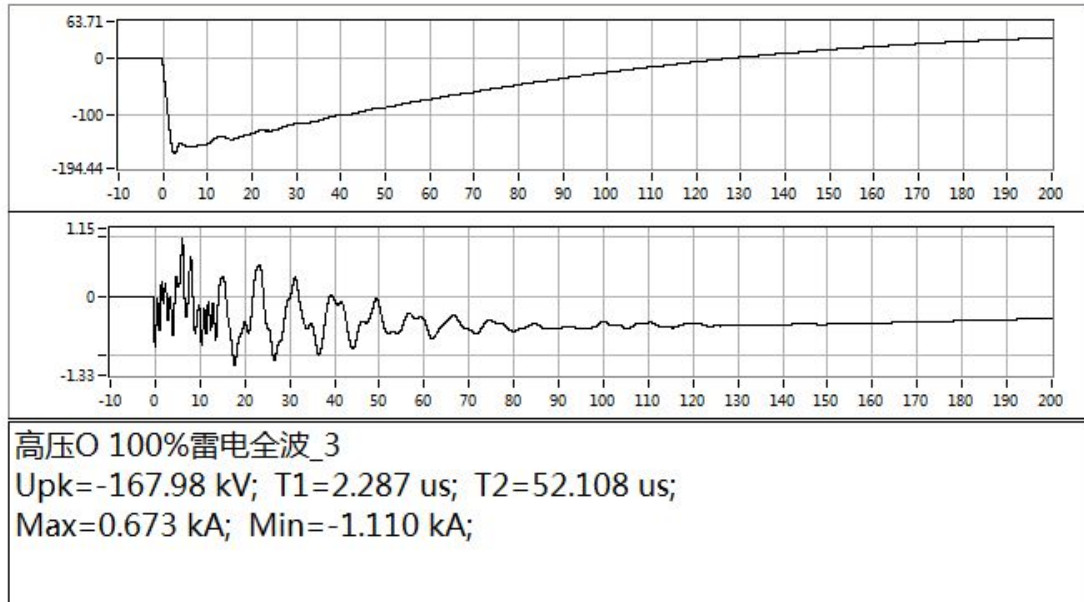
Max=0.998 kA; Min=-1.109 kA;

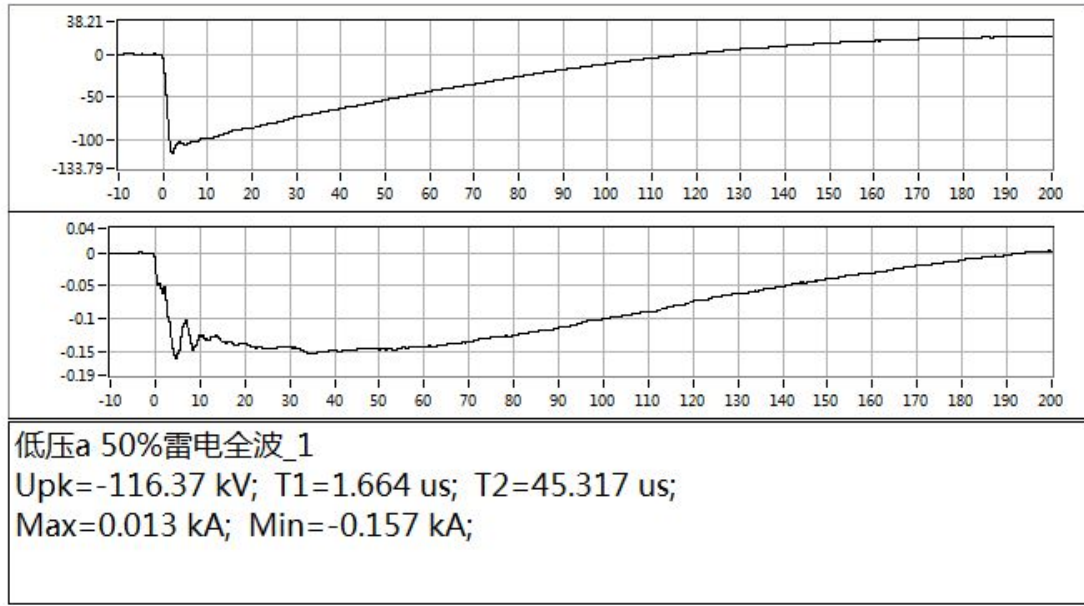


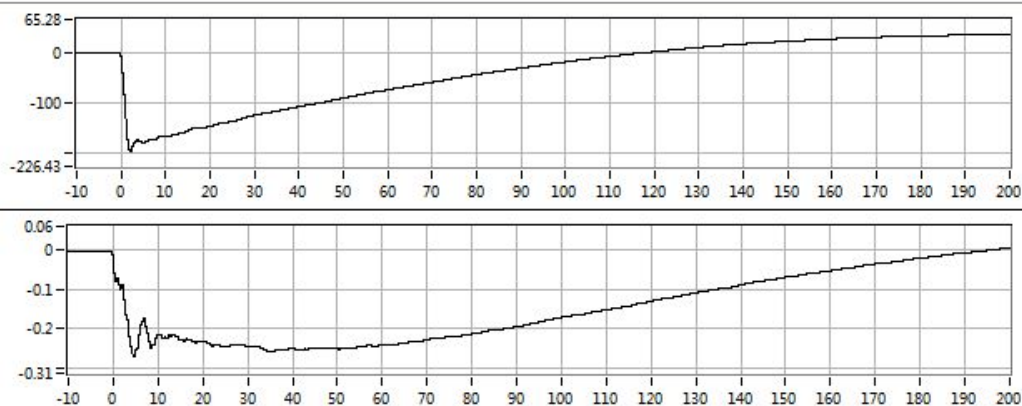
高压O 100%雷电全波_2

Upk=-168.86 kV; T1=2.376 us; T2=51.865 us;

Max=0.634 kA; Min=-1.079 kA;



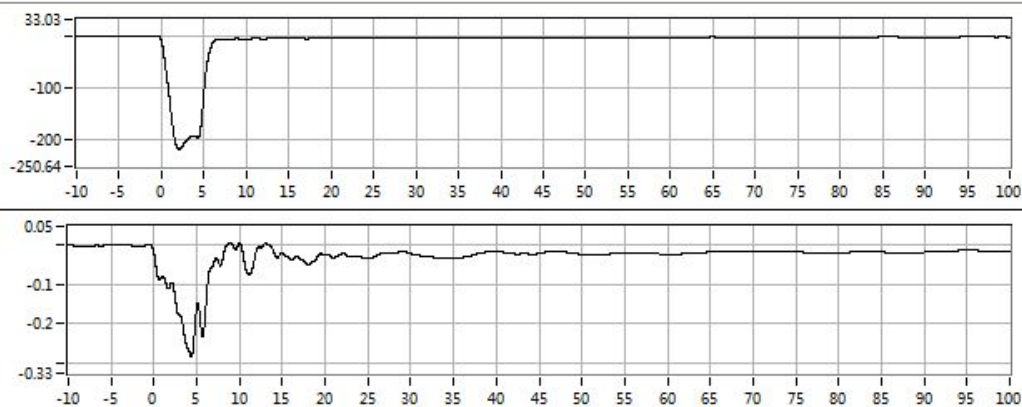




低压a 100%雷电全波_2

Upk=-196.79 kV; T1=1.659 us; T2=45.393 us;

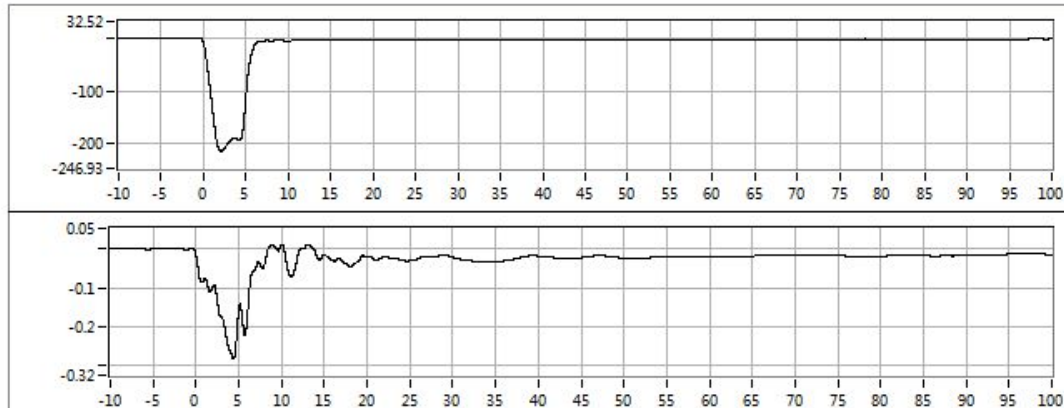
Max=0.021 kA; Min=-0.266 kA;



低压a 100%雷电截波_3

Upk=-218.01 kV; T1=1.660 us; Tc=4.616 us; Oz=0.09%;

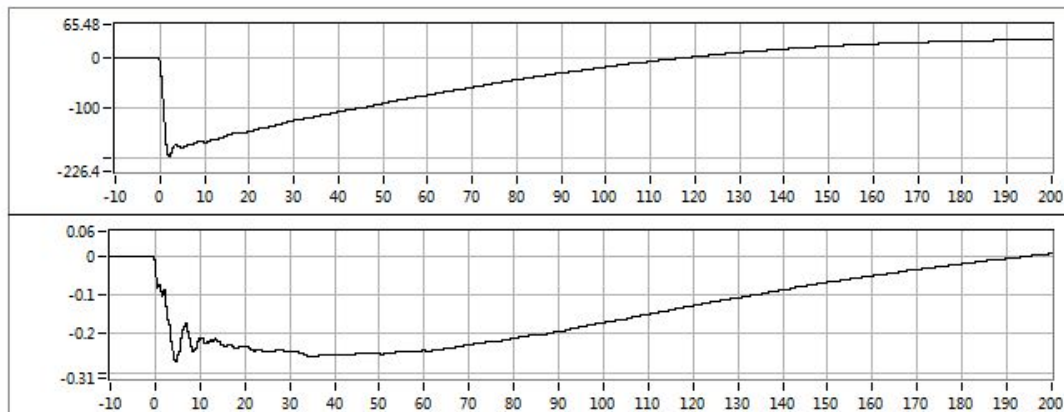
Max=0.004 kA; Min=-0.268 kA;



低压a 100%雷电截波_4

Upk=-214.71 kV; T1=1.656 us; Tc=4.632 us; Oz=0.06%;

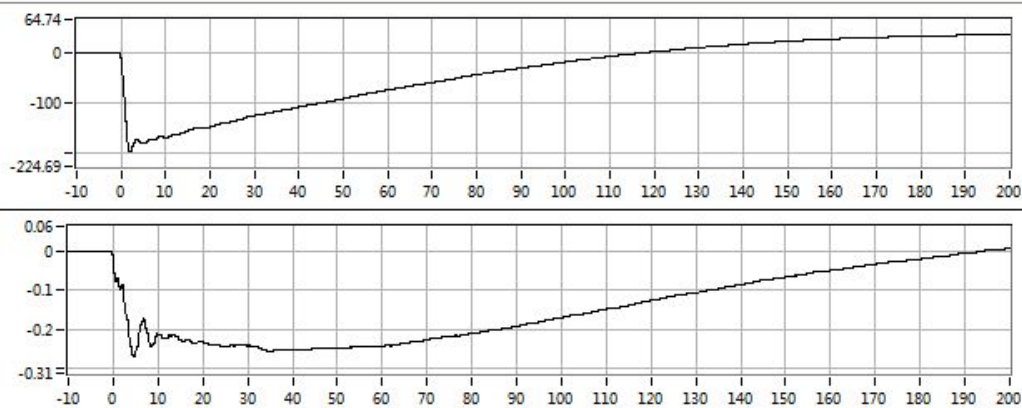
Max=0.009 kA; Min=-0.267 kA;



低压a 100%雷电全波_5

Upk=-196.82 kV; T1=1.643 us; T2=45.776 us;

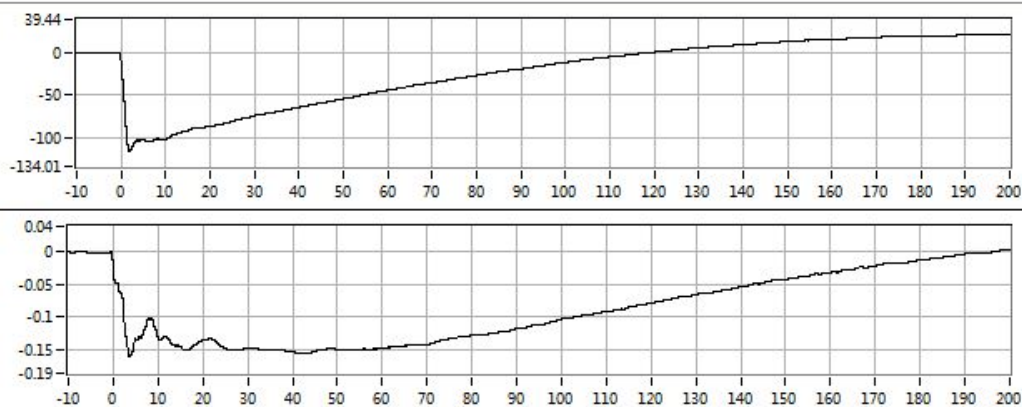
Max=0.024 kA; Min=-0.266 kA;



低压a 100%雷电全波_6

Upk=-195.13 kV; T1=1.647 us; T2=45.860 us;

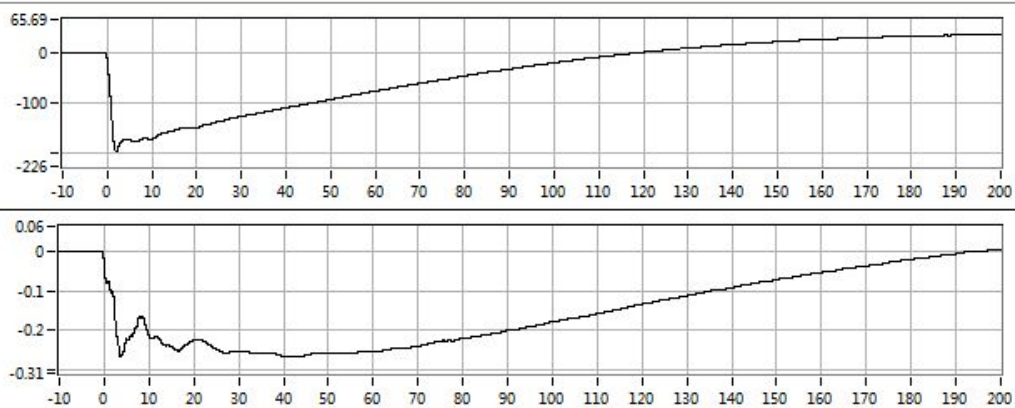
Max=0.023 kA; Min=-0.265 kA;



低压b 50%雷电全波_1

Upk=-116.77 kV; T1=1.656 us; T2=46.097 us;

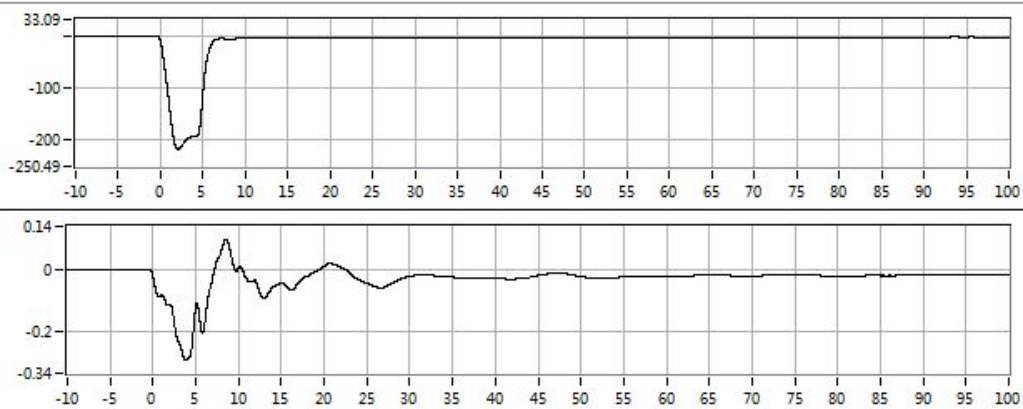
Max=0.014 kA; Min=-0.158 kA;



低压b 100%雷电全波_2

Upk=-196.29 kV; T1=1.664 us; T2=46.823 us;

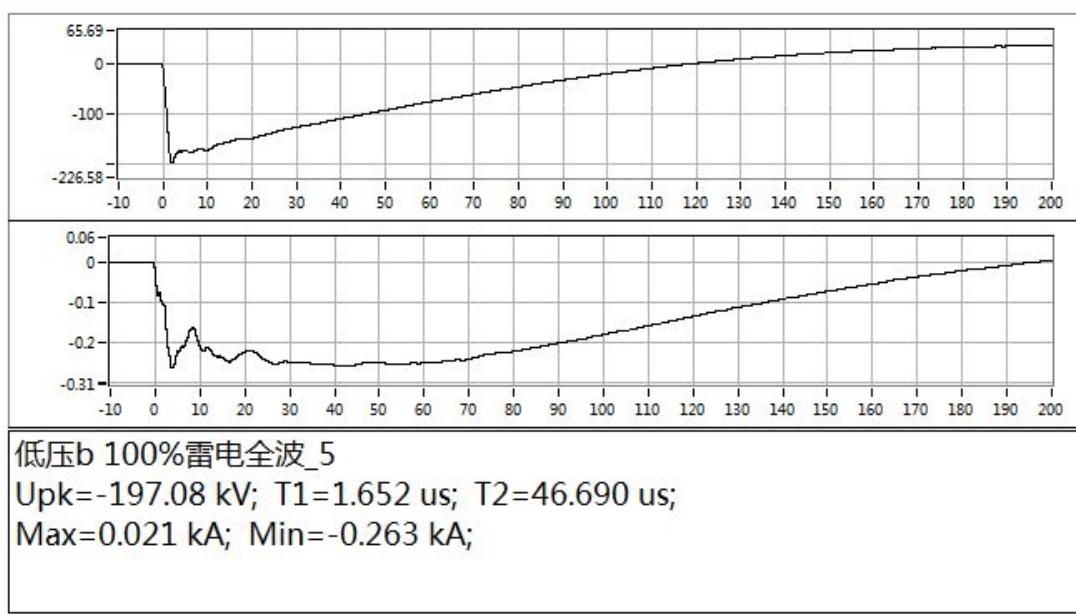
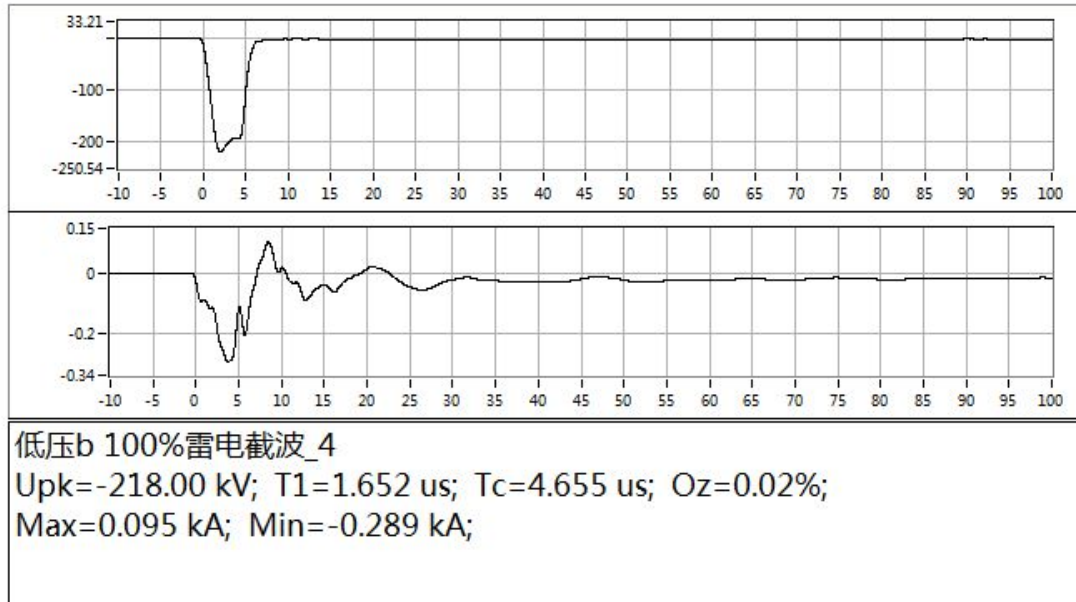
Max=0.022 kA; Min=-0.266 kA;

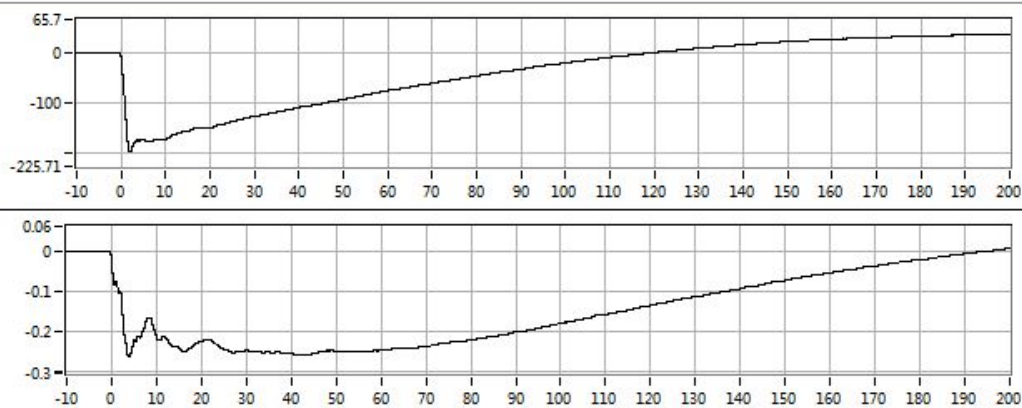


低压b 100%雷电截波_3

Upk=-217.94 kV; T1=1.667 us; Tc=4.631 us; Oz=0.14%;

Max=0.088 kA; Min=-0.289 kA;

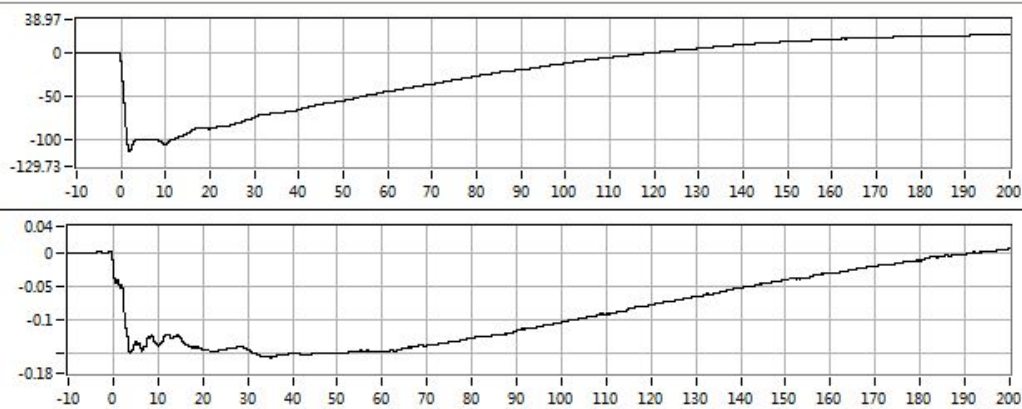




低压b 100%雷电全波_6

Upk=-196.32 kV; T1=1.656 us; T2=46.723 us;

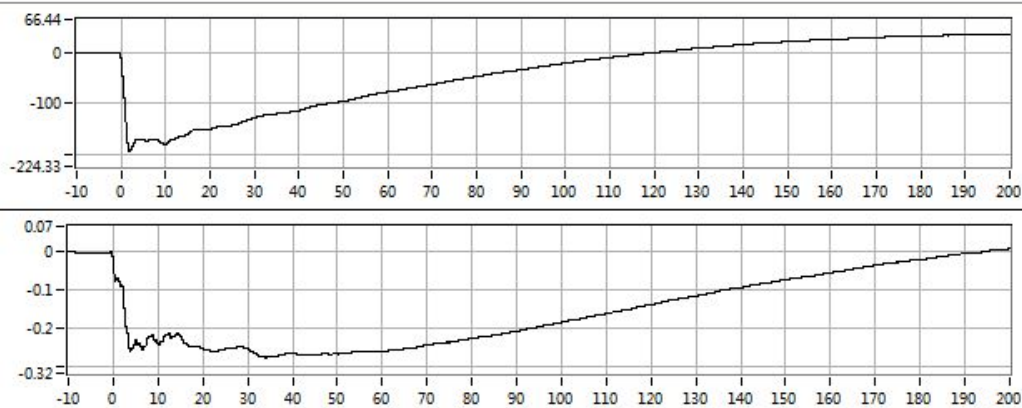
Max=0.022 kA; Min=-0.261 kA;



低压c 50%雷电全波_1

Upk=-113.08 kV; T1=1.648 us; T2=47.738 us;

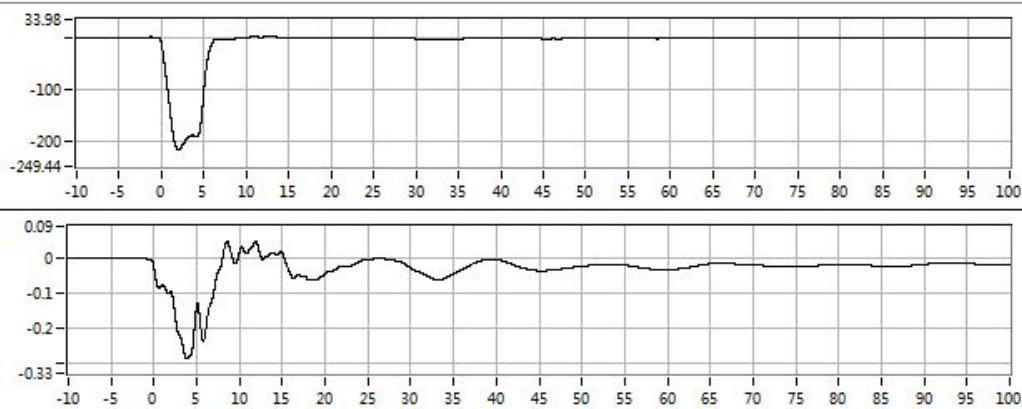
Max=0.016 kA; Min=-0.157 kA;



低压c 100%雷电全波_2

Upk=-195.22 kV; T1=1.652 us; T2=48.428 us;

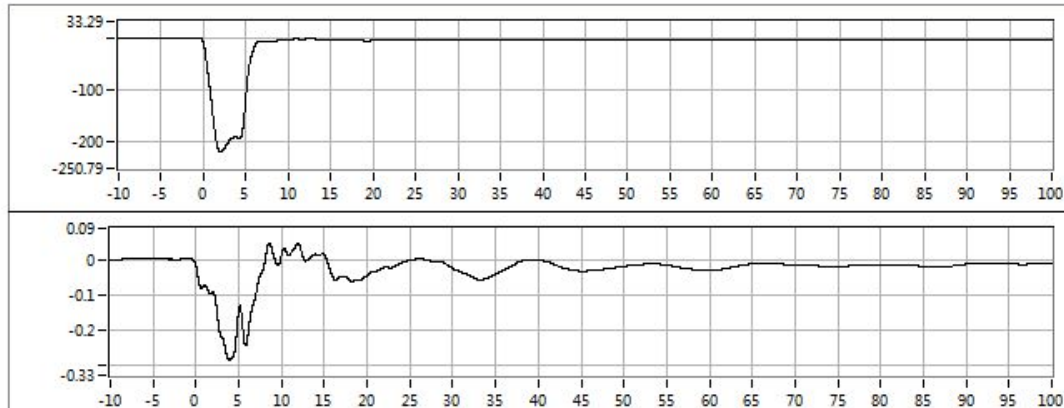
Max=0.026 kA; Min=-0.273 kA;



低压c 100%雷电截波_3

Upk=-216.89 kV; T1=1.653 us; Tc=4.679 us; Oz=0.67%;

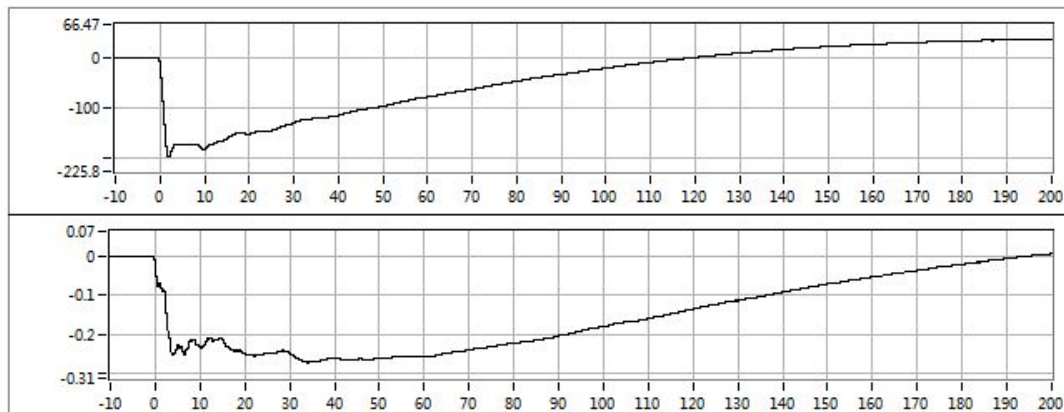
Max=0.038 kA; Min=-0.280 kA;



低压c 100%雷电截波_4

Upk=-218.04 kV; T1=1.643 us; Tc=4.677 us; Oz=0.28%;

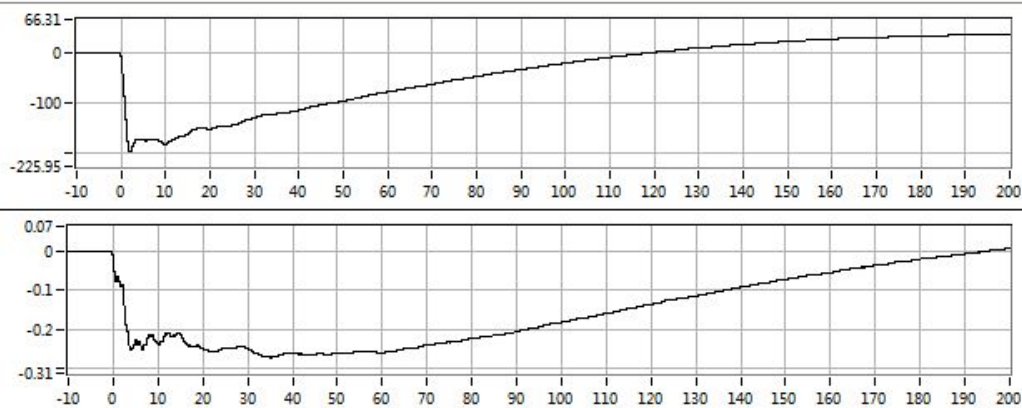
Max=0.035 kA; Min=-0.279 kA;



低压c 100%雷电全波_5

Upk=-196.18 kV; T1=1.634 us; T2=48.327 us;

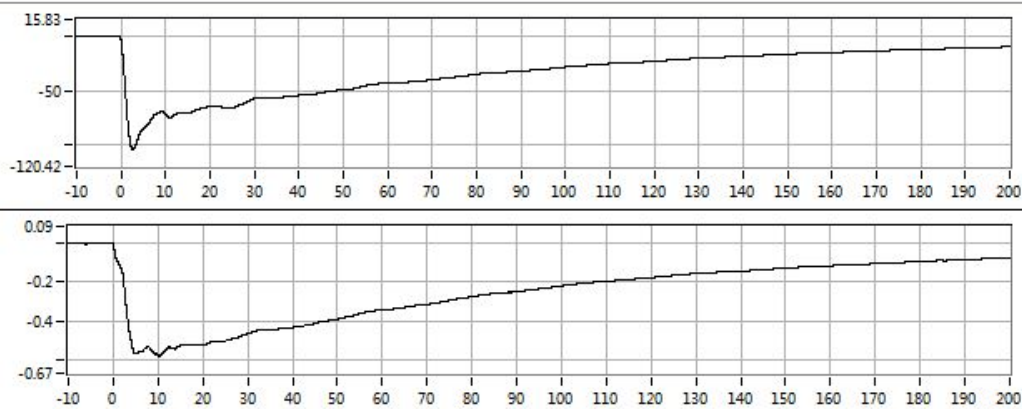
Max=0.025 kA; Min=-0.274 kA;



低压c 100%雷电全波_6

Upk=-196.41 kV; T1=1.647 us; T2=48.379 us;

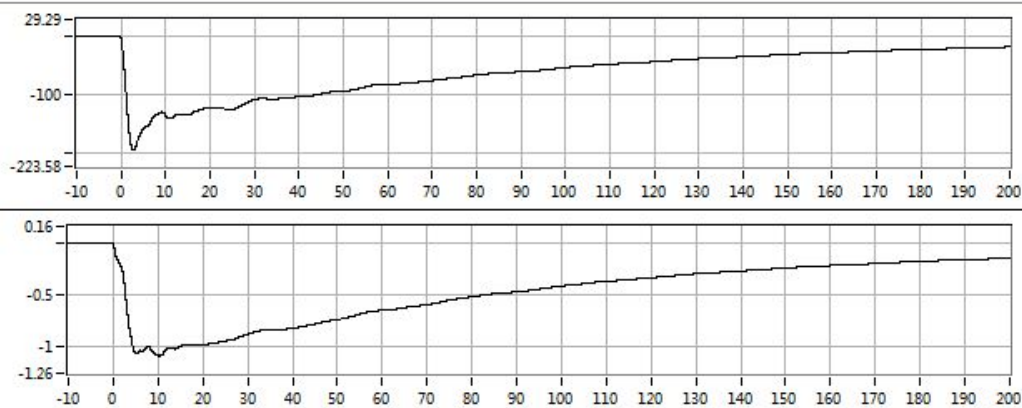
Max=0.024 kA; Min=-0.272 kA;



低压o 50%雷电全波_1

Upk=-104.76 kV; T1=2.095 us; T2=44.556 us;

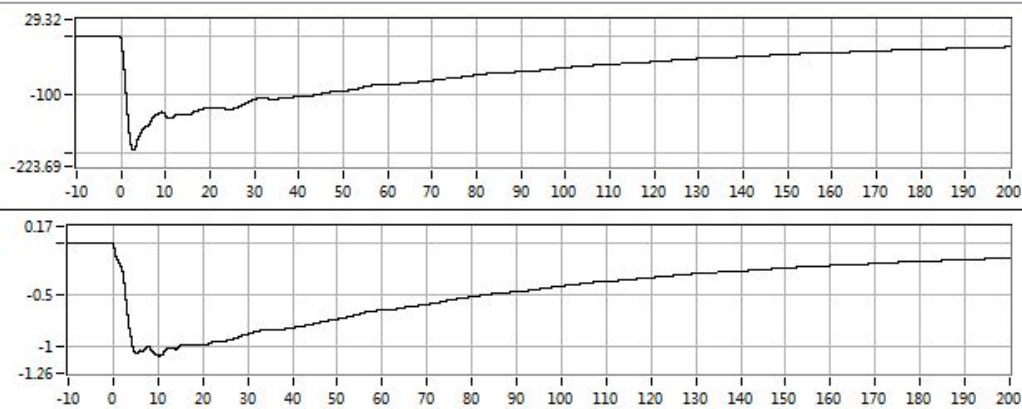
Max=0.002 kA; Min=-0.582 kA;



低压o 100%雷电全波_2

Upk=-194.43 kV; T1=2.107 us; T2=46.142 us;

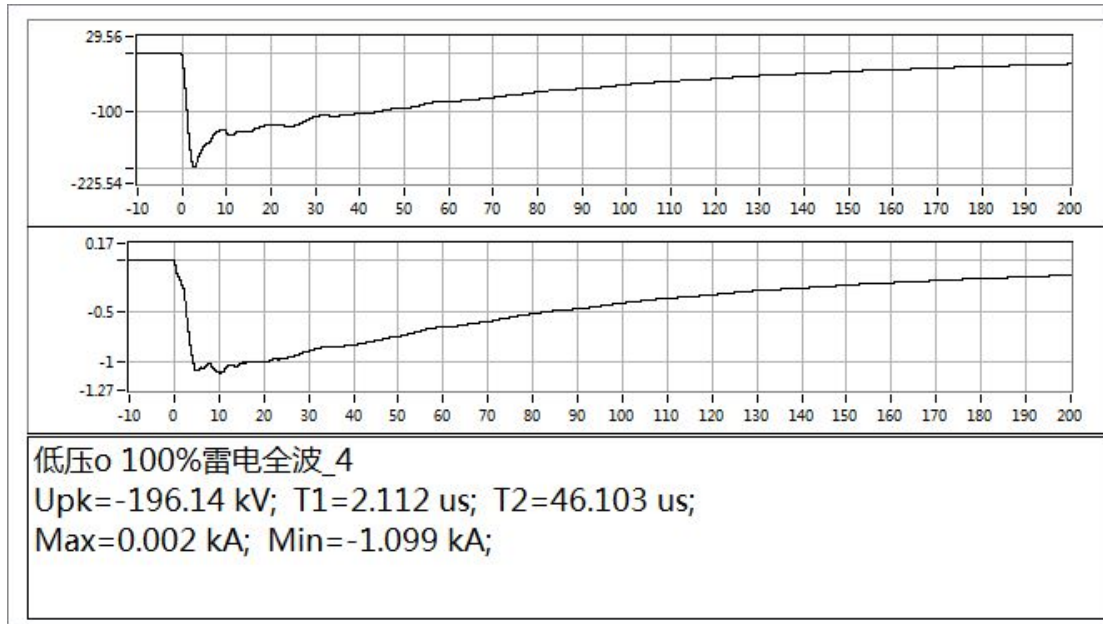
Max=0.001 kA; Min=-1.089 kA;



低压o 100%雷电全波_3

Upk=-194.56 kV; T1=2.119 us; T2=46.041 us;

Max=0.001 kA; Min=-1.092 kA;



Date

11.2 Applied-voltage tests**IEEE C57.12.90 Clause 10.6****IEEE C57.12.00 table 17, Clause 5.10.5.2****CSA C88 16.2 g), 16.3.2**

1.The test shall be performed at low frequency (< 500 Hz), and the duration of the test shall be 1 min.

2.The winding being tested shall have all its parts joined together and connected to the line terminal of the testing transformer.

3.All other terminals and parts (including core and tank) shall be connected to ground.

4.The voltage should be started at one quarter or less of the full value and be brought up gradually to full value. After being held for the time specified in 10.6.1, it should be reduced gradually before the circuit is opened.

5.Failure detection

Careful attention should be maintained for evidence of possible failure, such as an indication of smoke and bubbles rising in the insulating liquid, an audible sound such as a thump, or a sudden increase in test circuit current. Any such indication should be carefully investigated by observation, by repeating the test, or by other tests to determine whether a failure has occurred.

Measured Between		Test Voltage (kV)		Test Frequency (Hz)	Duration (s)	Result
H.V.~L.V & Earth & Stabilizing windings		95 @	KV	50	60	Pass
L.V ~H.V. & Earth & Stabilizing windings		50 @	KV	50	60	
Stabilizing windings ~H.V. & Earth		34	KV	50	60	
Test date:	2024/7/30		Test Sample		01	
Ambient Temp.:	37	℃	Relative Humidity:		63%	
Top Oil Temp. Indicator:	37	℃	Bottom Oil Temp. Indicator:		37	℃
Measuring instrument:						
temperature and humidity meter		J202404303427-0001				
Temperature Recorder		2JB24000718-0004				
Power-Frequency high voltage test equipment		24KA202406280026				
Stop Watch		2JB23008317-0012A				

@: Standard require 34KV (IEEE C57.12.00 Table 4 col 4), Customer required to test 95KV for HV winding and 85KV for LV winding.

Date

11.3 Induced-voltage test and Partial discharge measurement

IEEE C57.12.90, Clause 10.7
IEEE C57.12.00 table 17, 5.10.5.3
CSA C88 16.2 h), 16.3.3

1. Test duration

The induced-voltage test shall be applied for 7200 cycles, or 60 s, whichever is shorter.

2. Test frequency

As an induced-voltage test applies greater-than-rated volts per turn to the transformer, the frequency of the impressed voltage shall be high enough to limit the flux density in the core to that permitted by 4.1.6.1 of IEEE Std C57.12.00-2015. The minimum test frequency to meet this condition is given in Equation (27):

$$\text{Minimum test frequency} = \frac{E_t}{1.1 \times E_r} \times \text{rated frequency} \quad (27)$$

where

E_t is the induced voltage across winding (V)

E_r is the rated voltage across winding (V)

1. A voltage of 1.58 times maximum nameplate rating shall be applied for a minimum of 5 min.								
Tap Position	Tested Winding	Supplied Terminal	Supplied Voltage	Induced Voltage	Induced Multiple	Test	Duration	Result
			(kV)	(kV)		Frequency		
						(Hz)		
17b	H.V.	L.V.	54.51	181.7	1.58	200	300	PASS
2. Measured the stable partial discharge.								
until a stable partial discharge (pc)								
H1			H2			H3		
32			50			63		
3. A voltage of 1.8 times the maximum rating plate voltage shall then be applied for 7200 cycles.								
17b	H.V.	L.V.	62.1	207	1.80	200	36	PASS
17b	H.V.	L.V.	54.51	181.7	1.58	200	3600	PASS
The voltage shall then be reduced directly to the one-hour level and held for 1 h. partial discharge measurements shall be made at 5 min intervals.								
Test time	0 min			5 min			10 min	
	H1	H2	H3	H1	H2	H3	H1	H2
Value(PC)	34	52	62	35	60	61	35	60
Test time	15 min			20 min			25 min	
	H1	H2	H3	H1	H2	H3	H1	H2
Value(PC)	34	52	62	35	60	61	35	60
Test time	30 min			35 min			40 min	
	H1	H2	H3	H1	H2	H3	H1	H2
Value(PC)	34	52	62	35	60	61	35	60
Test time	45 min			50 min			55 min	
	H1	H2	H3	H1	H2	H3	H1	H2
Value(PC)	34	52	62	35	60	61	35	60
Test time	60 min			Pass				
	H1	H2	H3					
Value(PC)	34	52	62					
a) The magnitude of the partial discharge level does not exceed 250 pC during the 1-h test period.								
b) The increase in partial discharge levels during the 1-h period does not exceed 50 pC.								
c) The partial discharge levels during the 1-h period do not exhibit any steadily rising trend, and no sudden sustained increase in the levels occurs during the last 20 min of the test.								

Test date:	2024/7/30		Test Sample	01	
Ambient Temp.:	33	℃	Relative Humidity:	63%	
Top Oil Temp. Indicator:	33	℃	Bottom Oil Temp. Indicator:	33	℃
Measuring instrument:					
temperature and humidity meter		J202404303427-0001			
Temperature Recorder		2JB24000718-0004			
Multi-channel power analyzer		24KA202406280013			
Current transformer		24KA202406280019	24KA202406280021	24KA202406280022	
Voltage transformer		24KA202406280020	24KA202406280023	24KA202406280024	
Partial discharge Tester		24KA202406280015			
Stop Watch		2JB23008317-0012A			

Date

11.4 Insulation resistance

IEEE C57.12.00, Table 17
CSA C88, 16.2 O), 16.6
IEEE C57.12.90, Clause 10.11

1. Insulation resistance tests shall be made when specified. Insulation resistance tests are made to determine the insulation resistance from individual windings to ground or between individual windings. Insulation resistance tests are commonly measured in megohms or may be calculated from measurements of applied voltage and leakage current.

NOTE 1—The insulation resistance of electrical apparatus is of doubtful significance compared to the dielectric strength. It is subject to wide variation in design, temperature, dryness, and cleanliness of the parts. When the insulation resistance falls below prescribed values, it can, in most cases of good design and where no defect exists, be brought up to the required standard by cleaning and drying the apparatus. Therefore, the insulation resistance may be useful to indicate whether the apparatus is in suitable condition for application of dielectric test.

NOTE 2—Insulation resistances may vary with applied voltage, and any comparison should be made with measurements at the same voltage.

NOTE 3—Under no circumstances should tests be made while the transformer is under vacuum.

2. The test specimen shall have the following:

- a) All windings immersed in insulating liquid
- b) All windings short-circuited
- c) All bushings in place
- d) Temperature of windings and insulating liquid near (the reference temperature of) 20 °C.

3. The dc voltage applied for measuring insulation resistance to ground shall not exceed a value equal to the rms low-frequency applied voltage allowed in 10.6.

NOTE 1—Partial discharges should not be present during insulation resistance tests because they may result in erroneous values of insulation resistance.

NOTE 2—When measurements are to be made using dc voltages exceeding the rms operating voltage of the winding involved (or 1000 V for a solidly grounded wye winding), a relief gap may be employed to protect the insulation.

4. Insulation-resistance tests shall be made with all circuits of equal voltage above ground connected together. Circuits or groups of circuits of different voltage above ground shall be tested separately. Examples of procedures include the following:

- a) High voltage to low voltage and ground, low voltage to high voltage and ground.
- b) Voltage should be increased in increments, typically 1 kV to 5 kV, and held for 1 min while the current is read.
- c) The test should be discontinued immediately if the current begins to increase without stabilizing.

Date

d) After the test has been completed, all terminals should be grounded for enough time to allow any trapped charges to decay to a negligible value.

Measured Between		Insulation Resistance (MΩ)					test Voltage (V)
		Time (600S)	Time (60S)	R (15S)	R60/R15	R600/R60	2500`
High to low, tertiary, and ground		9560	5850	4330	1.35	1.63	
Low to high, tertiary, and ground		11090	7200	4640	1.55	1.54	
Tertiary to high, low, and ground		10500	5800	3120	1.86	1.81	
High and low to tertiary and ground		9090	5750	4470	1.29	1.58	
High and tertiary to low and ground		5740	3560	2330	1.53	1.61	
Low and tertiary to high and ground		8800	4910	2900	1.69	1.79	
High, low, and tertiary to ground		9240	5030	3350	1.29	1.84	
Core to ground(60S)		1500 MΩ(60s)					
Pass							
Test date:		2024/7/11		Test Sample		01	
Ambient Temp.:		37	℃	Relative Humidity:		69%	
Top Oil Temp. Indicator:		37	℃	Bottom Oil Temp. Indicator:		37℃	
Criterion:							
Test voltage: 5 KV							
Measuring instrument:							
temperature and humidity meter		J202404303427-0001					
Temperature Recorder		24KA202406280014					
High voltage insulation resistance tester		24KA202406280010					
Stop Watch		2JB23008317-0012A					

11.5 Insulation power-factor tests**IEEE C57.12.00, 10.10****CSA C88-16, 20.4.6 b)**

The insulation power factor is the ratio of the power dissipated in the insulation in watts to the product of the effective voltage and current in voltamperes when tested under a sinusoidal voltage and prescribed conditions.

The methods described in this standard are applicable to distribution and power transformers of present-day design that are immersed in an insulating liquid.

Preparation for tests

- a) All windings immersed in insulating liquid.
- b) All windings short-circuited.
- c) All bushings in place.
- d) The average temperature of the windings and insulating liquid should be between 10 °C and 40 °C, but preferably as close to 20 °C as practical. The top liquid temperature shall be measured and recorded.

Insulation power-factor tests shall be made from windings to ground and between windings as shown below table

The methods described in this standard are applicable to distribution and power transformers of present-day design that are immersed in an insulating liquid.

Insulation power-factor tests							
applied voltage		10000				KV	
Measured Between		Cx(PF)	Tg @ %	Location	Nameplate	Cx(PF)	Tg @ %
High to low, tertiary, and ground		11620	0.21	H1	401	402.5	0.291
Low to high, tertiary, and ground		17570	0.198	H2	400	402.1	0.283
Tertiary to high, low, and ground		19440	0.176	H3	398	400.3	0.281
High and low to tertiary and ground		14820	0.237	X1	498	501	0.287
High and tertiary to low and ground		30900	0.198	X2	502	505	0.29
Low and tertiary to high and ground		22150	0.208	X3	498	501.5	0.293
High, low, and tertiary to ground		19250	0.229	Pass			
bushing power-factor tests							result
	Nameplate	Cx(PF)	Tg @ %		Nameplate	Cx(PF)	Tg @ %
H1	401	402.5	0.291	X1	498	501	0.287
H2	400	402.1	0.283	X2	502	505	0.29
H3	398	400.3	0.281	X3	498	501.5	0.293
Test date:		2024/7/13		Test Sample		01	
Ambient Temp.:		33.9	°C	Relative Humidity:		53%	
Top Oil Temp. Indicator:		33.9	°C	Bottom Oil Temp. Indicator:		32.8	°C
Measuring instrument:							
temperature and humidity meter				24KA202406280031			
Temperature Recorder				24KA202406280018			
Dielectric loss resistivity tester				24KA202406280016			

**12. 1 h excitation run at 110% of rated
voltage follow by exciting current and loss
at 90%, 100%, and 110% of rated voltage**

CSA C88 16.2i), 16.9

A 1 h excitation run at 110% of rated voltage after the completion of all dielectric tests on all transformers with a voltage classification at or above 123 kV. At the completion of this 1 h test, the exciting current and loss shall be measured at 90%, 100%, and 110% of rated voltage.							
Tested Frequency		50	Hz	Tested Ambient Temp		37	°C
Corrected Frequency		60	Hz	Corrected Ambient Temp		20	°C
90%	Vrms (V)			Irm (A)			
	H1	H2	H3	H1	H2	H3	I ₀ (%)
	25652	25465	25634	0.867	0.501	0.867	/
	Vavg (V)			P (w)			Corrected Loss
	H1	H2	H3	H1	H2	H3	
	25614	25425	25608	9389	6243	12086	
100%	Vrms (V)			Irm (A)			
	H1	H2	H3	H1	H2	H3	I ₀ (%)
	28776	28613	28786	1.102	0.635	1.039	0.17%
	Vavg (V)			P (w)			Corrected Loss
	H1	H2	H3	H1	H2	H3	
	28760	28605	28798	12460	7790	15702	
110%	Vrms (V)			Irm (A)			
	H1	H2	H3	H1	H2	H3	I ₀ (%)
	31631	31434	31613	3.34			/
	Vavg (V)			P (w)			Corrected Loss
	H1	H2	H3	H1	H2	H3	
	31676	31488	31696	15240	9636	20240	

Measuring instrument:			
temperature and humidity meter	J202404303427-0001		
Temperature Recorder	2JB24000718-0004		
Multi-channel power analyzer	24KA202406280013		
Current transformer	24KA202406280019	24KA202406280021	24KA202406280022
Voltage transformer	24KA202406280020	24KA202406280023	24KA202406280024
Stop Watch	2JB23008317-0012A		
DC resistance tester	24KA202406280012		

Test date:	2024/7/31	Test Sample	01
Ambient Temp.:	33 °C	Relative Humidity:	54%
Top Oil Temp. Indicator:	33 °C	Bottom Oil Temp. Indicator:	33 °C

13.Dissolved gas in oil analysis

IEEE C57.12.00, Table 17

4.3 变压器油试验 介电损失角正切(90℃): 0.086 %; 油击穿电压: 65.8 kV; 微量水: 10.8 mg/L; 试验前气相色谱分析(μL/L):								
项 目	H ₂	CO	CO ₂	CH ₄	C ₂ H ₆	C ₂ H ₄	C ₂ H ₂	总 烃
试验结果	0	2.59	118.40	0.18	0	0	0	0.18
冲击后感应前气相色谱分析(μL/L):								
项 目	H ₂	CO	CO ₂	CH ₄	C ₂ H ₆	C ₂ H ₄	C ₂ H ₂	总 烃
试验结果	0	3.12	169.61	0.27	0	0	0	0.27
感应耐压和局放试验后气相色谱分析(μL/L):								
项 目	H ₂	CO	CO ₂	CH ₄	C ₂ H ₆	C ₂ H ₄	C ₂ H ₂	总 烃
试验结果	0	3.68	216.40	0.36	0	0	0	0.36
长时空载前气相色谱分析(μL/L):								
项 目	H ₂	CO	CO ₂	CH ₄	C ₂ H ₆	C ₂ H ₄	C ₂ H ₂	总 烃
试验结果	0	4.14	223.23	0.38	0	0	0	0.38
长时空载后气相色谱分析(μL/L):								
项 目	H ₂	CO	CO ₂	CH ₄	C ₂ H ₆	C ₂ H ₄	C ₂ H ₂	总 烃
试验结果	0	5.34	245.71	0.41	0	0	0	0.41
温升前气相色谱分析(μL/L):								
项 目	H ₂	CO	CO ₂	CH ₄	C ₂ H ₆	C ₂ H ₄	C ₂ H ₂	总 烃
试验结果	0	7.77	151.57	0.24	0	0	0	0.24
温升后气相色谱分析(μL/L):								
项 目	H ₂	CO	CO ₂	CH ₄	C ₂ H ₆	C ₂ H ₄	C ₂ H ₂	总 烃
试验结果	0	41.23	298.26	1.10	0	0	0	1.10
变压器油牌号: 25 [#]								
项 目				结 果				
外 观				无色、透明				
机 械 杂 质				无				
游 离 炭				无				
闪 点 (闭 口) (°C)				142				
酸 值 反 应 (PH)				6.0				
酸 值 (KOHmg/g)				0.01				
倾 点 (°C)				< -20				

Test date:	2023/7/11		Test Sample	01	
Ambient Temp.:	34	℃	Relative Humidity:	69%	
Top Oil Temp. Indicator:	34	℃	Bottom Oil Temp. Indicator:	34	℃
Test date:	2023/7/31		Test Sample	01	
Ambient Temp.:	33	℃	Relative Humidity:	69%	
Top Oil Temp. Indicator:	33	℃	Bottom Oil Temp. Indicator:	33	℃

Measuring instrument:

temperature and humidity meter	J202404303427-0001
Temperature Recorder	24KA202406280014
Oil withstand voltage tester	24KA202406280030
Micro water Analyzer	24KA202406280033
Chromatographic analyzer	24KA202406280029
Oil dielectric loss resistivity tester	24KA202406280031

14. Leak and Pressure**IEEE C57.12.00, Table 17
CSA C88 16.2i), 16.9**

1.A pressure test shall be performed on all transformer tanks, sealed compartments, and liquid-filled accessories, using liquid maintained at a minimum temperature of 20 °C. Valves may be blanked off and temporary pressure relief devices may be used for this test. The test pressure shall be 125% of the tank maximum operating pressure stated in the transformer rating plate. When testing the main tank, the pressure shall be applied at the tank cover level.

2.No visible leakage or permanent compartment deformation shall occur with the test conditions maintained for a 24-hour period. Powdered chalk or other suitable means shall be used in leak detection.

3.The transformer shall have sufficient strength to withstand a range of gauge pressures between -35 kPa to +50 kPa, without leakage or permanent distortion of the tank exceeding 0.25% of the diagonal dimension of the surface so affected.

Nameplate rating	Position	Pressure (K Pa)	time	Pressure (K Pa)	result
		begin		End	
40Kpa	enclosure	50	24h	48.7	Pass
	OLTC	50		48.5	
Test date:	2024/7/14		Test Sample	01	
Ambient Temp.:	34	℃	Relative Humidity:	53%	
Top Oil Temp. Indicator:	34	℃	Bottom Oil Temp. Indicator:	34	℃
Test date:	2024/7/15		Test Sample	01	
Ambient Temp.:	33	℃	Relative Humidity:	53%	
Top Oil Temp. Indicator:	33	℃	Bottom Oil Temp. Indicator:	33	℃
Test date:	2024/7/16		Test Sample	01	
Ambient Temp.:	33.2	℃	Relative Humidity:	53%	
Top Oil Temp. Indicator:	33.2	℃	Bottom Oil Temp. Indicator:	33.2	℃

Measuring instrument:

temperature and humidity meter	24KA202406280031
Temperature Recorder	24KA202406280018
Pressure gauge	2JB24001055-0003
Stop Watch	2JB23008317-0012A

15.Lifting and moving devices**IEEE C57.12.00, Table 17**

Test date:	2024/7/14		Test Sample	01	
Ambient Temp.:	27	°C	Relative Humidity:	53%	
Top Oil Temp. Indicator:	27	°C	Bottom Oil Temp. Indicator:	27	°C

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1 计算要求及模型参数 Calculation requirements and model parameters

1.1 计算要求 Calculation requirements

变压器起吊重量: 4 倍重力加速度起吊

The lifting weight of transformer required for calculation: 4 times of gravity acceleration

计算目的: 在上述重量起吊时, 吊板及其他零件不得有损伤和不允许的永久变形。

Calculation purpose: when lifting with the above weight, the lifting plate and other parts shall be free of damage and impermissible permanent deformation.

1.2 模型参数 Model parameters

115kV-34.5-13.2 变压器结构件简化模型结构如图 1 所示。

The simplified model structure of 115kV-34.5-13.2 transformer structure is shown in Figure 1.

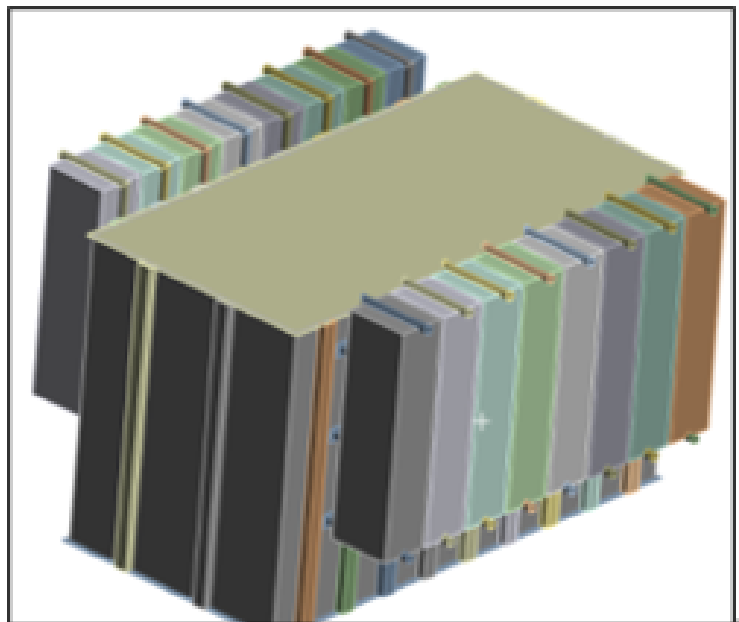


图 1 变压器三维模型示意图

Fig. 1 Schematic Diagram of Three-dimensional Model of Transformer

• 1.3 变压器材料特性 Transformer material characteristics

顶盖、箱壁、吊环、底座等材料为 Q235 钢板，其主要参数如下：

The cover, tank wall, rings, seat and other materials are Q235 steel plates, and the main parameters are as follows:

弹性模量 Elastic modulus: 218000MPa

泊松比 Poisson's ratio: 0.3

密度 density: 7850kg/m³

屈服强度 yield strength: 235MPa

抗拉强度 tensile strength: 460MPa

• 1.4 模型有限元网格划分 Finite element mesh generation of model

模型有限元网格划分如图 2 所示。

The finite element mesh generation of the model is shown in Figure 2.

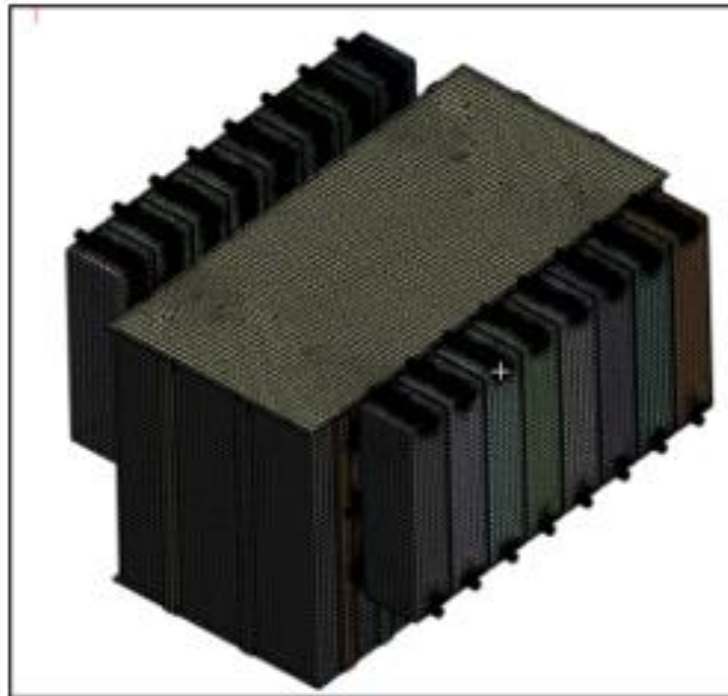


图 2 网格划分示意图

Fig. 2 Schematic Diagram of Grid Division

2 变压器起吊强度计算 Calculation of transformer lifting strength

2.1 起吊重量 Lifting weight

变压器起吊重量: 4 倍重力加速度起吊, 加速度= $4 \times 9806 \text{ m/s}^2 = 39224 \text{ m/s}^2$. 加载如图 3 所示.

Lifting weight of transformer: 4 times of gravity acceleration, acceleration value= $4 \times 9806 \text{ m/s}^2 = 39224 \text{ m/s}^2$. Load loading is shown in Figure 3.

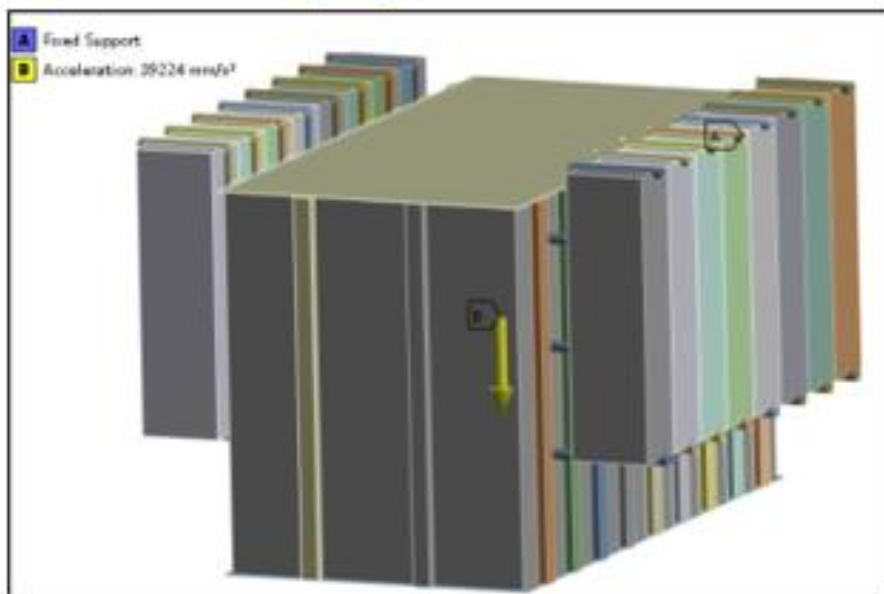


图 3 载荷及约束位置示意图

Figure 3 Schematic Diagram of Load and Constraint Position

2.2 计算结果 Calculation results

变压器的等效应力云图如图 4-6 所示, 单位: MPa.

The equivalent stress nephogram of transformer is shown in Figure 4-8, unit: MPa.

变压器的变形云图如图 7 所示, 单位: mm.

The deformation cloud diagram of transformer is shown in Fig. 9, unit: mm.

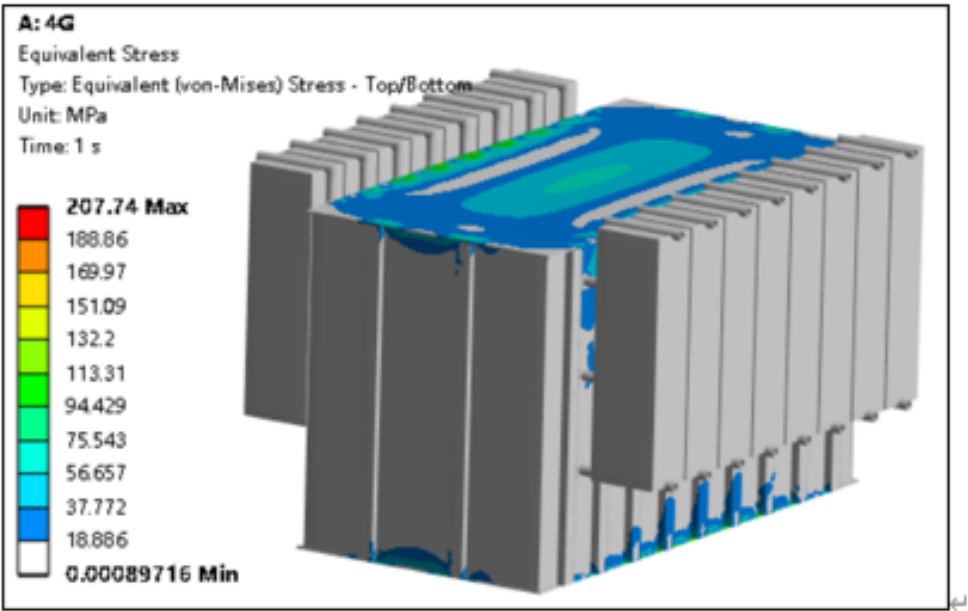


图 4 变压器总体应力云图（单位：MPa）
Fig. 4 Stress nephogram of Transformer (Unit: MPa)

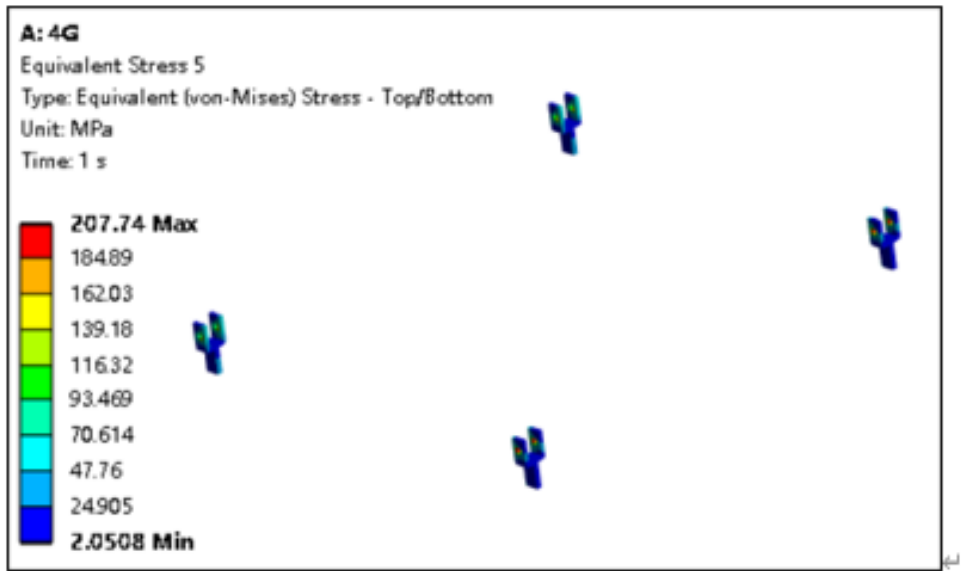


图 5 吊耳应力云图（单位：MPa）
Fig. 5 Stress nephogram of Lifting Lug (Unit: MPa)

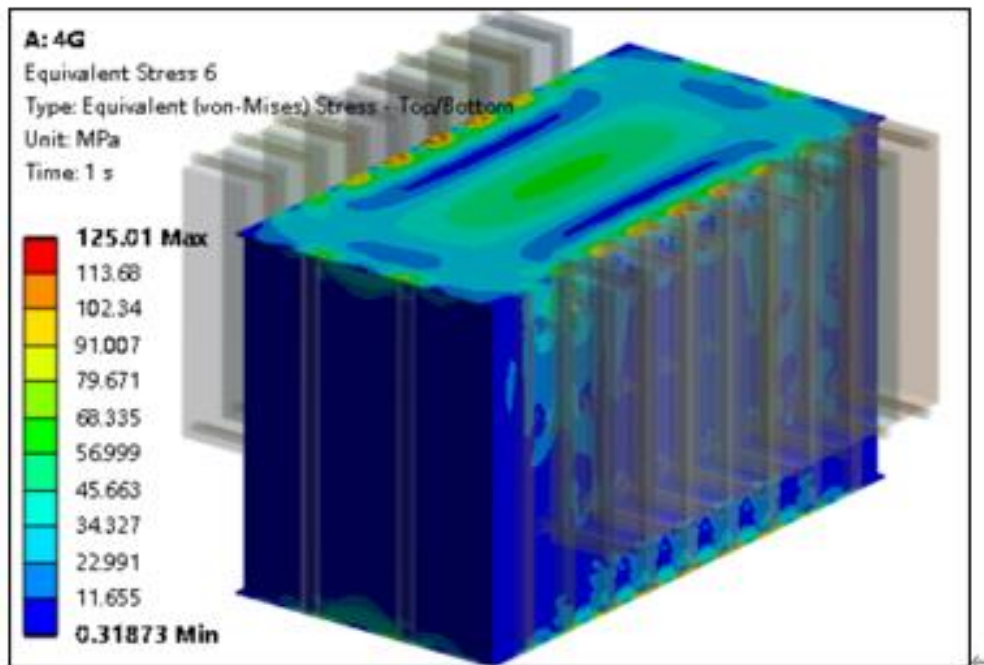


图 6 箱壁及箱底应力云图 (单位: MPa) ←

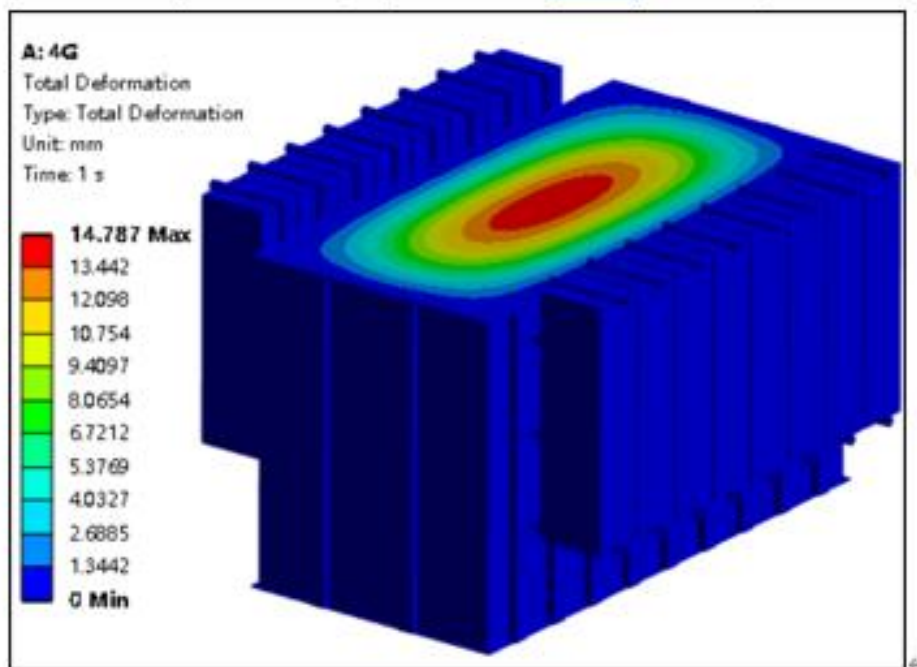
Fig. 6 Stress nephogram of tank wall (unit: MPa) ←

图 7 变压器总变形云图 (单位: mm) ←

Figure 7 Total Deformation nephogram of Transformer (Unit: mm) ←

表 1 变压器起吊强度计算结果汇总 Summary of calculation results

部件部件	最大应力 Maximum stress	许用应力 Allowable stress	安全系数 safety factor	变形量 Amount of deformation
总体 total	207.74MPa	235MPa	1.13	14.79mm
吊耳 lifting lug	207.74MPa	235MPa	1.13	0.31mm
油箱壳体 tank	125.01 MPa	235MPa	1.88	14.79mm

综上，4 倍重力起吊时，变压器的最大应力在 Q235 材料的屈服强度内，吊板及其他零件不得有损伤和不允许的永久变形。

To sum up, when lifting with 4 times of gravity, the maximum stress of the transformer is within the yield strength of Q235 material, and the hanging plate and other parts shall be free of damage and impermissible permanent deformation.

16. Test on Current transformer

For reference only

1. Product specification and quantity

No.	Type	Current Ratio A	Performance parameter	Quantity
1	LRB-115	200-300-800-1200/5	20VA 40VA 120VA 200VA C800	12
2	LR-115	600/5	45VA0.3B1.8	3
3	LRB-115	600/5	25VA C100	1
4	LR-115	600/1	45VA0.3B1.8	1
5	LRB-115	200-300-800-1200/5	20VA 40VA 120VA 200VA C800	1
6	LRB-34.5	400-1200-1500-2000/5	30VA 100VA 120VA 200VA C800	6
7	LRB-34.5	2000/5	200VA C800	3
8	LR-34.5	1500/5	45VA0.3B1.8	3
9	LR-34.5	1500/1	45VA0.3B1.8	1
10	LRB-13.2	200-300-800-1200/5	20VA 40VA 120VA 200VA C800	1
11	LRB-13.2	1200/5	200VA C800	1

2 Error test data and voltammetry characteristics

2.1.1 LRB-115 (200-300-800-1200/5 200VAC800) Error test data

Product serial number	Secondary terminal mark	Current ratio A	Secondary Burden VA	Percentage of rated primary current %	Error error	
					Current error ±%	Phase differenc e ±°
J2405037	1X1-1X2	200/5	20	100	-1.988	19.03
	1X1-1X3	300/5	40	100	0.192	16.38
	1X1-1X4	800/5	120	100	-0.356	6.316
	1X1-1X5	1200/5	200	100	0.080	4.414
J2405038	1X1-1X2	200/5	20	100	0.172	18.87
	1X1-1X3	300/5	40	100	0.104	14.12
	1X1-1X4	800/5	120	100	-0.324	5.468
	1X1-1X5	1200/5	200	100	-0.021	3.826
J2405039	1X1-1X2	200/5	20	100	-1.945	21.32
	1X1-1X3	300/5	40	100	0.168	15.97
	1X1-1X4	800/5	120	100	-0.300	6.206
	1X1-1X5	1200/5	200	100	0.070	4.248
J2405040	2X1-2X2	200/5	20	100	0.133	17.45
	2X1-2X3	300/5	40	100	0.085	13.18
	2X1-2X4	800/5	120	100	-0.324	4.995

Date

	2X1-2X5	1200/5	200	100	-0.021	3.482
J2405041	2X1-2X2	200/5	20	100	0.108	17.37
	2X1-2X3	300/5	40	100	0.046	13.13
	2X1-2X4	800/5	120	100	-0.345	5.085
	2X1-2X5	1200/5	200	100	0.061	3.518
J2405042	2X1-2X2	200/5	20	100	-2.002	19.28
	2X1-2X3	300/5	40	100	0.152	14.23
	2X1-2X4	800/5	120	100	-0.351	5.402
	2X1-2X5	1200/5	200	100	0.068	3.719
J2405043	3X1-3X2	200/5	20	100	-1.791	21.16
	3X1-3X3	300/5	40	100	-1.536	15.74
	3X1-3X4	800/5	120	100	-0.317	5.973
	3X1-3X5	1200/5	200	100	0.070	4.176
J2405044	3X1-3X2	200/5	20	100	0.130	18.0
	3X1-3X3	300/5	40	100	0.080	13.5
	3X1-3X4	800/5	120	100	-0.300	5.165
	3X1-3X5	1200/5	200	100	-0.044	4.271
J2405045	3X1-3X2	200/5	20	100	-1.512	21.28
	3X1-3X3	300/5	40	100	0.107	16.14
	3X1-3X4	800/5	120	100	-0.252	6.346
	3X1-3X5	1200/5	200	100	-0.011	4.549
J2405046	4X1-4X2	200/5	20	100	0.161	20.21
	4X1-4X3	300/5	40	100	0.171	15.45
	4X1-4X4	800/5	120	100	-0.294	6.047
	4X1-4X5	1200/5	200	100	-0.002	4.268
J2405047	4X1-4X2	200/5	20	100	-1.805	17.96
	4X1-4X3	300/5	40	100	-0.121	13.59
	4X1-4X4	800/5	120	100	-0.342	5.302
	4X1-4X5	1200/5	200	100	-0.013	3.685
J2405048	4X1-4X2	200/5	20	100	0.176	19.49
	4X1-4X3	300/5	40	100	0.077	14.71
	4X1-4X4	800/5	120	100	-0.239	5.867

	4X1-4X5	1200/5	200	100	0.004	4.152
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2.1.2 LRB-115 (200-300-800-1200/5 200VAC800) Excitation data

Exciting current A Induced voltage V			0.05A	0.1A	0.25A	1A	5A
ID	J2405037	1X1-1X2	4.1	10	36	128	158
		1X1-1X3	10	27	105	211	243
		1X1-1X4	149	375	523	619	638
		1X1-1X5	472	685	792	815	820
	J2405038	1X1-1X2	4.5	11	38	138	158
		1X1-1X3	11	29	125	223	243
		1X1-1X4	175	469	575	632	665
		1X1-1X5	623	808	896	964	996
	J2405039	1X1-1X2	4.0	10	33	134	157
		1X1-1X3	10.2	25	103	211	224
		1X1-1X4	129	377	425	434	436
		1X1-1X5	536	754	867	941	975
	J2405040	2X1-2X2	4.3	11	37	129	161
		2X1-2X3	11	27	107	221	230
		2X1-2X4	172	418	571	638	668
		2X1-2X5	545	780	902	972	1006
	J2405041	2X1-2X2	4.1	9.8	31	137	160
		2X1-2X3	10	24	89	211	224
		2X1-2X4	131	418	475	485	489
		2X1-2X5	577	792	896	954	970
	J2405042	2X1-2X2	3.9	10	35	140	160
		2X1-2X3	11	27	111	223	242
		2X1-2X4	165	463	577	631	656
		2X1-2X5	606	805	897	960	996
	J2405043	3X1-3X2	4.0	10	34	137	157
		3X1-3X3	10	25	111	218	240
		3X1-3X4	147	434	558	619	661
		3X1-3X5	554	768	873	945	969

	J2405044	3X1-3X2	4.3	11	37	139	161
		3X1-3X3	10	27	120	220	233
		3X1-3X4	167	450	570	634	667
		3X1-3X5	589	787	895	967	1008
	J2405045	3X1-3X2	4.9	12	43	131	158
		3X1-3X3	12	32	124	212	241
		3X1-3X4	176	412	535	617	664
		3X1-3X5	526	725	852	927	945
	J2405046	4X1-4X2	4.7	12	37	134	158
		4X1-4X3	12	29	119	208	219
		4X1-4X4	165	425	545	620	651
		4X1-4X5	549	743	860	949	970
	J2405047	4X1-4X2	4.5	11	39	134	160
		4X1-4X3	11	29	126	217	245
		4X1-4X4	183	420	546	630	669
		4X1-4X5	550	735	868	960	980
	J2405048	4X1-4X2	4.9	12	39	135	157
		4X1-4X3	12	30	125	216	239
		4X1-4X4	175	432	551	617	651
		4X1-4X5	558	757	868	939	957

2.2.1 LR-115 (600/5 45VA0.3B1.8) Error test data

Product serial number	Secondary terminal mark	Current ratio A	Secondary Burden VA	Percentage of rated primary current %	Error error	
					Current error ±%	Phase difference ±°
J2405049	5X1-5X2	600/5	45	5	-0.08	5.06
				10	-0.02	4.71
				20	0.01	3.88
				100	0.04	1.32
				120	0.04	1.98
			11.25	120	0.08	1.02
J2405050	5X1-5X2	600/5	45	5	-0.09	5.07
				10	-0.02	4.73
				20	0.01	3.90
				100	0.05	1.33
				120	0.05	1.98
			11.25	120	0.09	1.03
J2405051	5X1-5X2	600/5	45	5	-0.08	5.07
				10	-0.03	4.74

				20	0.02	3.90
				100	0.06	1.33
				120	0.06	1.99
			11.25	120	0.10	1.04

2.3.1 LRB-115 (600/5 25VAC100) Error test data

Product serial number	Secondary terminal mark	Current ratio A	Secondary Burden VA	Percentage of rated primary current %	Error error	
					Current error ±%	Phase difference ±°
J2405052	6X1-6X2	600/5	25	100	0.049	10.54

3.3.2 LRB-115 (600/5 25VAC100) Excitation data

Exciting current A Induced voltage V			0.05A	0.1A	0.25A	1A	5A
ID	J2405052	6X1-6X2	14	38	97	129	139

2.4.1 LR-115 (600/1 45VA0.3B1.8) Error test data

Product serial number	Secondary terminal mark	Current ratio A	Secondary Burden VA	Percentage of rated primary current %	Error error	
					Current error ±%	Phase difference ±°
J2405053	6X1-6X2	600/1	45	5	-0.12	5.33
				10	-0.07	4.77
				20	-0.03	3.82
				100	0.01	2.63
				120	-0.03	2.79
				11.25	-0.01	2.73

3.5.1 LRB-115 (200-300-800-1200/5 200VAC800) Error test data

Product serial number	Secondary terminal mark	Current ratio A	Secondary Burden VA	Percentage of rated primary current %	Error error	
					Current error ±%	Phase difference ±°
J2405054	1X1-1X2	200/5	20	100	-2.024	36.38
	1X1-1X3	300/5	40	100	-1.770	28.48
	1X1-1X4	800/5	120	100	-0.465	6.068
	1X1-1X5	1200/5	200	100	0.001	4.156

2.5.2 LRB-115 (200-300-800-1200/5 200VAC800) Excitation data

Exciting current A Induced voltage V			0.05A	0.1A	0.25A	1A	5A
ID	J2405054	1X1-1X2	2.9	7.6	24	120	149
		1X1-1X3	7.6	19	63	203	227
		1X1-1X4	76	348	423	434	437

		1X1-1X5	452	694	826	895	938
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2.6.1 LRB-34.5 (400-1200-1500-2000/5 200VAC800) Error test data

Product serial number	Secondary terminal mark	Current ratio A	Secondary Burden VA	Percentage of rated primary current %	Error error	
					Current error ±%	Phase difference ±°
J2405055	1X1-1X2	400/5	30	100	-0.027	6.397
	1X1-1X3	1200/5	100	100	-0.027	2.160
	1X1-1X4	1500/5	120	100	0.004	1.663
	1X1-1X5	2000/5	200	100	0.014	1.364
J2405056	1X1-1X2	400/5	30	100	0.016	8.776
	1X1-1X3	1200/5	100	100	0.055	3.003
	1X1-1X4	1500/5	120	100	-0.012	2.352
	1X1-1X5	2000/5	200	100	-0.001	1.913
J2405057	1X1-1X2	400/5	30	100	-0.034	6.140
	1X1-1X3	1200/5	100	100	-0.023	2.093
	1X1-1X4	1500/5	120	100	0.004	1.623
	1X1-1X5	2000/5	200	100	0.012	1.305
J2405058	2X1-2X2	400/5	30	100	-0.016	7.616
	2X1-2X3	1200/5	100	100	0.023	3.111
	2X1-2X4	1500/5	120	100	0.020	2.021
	2X1-2X5	2000/5	200	100	-0.009	1.630
J2405059	2X1-2X2	400/5	30	100	-0.365	6.796
	2X1-2X3	1200/5	100	100	-0.005	2.315
	2X1-2X4	1500/5	120	100	0.015	1.778
	2X1-2X5	2000/5	200	100	0.025	1.466
J2405060	2X1-2X2	400/5	30	100	0.009	6.533
	2X1-2X3	1200/5	100	100	-0.006	2.215
	2X1-2X4	1500/5	120	100	0.014	1.700
	2X1-2X5	2000/5	200	100	0.024	1.398

2.6.2 LRB-34.5 (400-1200-1500-2000/5 200VAC800) Excitation data

Exciting current A Induced voltage V			0.05A	0.1A	0.25A	1A	5A
ID	J2405055	1X1-1X2	22	51	58	188	202

		1X1-1X3	430	507	555	589	601
		1X1-1X4	582	652	701	745	762
		1X1-1X5	827	896	950	999	1016
	J2405056	1X1-1X2	19	49	160	193	205
		1X1-1X3	427	526	572	604	615
		1X1-1X4	598	676	722	762	775
		1X1-1X5	860	924	976	1020	1036
	J2405057	1X1-1X2	22	50	156	188	201
		1X1-1X3	423	504	554	589	605
		1X1-1X4	577	650	701	742	754
		1X1-1X5	818	894	949	999	1016
	J2405058	2X1-2X2	17	40	147	191	203
		2X1-2X3	397	518	565	596	610
		2X1-2X4	582	669	714	750	761
		2X1-2X5	847	918	965	1010	1026
	J2405059	2X1-2X2	26	60	160	188	201
		2X1-2X3	429	507	555	590	602
		2X1-2X4	580	653	703	742	753
		2X1-2X5	823	896	950	997	1012
	J2405060	2X1-2X2	25	58	162	191	205
		2X1-2X3	435	513	563	599	609
		2X1-2X4	586	661	713	754	767
		2X1-2X5	833	908	965	1012	1029

2.7.1 LRB-34.5 (2000/5 200VAC800) Error test data

Product serial number	Secondary terminal mark	Current ratio A	Secondary Burden VA	Percentage of rated primary current %	Error error	
					Current error ±%	Phase difference ±°
J2405061	3X1-3X2	2000/5	200	100	-0.012	1.662
J2405062	3X1-3X2	2000/5	200	100	-0.014	1.636
J2405063	3X1-3X2	2000/5	200	100	-0.015	1.643

2.7.2 LRB-34.5 (2000/5 200VAC800) Excitation data

Exciting current A	0.05A	0.1A	0.25A	1A	5A
Induced voltage V					

ID	J2405061	3X1-3X2	845	907	951	995	1011
	J2405062	3X1-3X2	842	906	949	987	1002
	J2405063	3X1-3X2	847	910	955	1001	1018

2.8.1 LR-34.5 (1500/5 45VA0.3B1.8) Error test data

Product serial number	Secondary terminal mark	Current ratio A	Secondary Burden VA	Percentage of rated primary current %	Error error	
					Current error ±%	Phase difference ±°
J2405064	4X1-4X2	1500/5	45	5	-0.11	1.76
				10	-0.05	1.63
				20	-0.02	1.38
				100	-0.02	0.58
				120	-0.02	1.11
			11.25	120	-0.01	0.39
J2405065	4X1-4X2	1500/5	45	5	-0.12	1.77
				10	-0.06	1.65
				20	-0.03	1.40
				100	-0.02	0.60
				120	-0.02	1.13
			11.25	120	-0.01	0.41
J2405066	4X1-4X2	1500/5	45	5	-0.12	1.78
				10	-0.07	1.68
				20	-0.04	1.42
				100	-0.02	0.62
				120	-0.02	1.12
			11.25	120	-0.01	0.42

2.9.1 LR-34.5 (1500/1 45VA0.3B1.8) Error test data

Product serial number	Secondary terminal mark	Current ratio A	Secondary Burden VA	Percentage of rated primary current %	Error error	
					Current error ±%	Phase difference ±°
J2405067	5X1-5X2	1500/1	45	5	-0.11	5.396
				10	-0.08	4.80
				20	-0.04	3.86
				100	0.01	2.66
				120	0.01	2.80
			11.25	120	-0.01	2.31

2.10.1 LRB-13.2 (200-300-800-1200/5 200VAC800) Error test data

Product serial number	Secondary terminal mark	Current ratio A	Secondary Burden VA	Percentage of rated primary current %	Error error	
					Current error ±%	Phase difference ±°
J2405068	1X1-1X2	200/5	20	100	-2.003	19.9
	1X1-1X3	300/5	40	100	0.133	14.7

	1X1-1X4	800/5	120	100	-0.309	5.572
	1X1-1X5	1200/5	200	100	0.060	3.861

2.10.2 LRB-13.2 (200-300-800-1200/5 200VAC800) Excitation data

Exciting current A Induced voltage V			0.05A	0.1A	0.25A	1A	5A
ID	J2405068	1X1-1X2	3.8	9.6	33	129	149
		1X1-1X3	10	25	109	202	216
		1X1-1X4	151	405	526	582	621
		1X1-1X5	529	720	824	889	931

2.11.1 LRB-13.2 (1200/5 200VAC800) Error test data

Product serial number	Secondary terminal mark	Current ratio A	Secondary Burden VA	Percentage of rated primary current %	Error error	
					Current error ±%	Phase difference ±°
J2405069	1X1-1X2	1200/5	200	100	-0.015	3.561

2.11.2 LRB-13.2 (1200/5 200VAC800) Excitation data

Exciting current A Induced voltage V			0.05A	0.1A	0.25A	1A	5A
ID	J2405069	1X1-1X2	612	749	826	886	919

END OF DATASHEET PACKAGE. THIS PAGE INTENTIONALLY LEFT BLANK